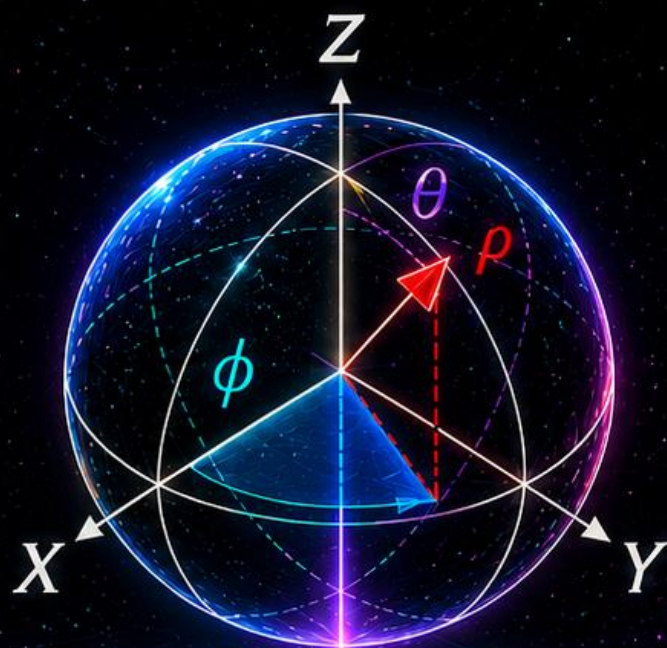




INTRODUCTION TO CIRCUIT QUANTUM ELECTRODYNAMICS (cQED)

ZLATKO K. MINEV

GOOGLE QUANTUM AI

FORMER AFFILIATION:

IBM QUANTUM • YALE UNIVERSITY

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The Light of Atoms

K 19
Potassium
Violet / Purple

Cu 29
Copper
Blue

Na 11
Sodium
Bright Yellow

Ca 20
Calcium
Orange

Sr 38
Strontium
Red / Crimson

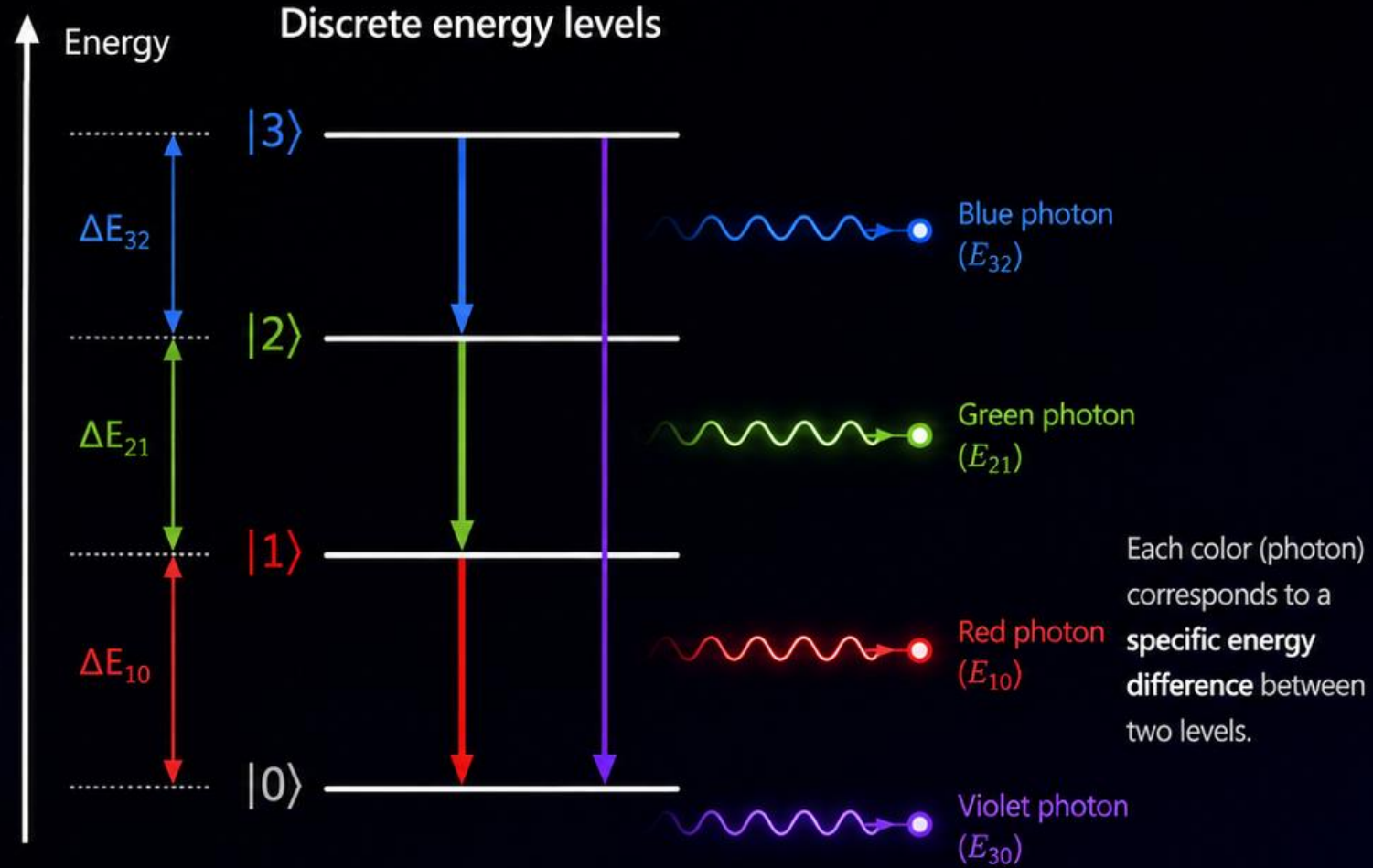
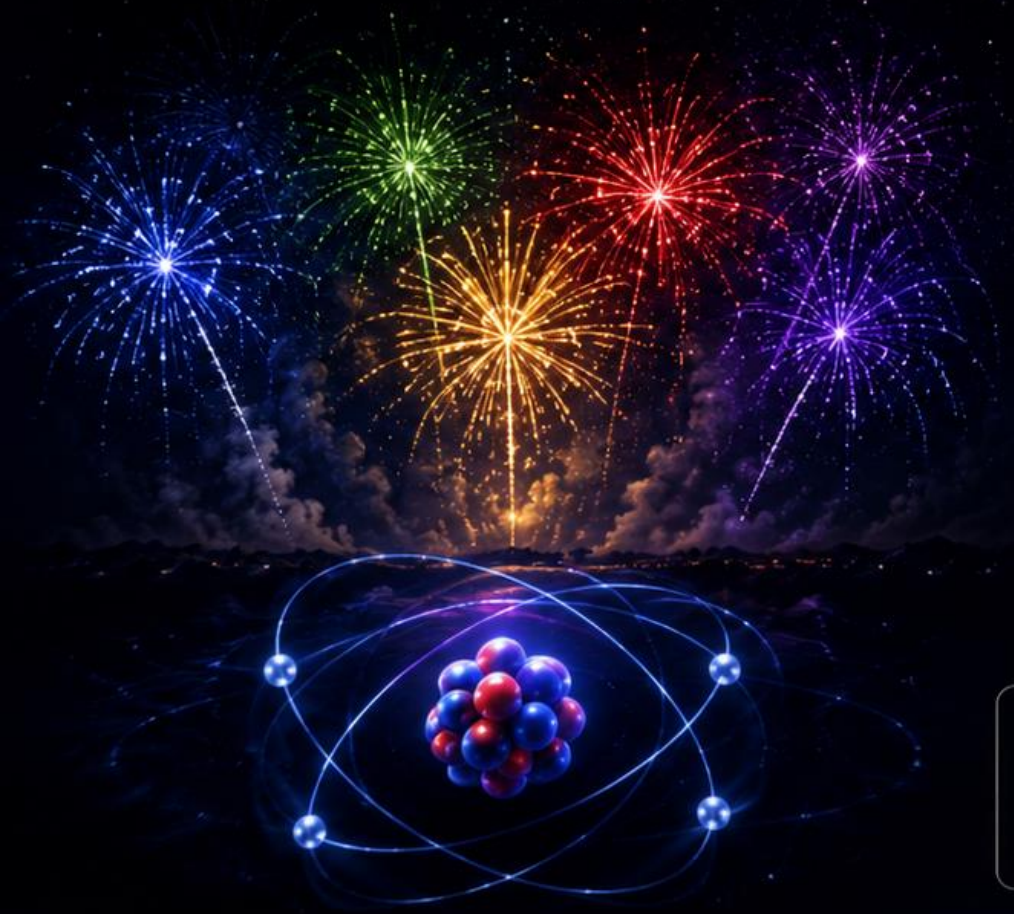
Ba 56
Barium
Green

Ti 22
Titanium
White /
Silver (Sparks)

Each color is one atom's quantum fingerprint

Why Colors Are Discrete

Electrons occupy only certain energy levels.
When they drop between levels,
they emit photons with **specific energies**.



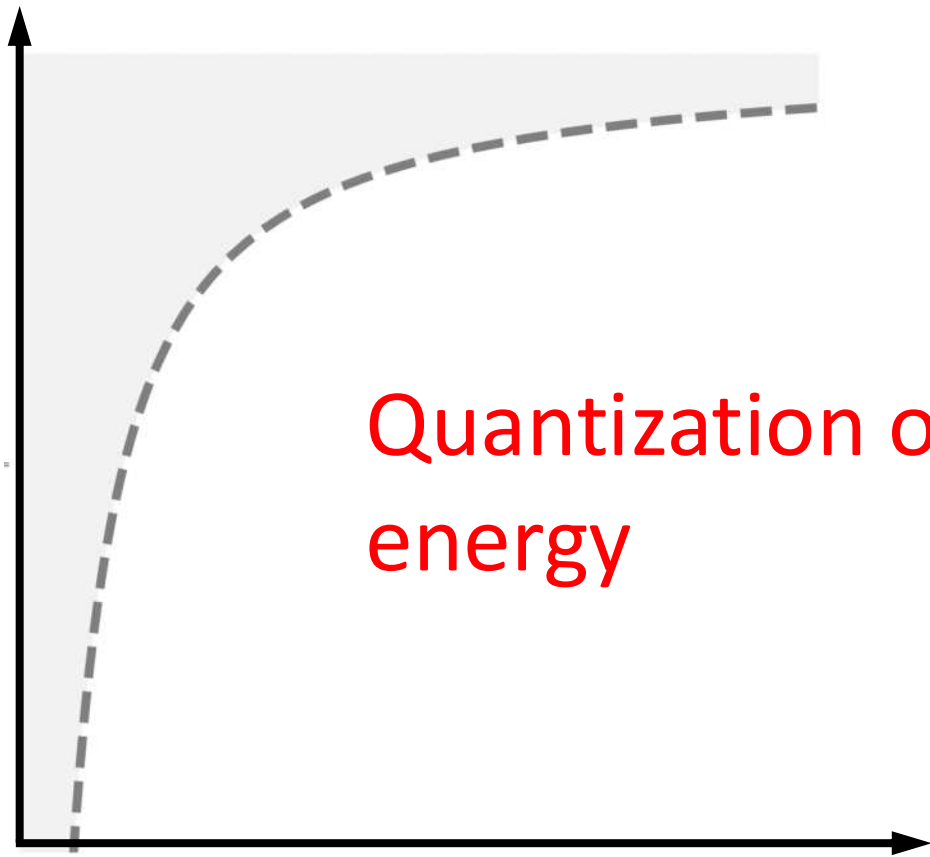
$$E_{\text{photon}} = h\nu = \Delta E$$

The energy of the emitted photon equals the energy difference between the levels.

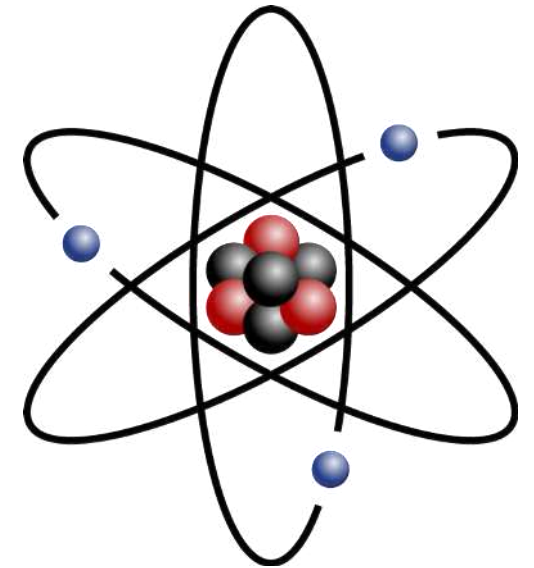
Atomic energy levels and transitions

Electron potential energy
landscape

$$V(r) \propto -\frac{1}{r}$$



Radial distance from nucleus r



Qubit from Atom

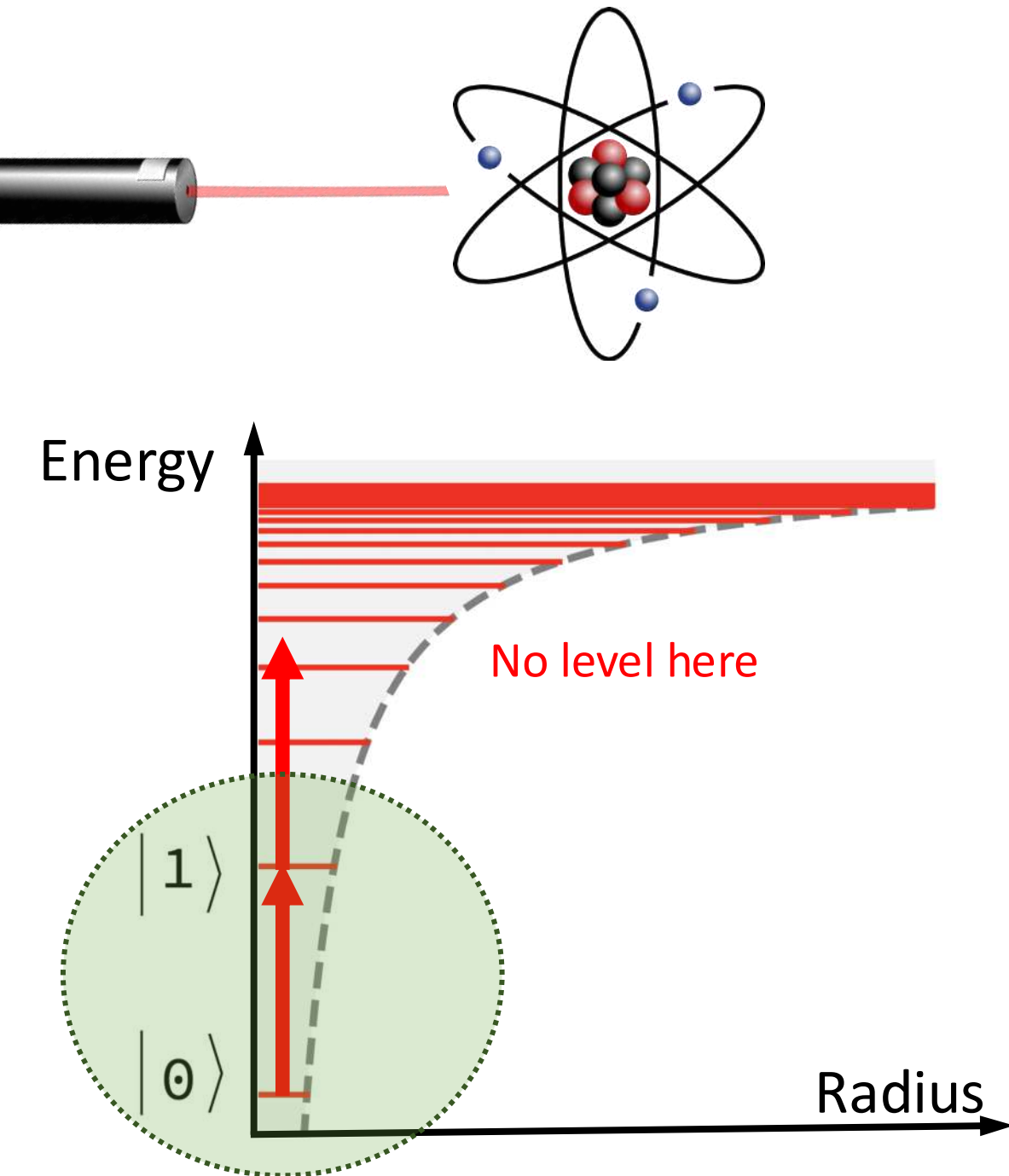
**Anharmonic degree of freedom
and spin**

**Isolated from environment and
thermal bath**

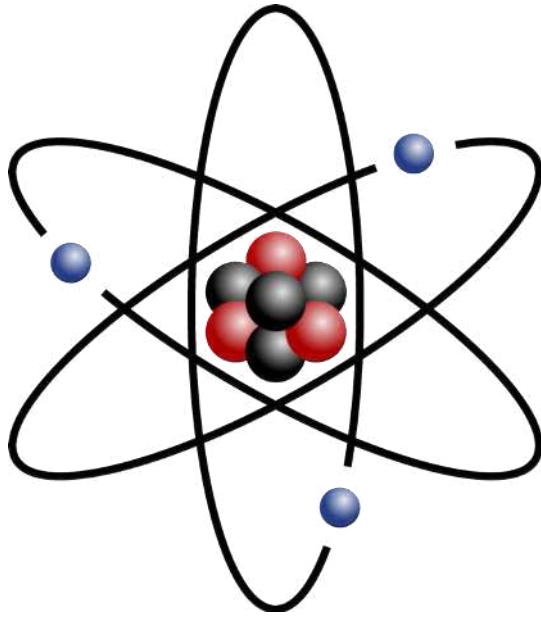
Low-loss

**Level diagram allows for qubit-
specific control and readout**

**There are always more than
two levels**

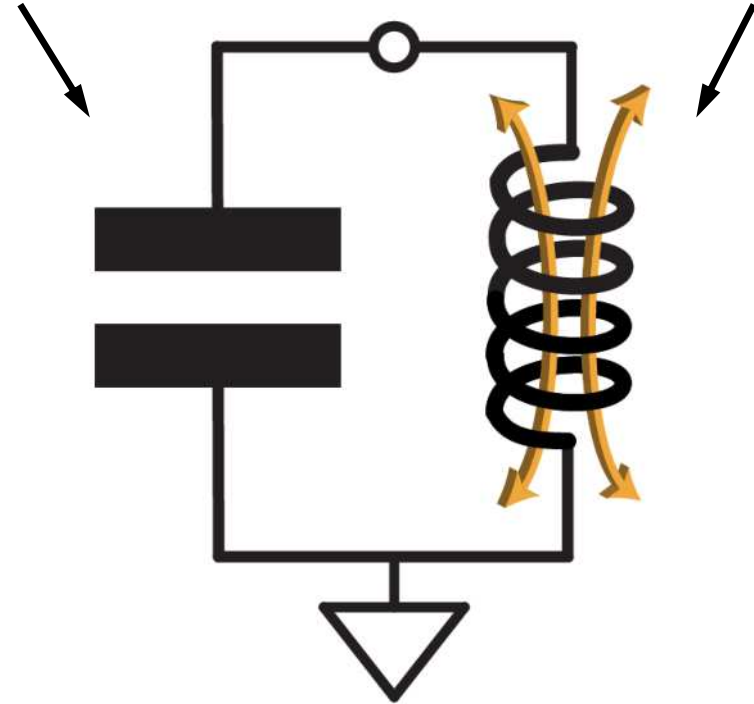


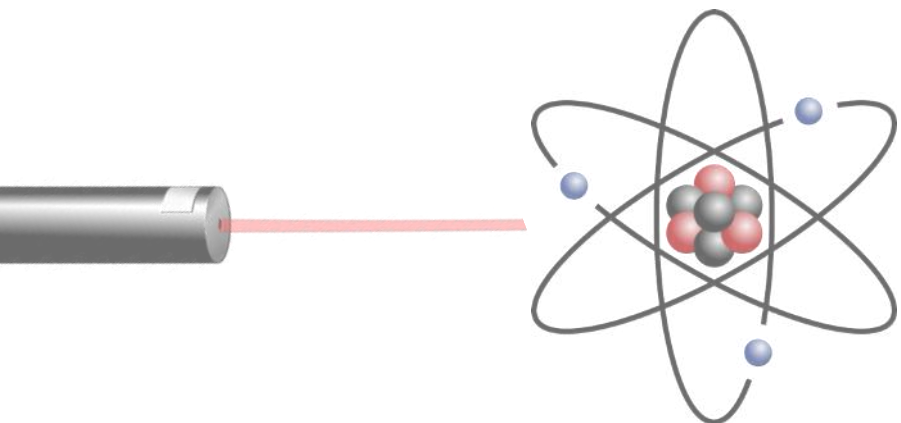
Artificial atoms



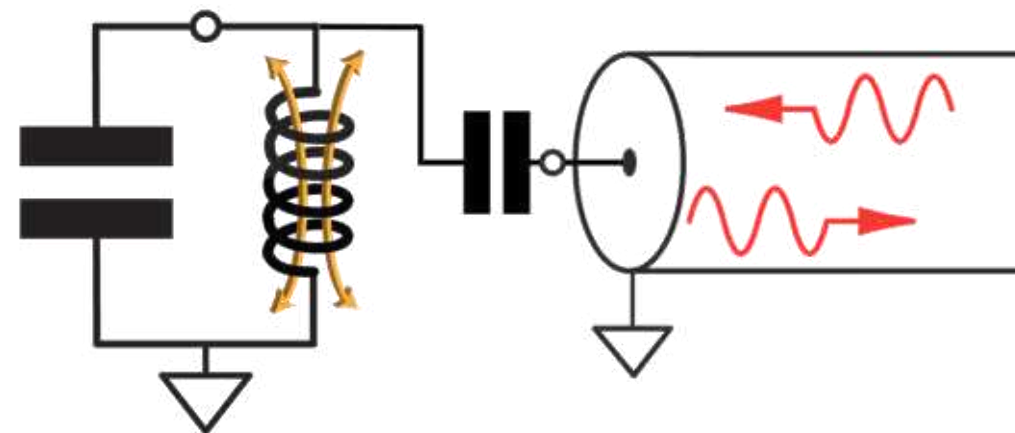
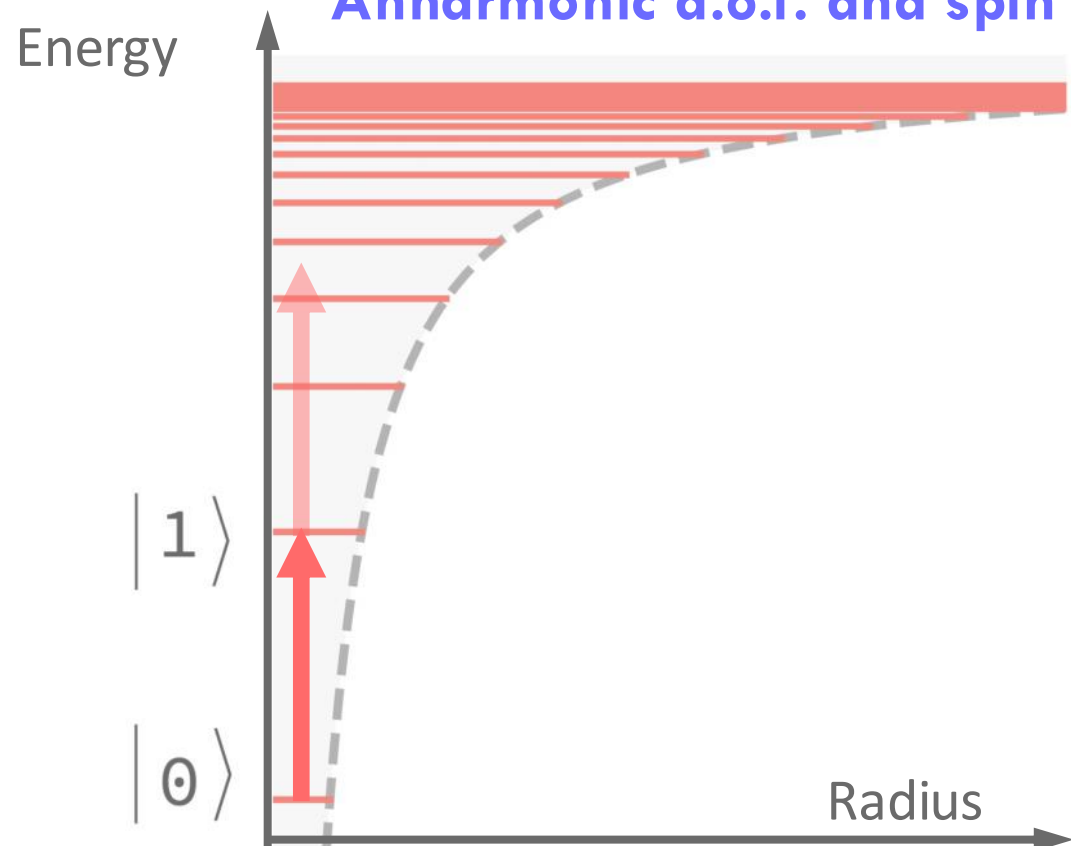
capacitor

inductor

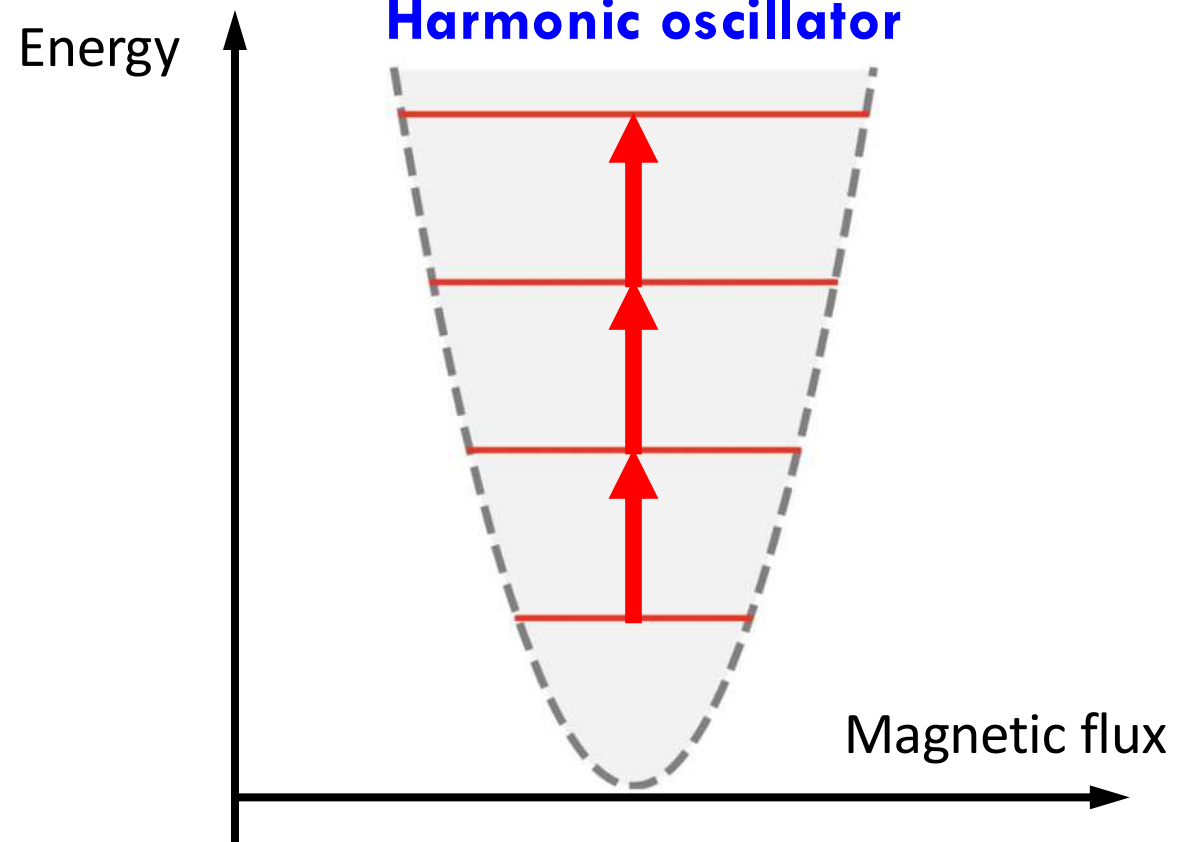


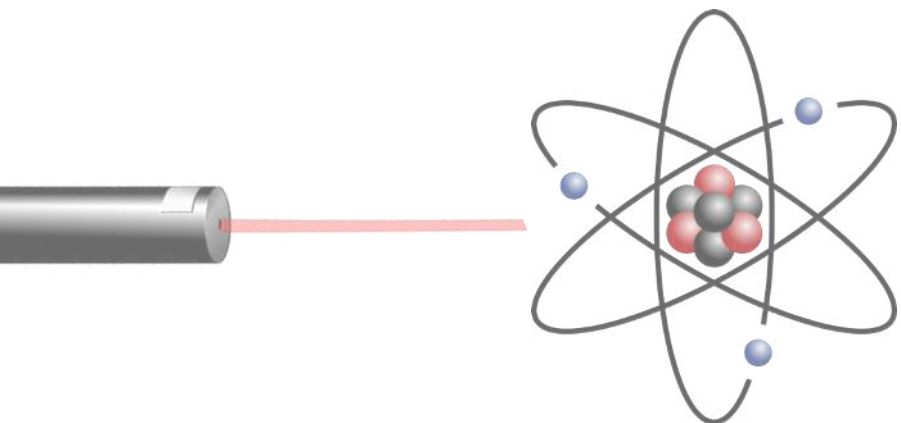


Anharmonic d.o.f. and spin

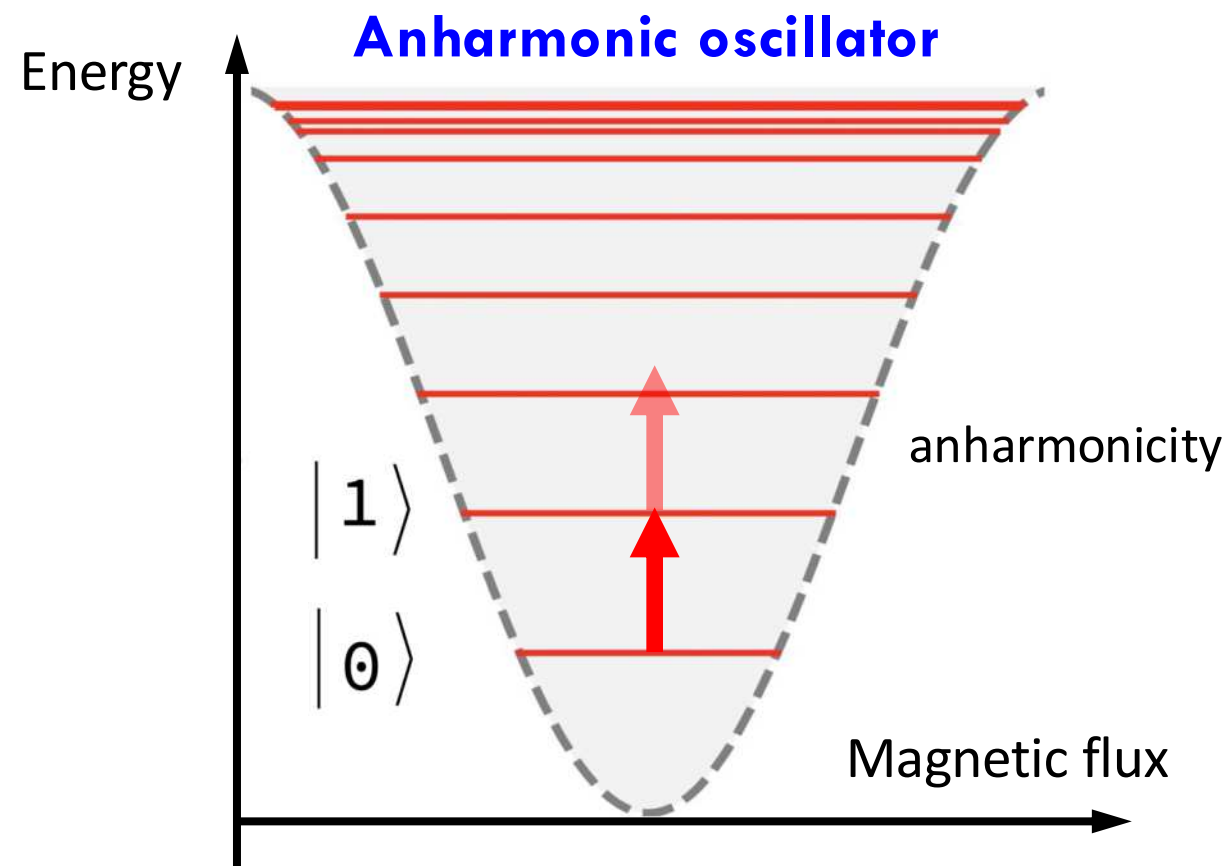
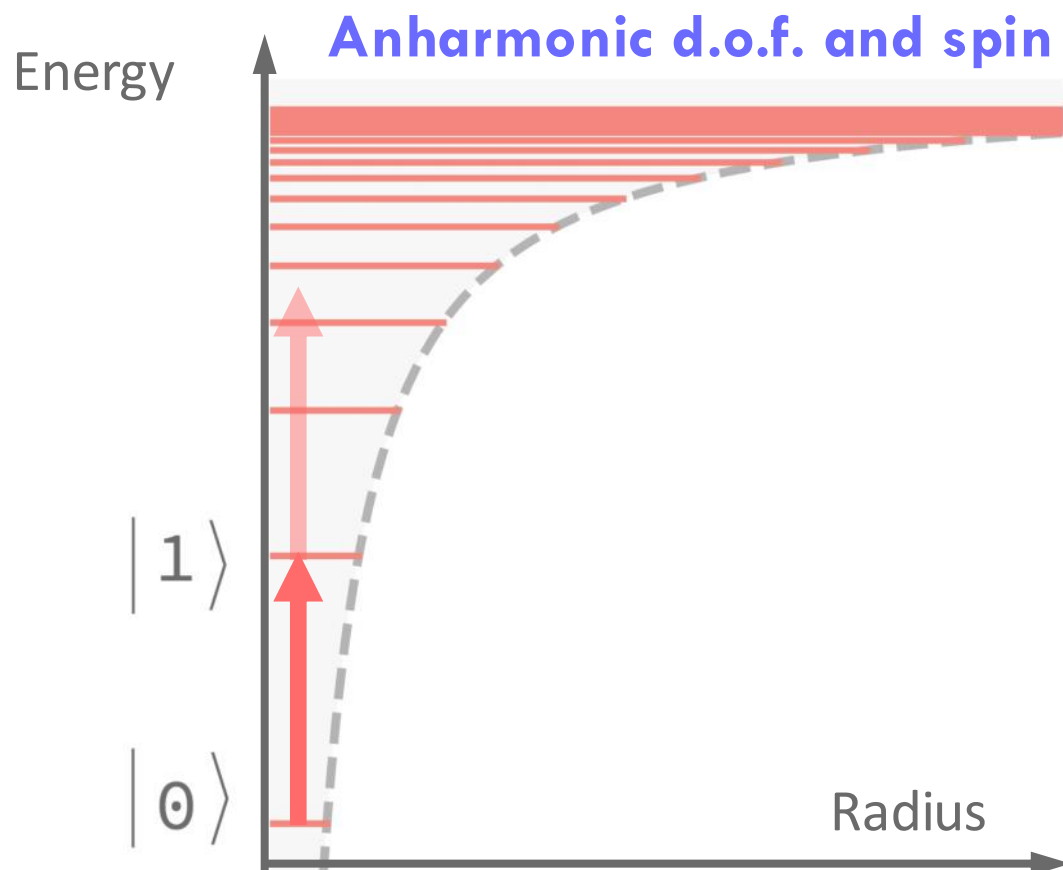
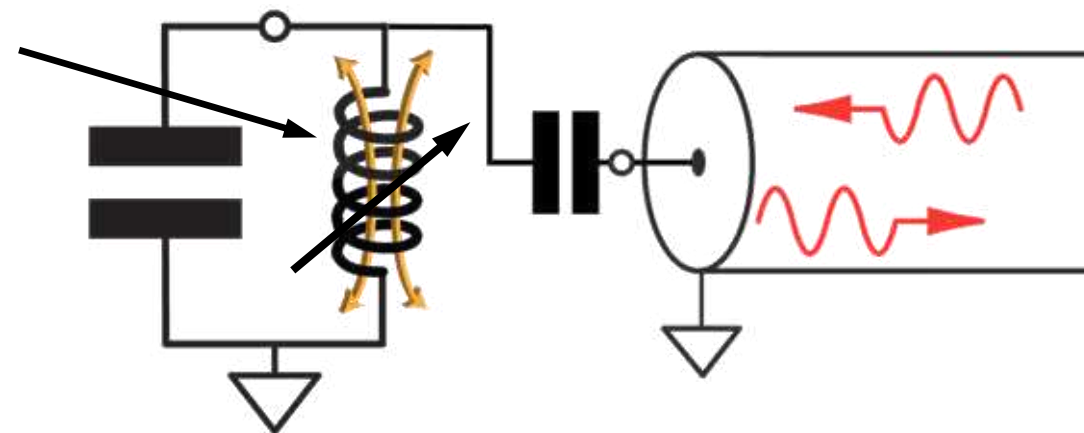


Harmonic oscillator



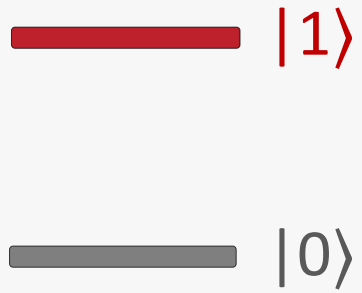


non-linear
inductor

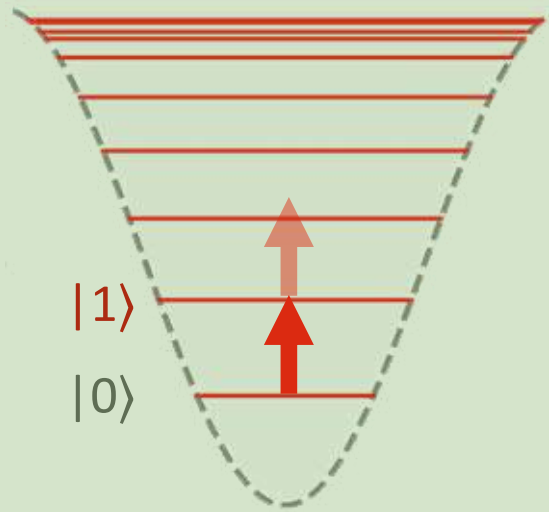


Big-picture connections

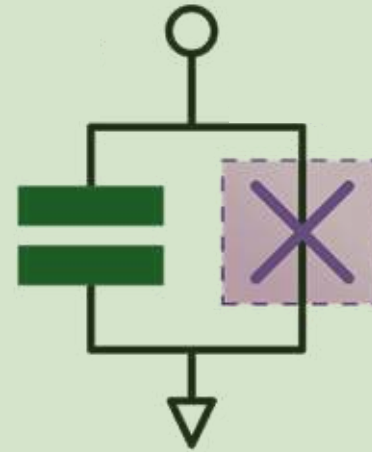
**Idealization of
qubit**



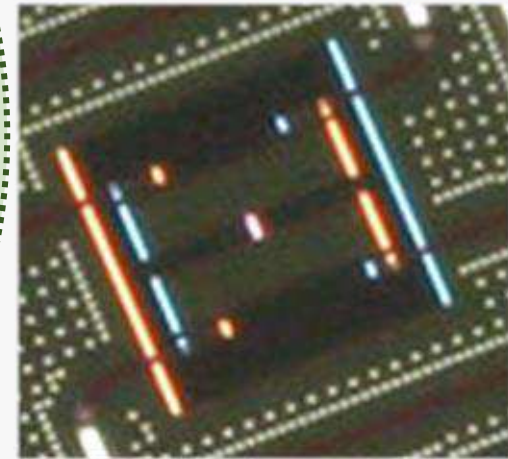
**Anharmonic
oscillator**



**Physical circuit
model**



Physical layout



Idealization

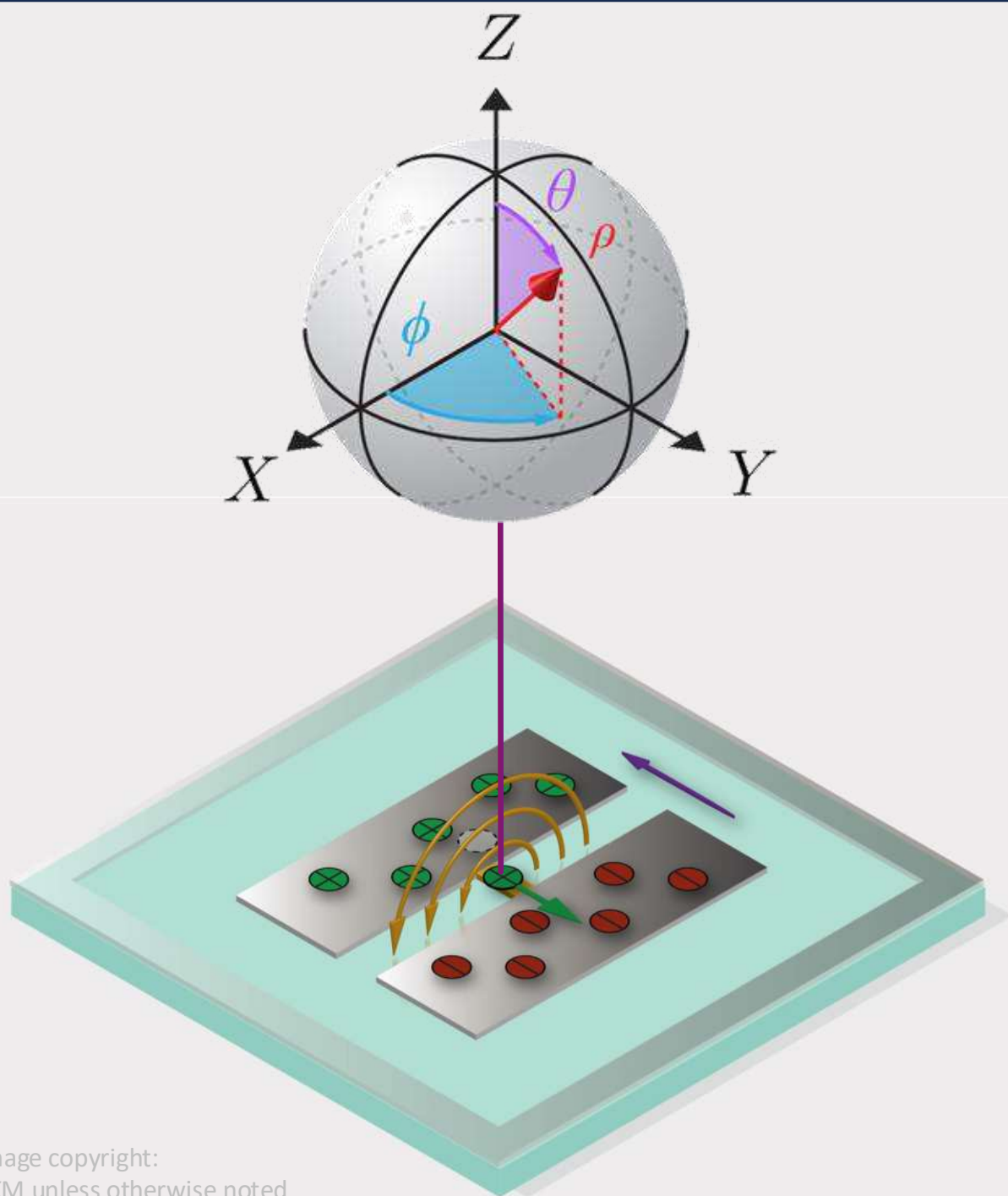


Physical reality

Qubit

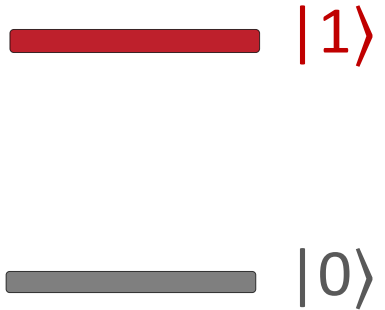
From Idea to Reality

THE BIG PICTURE
before calculations

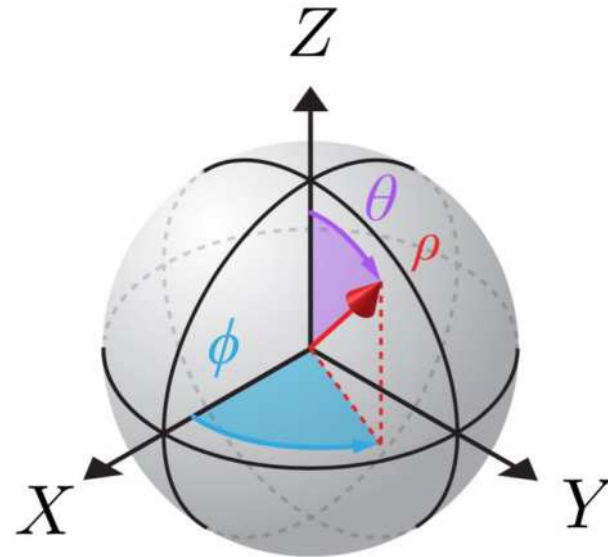


Qubit: idea

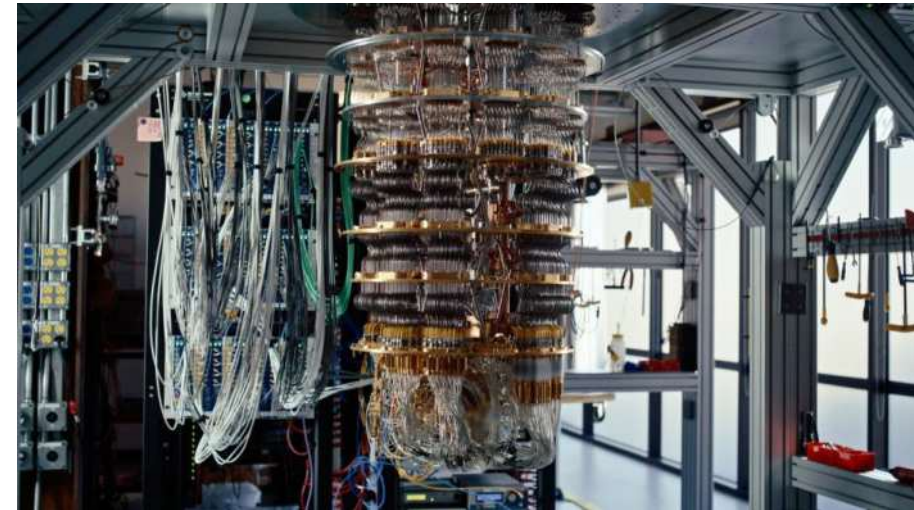
Energy levels



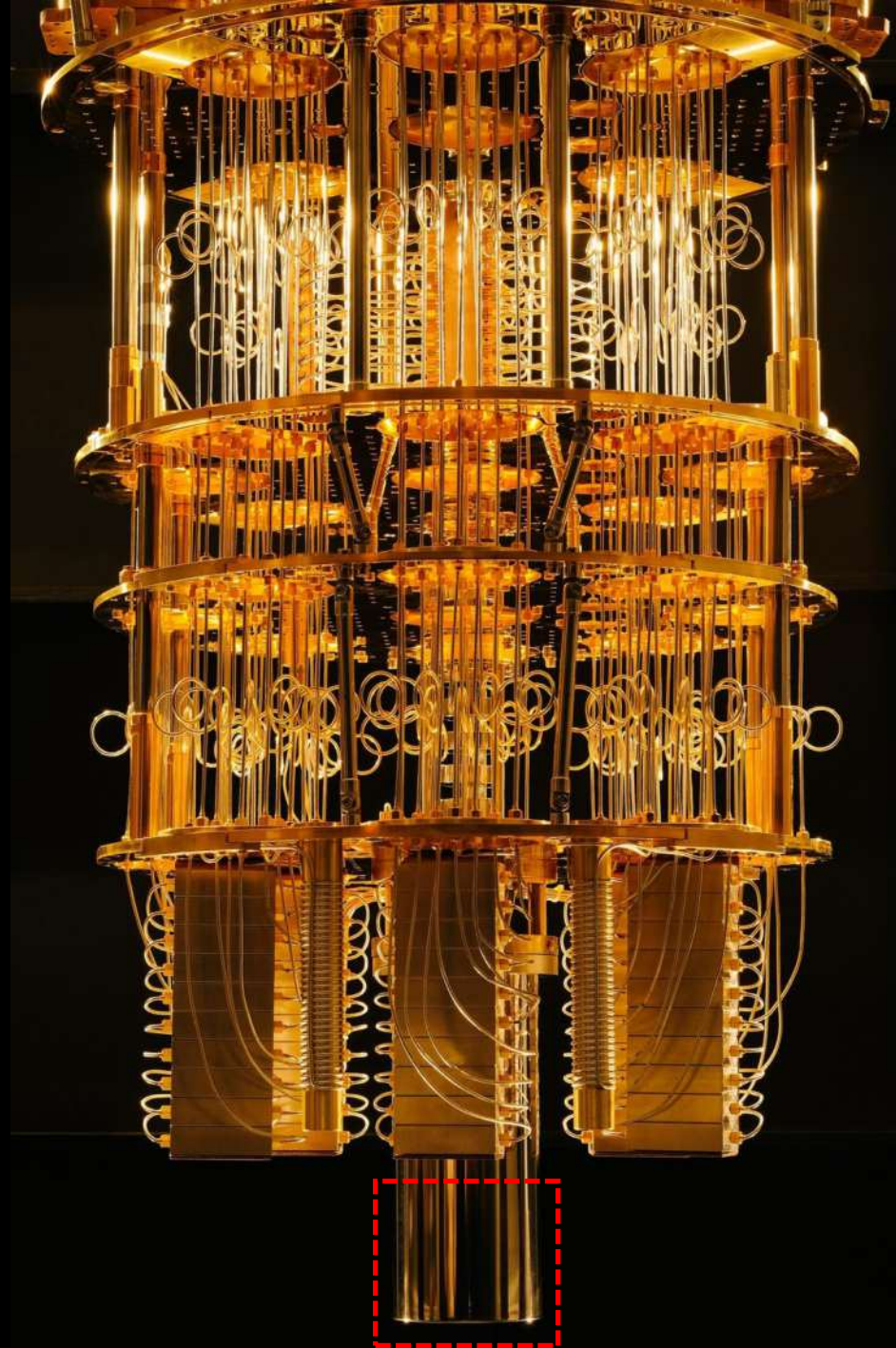
Hilbert space



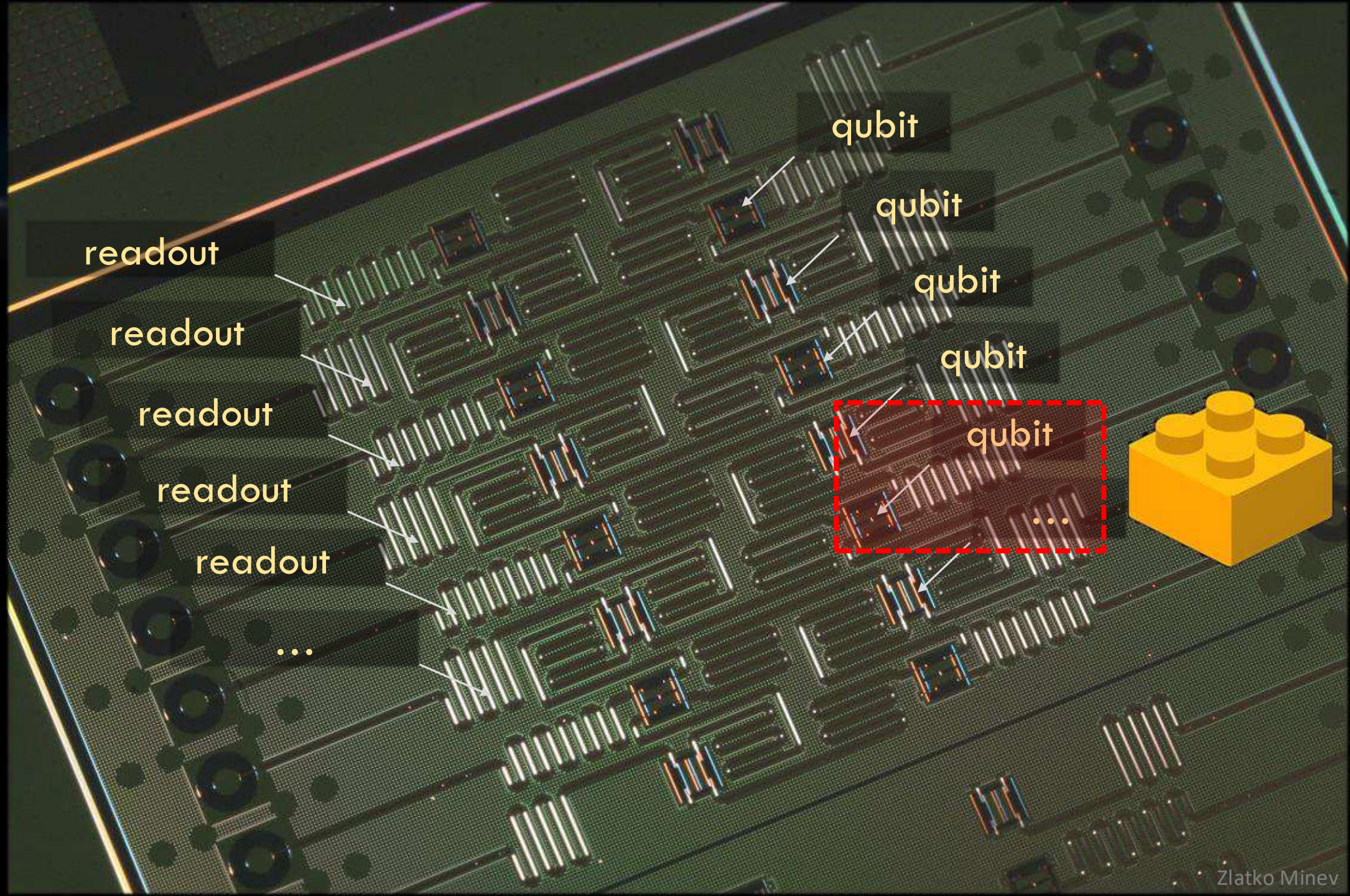
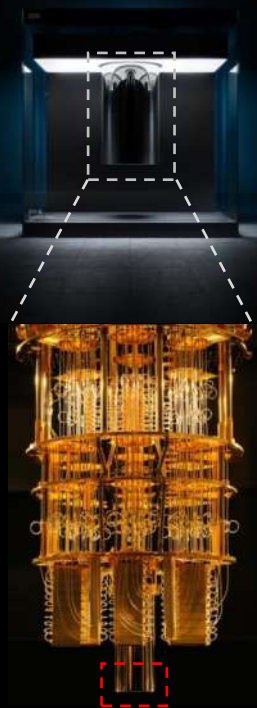
Quantum cloud?



* Images: Left/Middle: Minev, arXiv:1902.10355; Right: Google Quantum AI

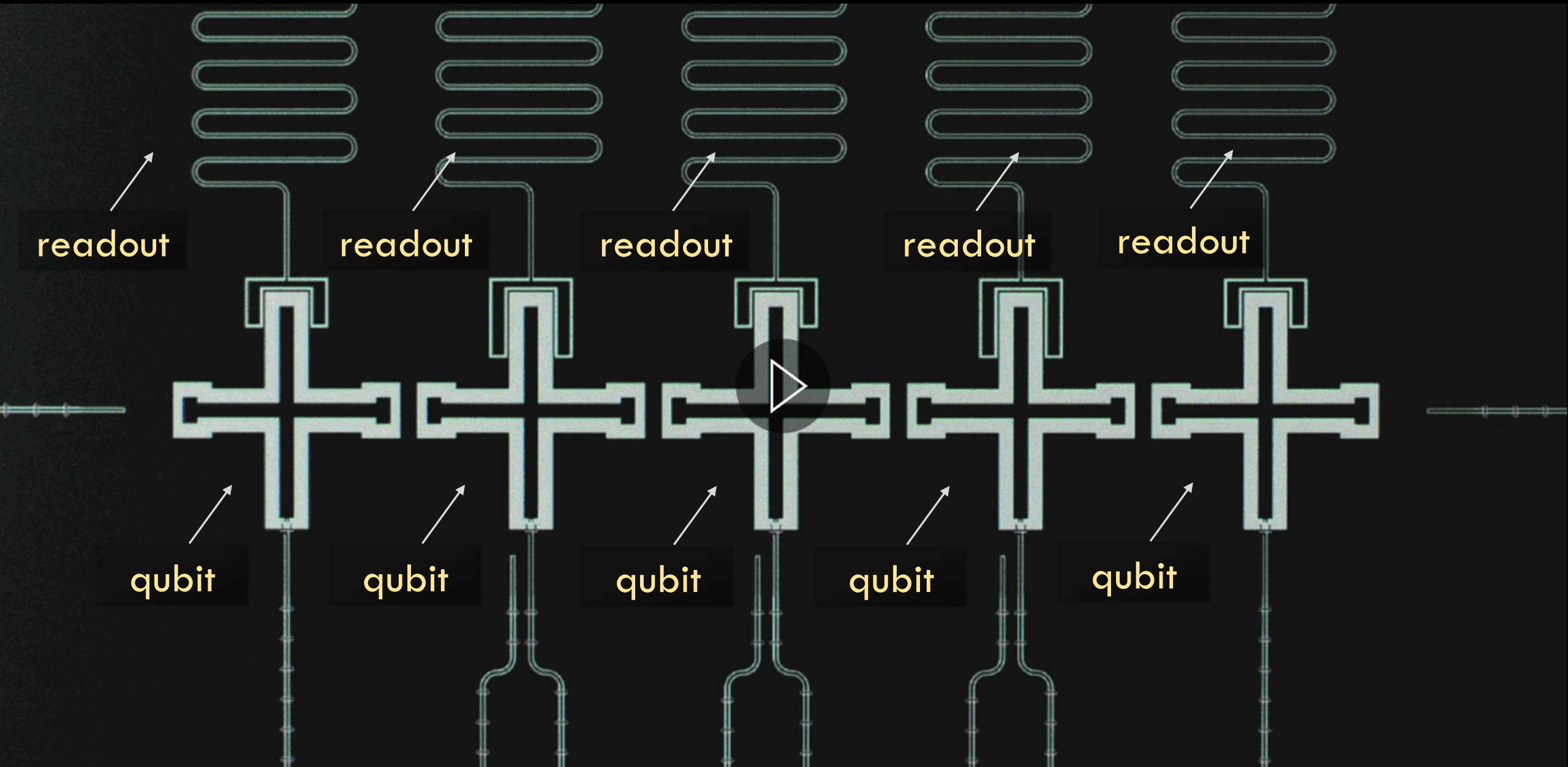


Operation at
15 mK (-273.13 °C)

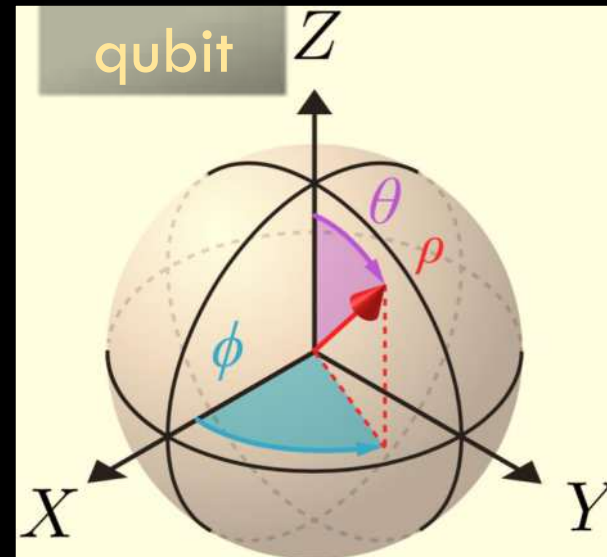
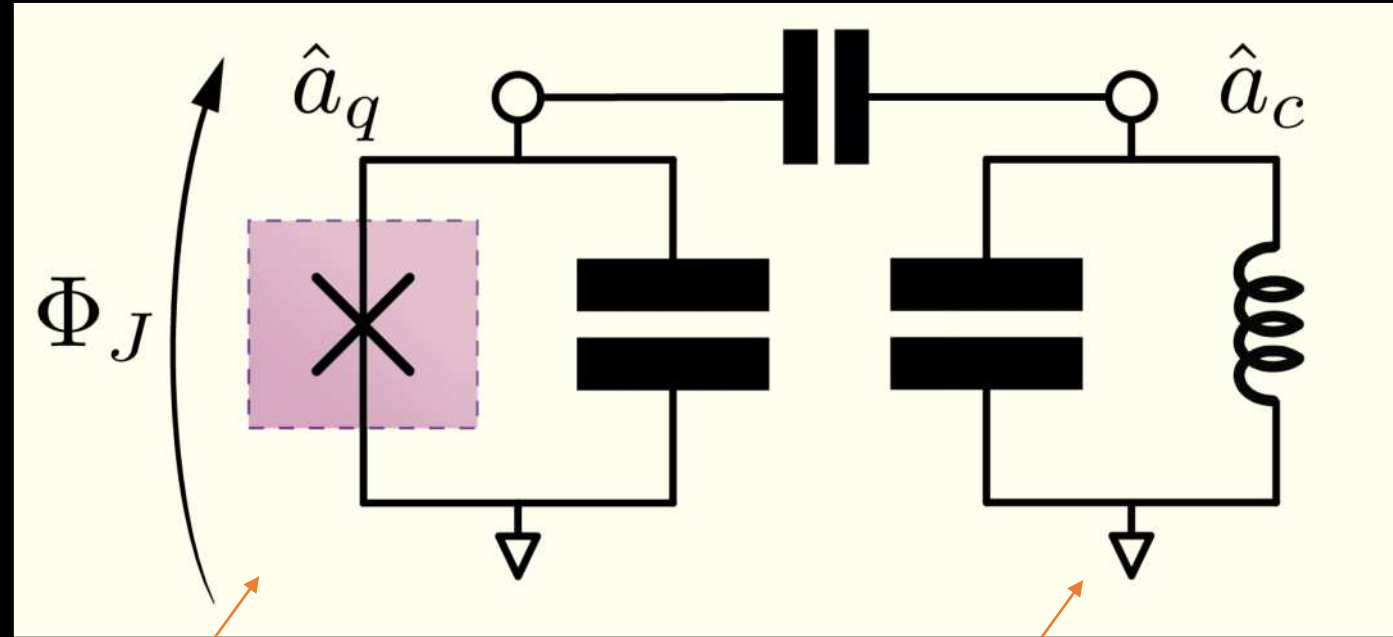
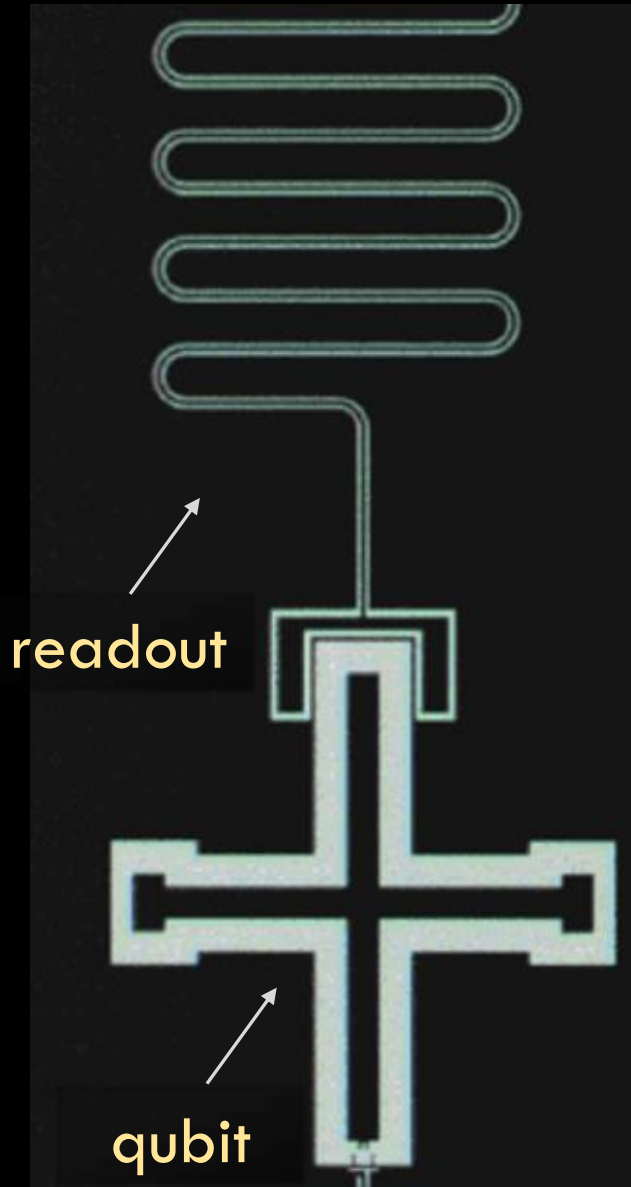


Example
IBM Architecture

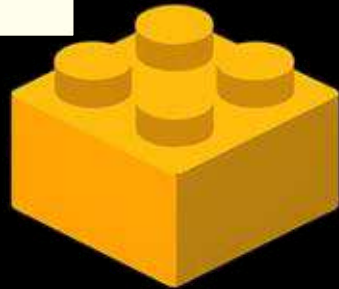
Example Google Architecture



Physical and abstract circuits and Hilbert space

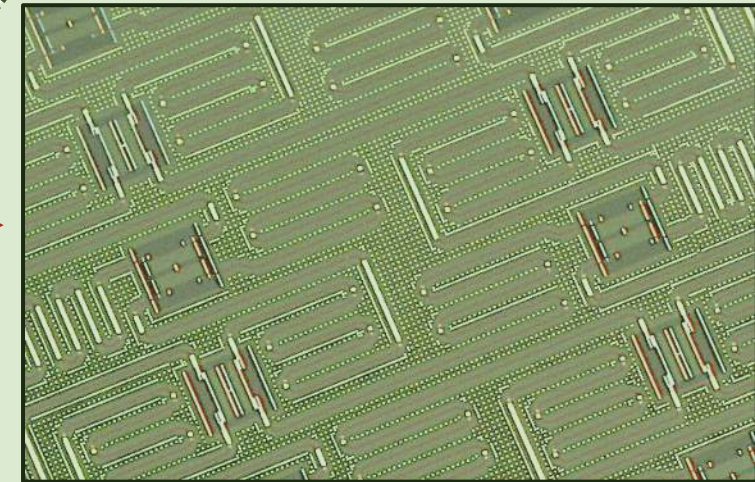
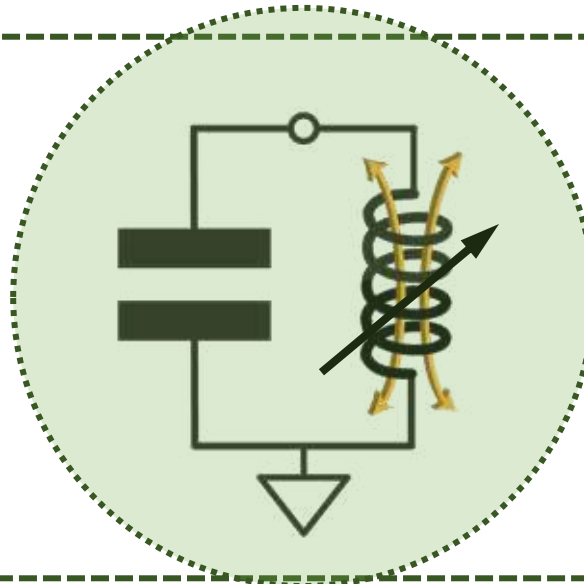
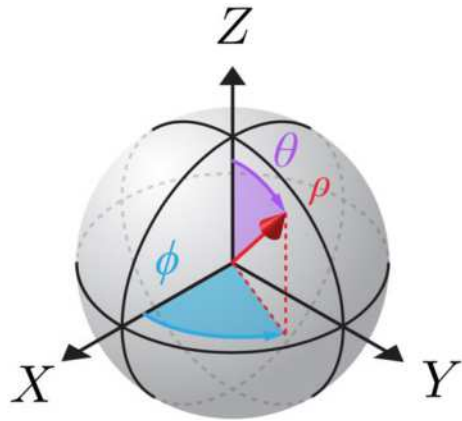
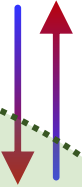
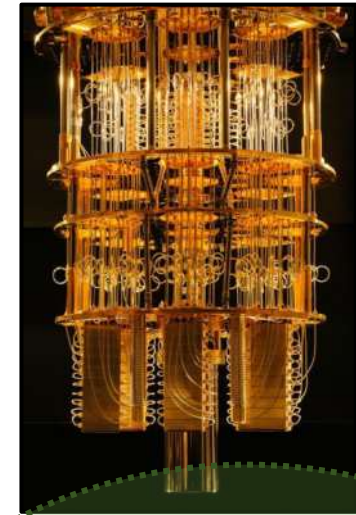


readout



$$\hat{H} \approx \omega_0 \hat{a}^\dagger \hat{a} - \frac{\alpha}{2} \hat{a}^{\dagger 2} \hat{a}^2$$

cQED qubit in the cloud: Summary of flow



Circuit Quantum Electrodynamics (cQED)

Macro
overview

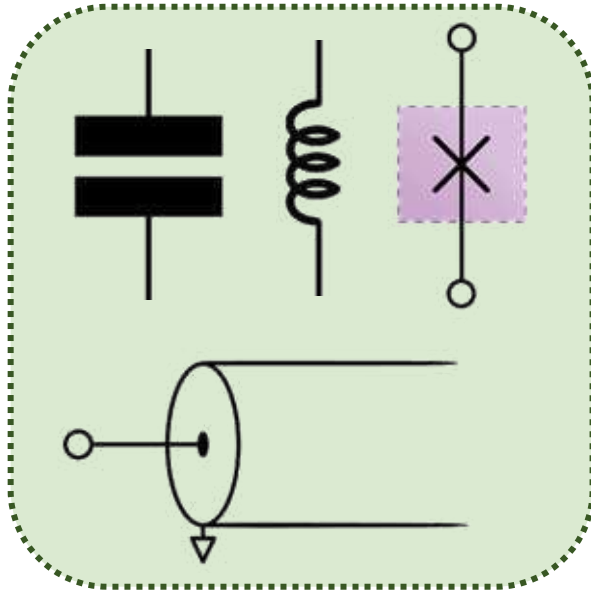
There are two kinds of physicists:

Those who believe all of physics is *spins*.

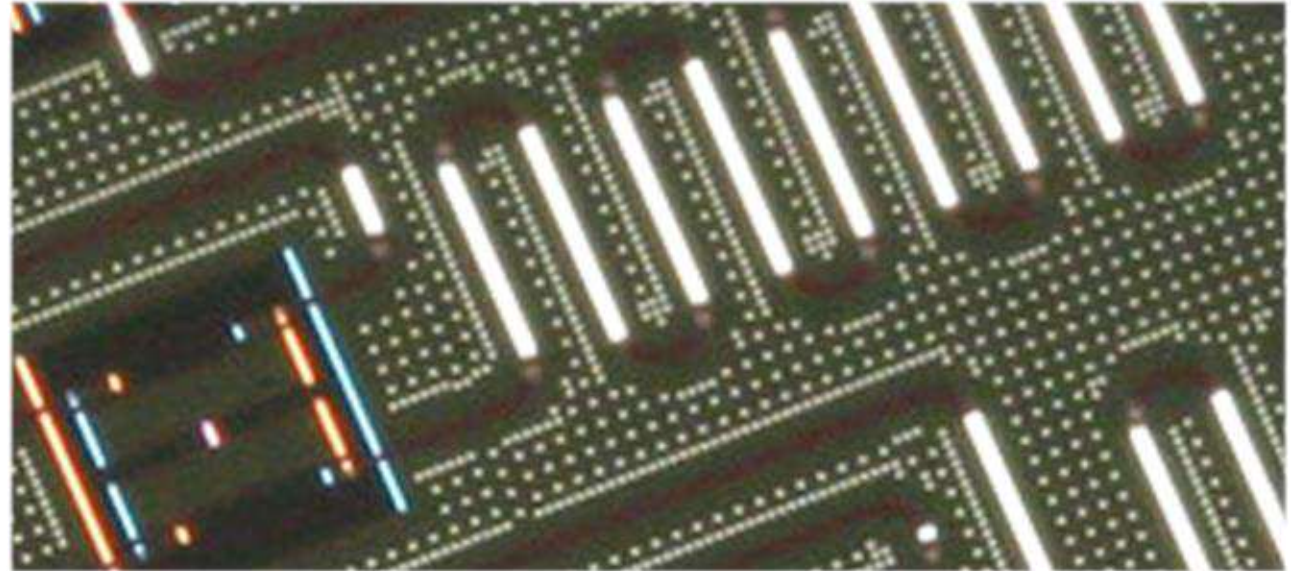
Those who believe all of physics is *oscillators*.

cQED Ingredients

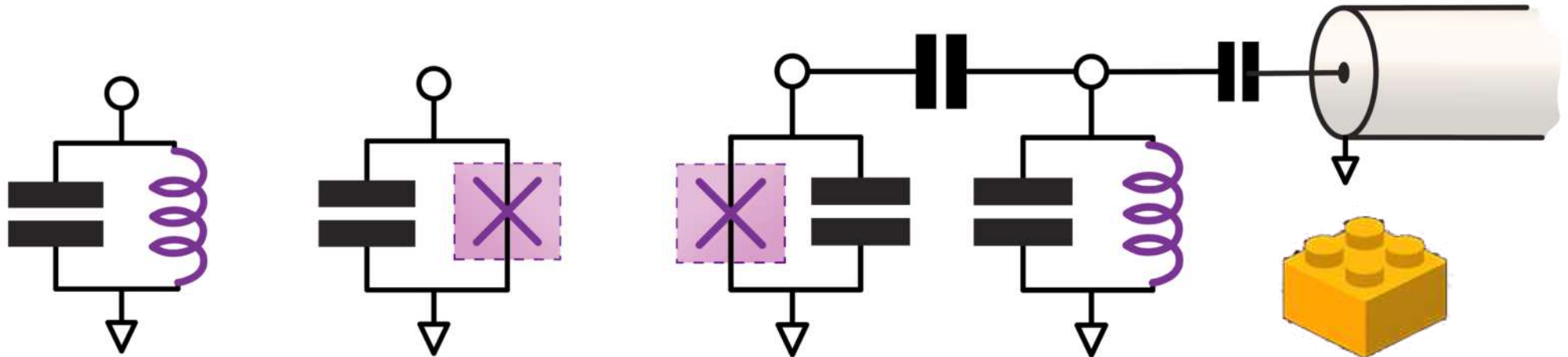
Circuit elements



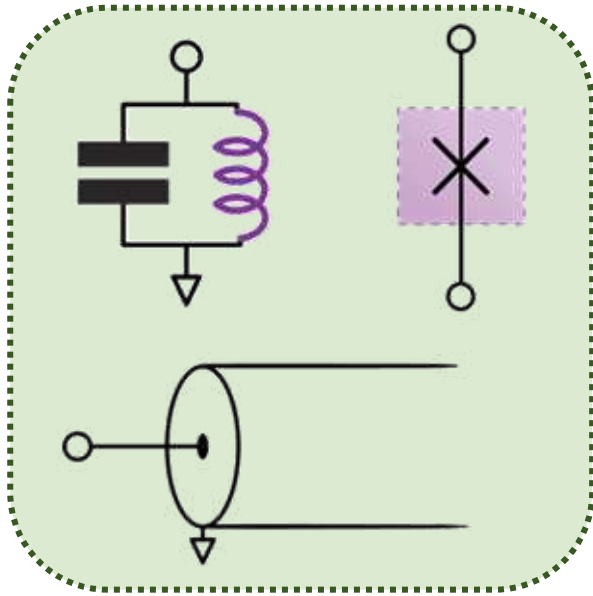
Microwave oscillators



Combine



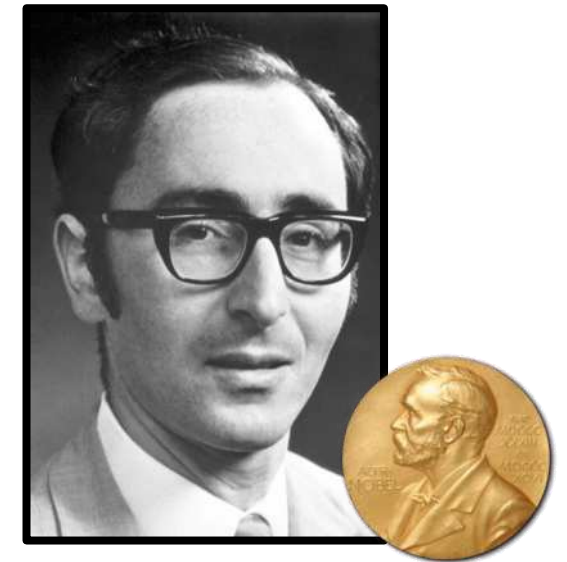
cQED Ingredients

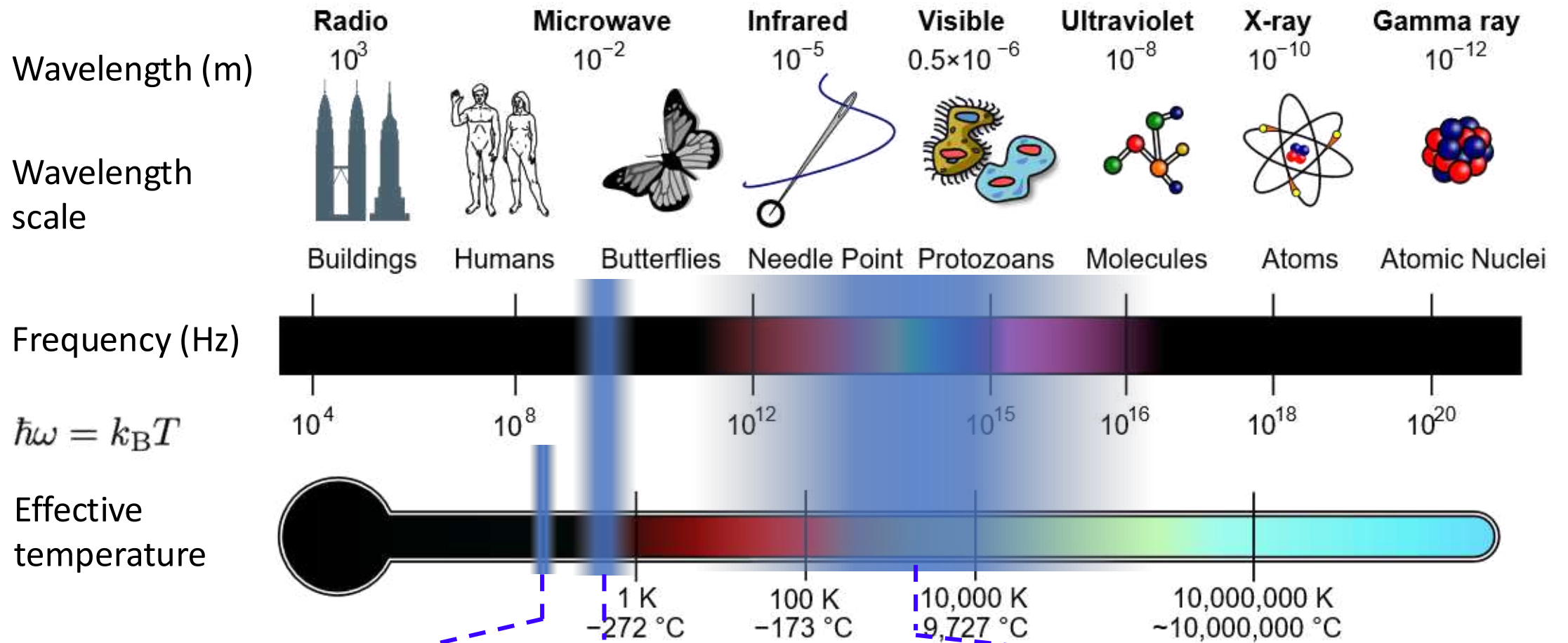


- Small dissipation
- Isolation from environment
- Low temperature
- Nonlinearity
- Large vacuum fluctuations

Superconductivity

- Nominally zero intrinsic dissipation and heat
- Nominal temperature far below energy level splitting
- Non-linear, robust Josephson tunnel junction effect



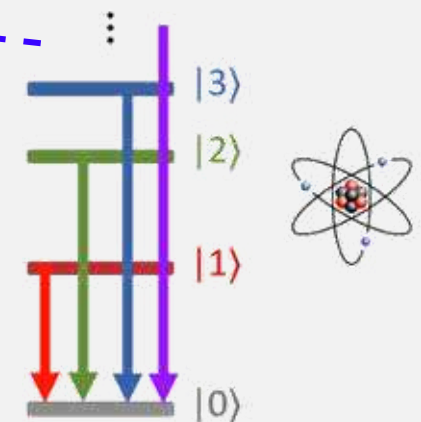
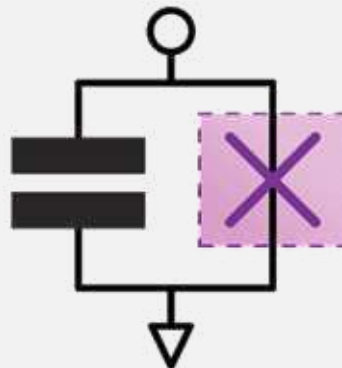


Order of magnitude



10^{-2} K

0.5 K, 10 GHz, 3 cm



Spectrum image: Inductiveload, NASA

A few introductory reviews

And many more... check online or ask us for specific topic

Qiskit Textbook (2020; more chapters coming)

Blais, A., Grimsmo, A. L., Girvin, S. M., & Wallraff, A. (2020)
Circuit Quantum Electrodynamics (arXiv:2005.12667)

Kjaergaard, M., Schwartz, ... Oliver, W. D. (2020)
Superconducting Qubits: Current State of Play
Annual Reviews of Condensed Matter Physics 11, 369-395

Krantz, P., Kjaergaard, M., Yan, F., ... & Oliver, W. D. (2019)
A quantum engineer's guide to superconducting qubits
Applied Physics Reviews, 6(2), 021318

Corcoles, A. D., Kandala, A., ... Gambetta, J. M. (2019)
Challenges and Opportunities of Near-Term Quantum Computing Systems. *Proceedings of the IEEE*, 1–15.

Wendin, G. (2017)
Quantum information processing with superconducting circuits. *Reports on Progress in Physics*, 80(10), 106001

Gambetta, J. M., Chow, J. M., & Steffen, M. (2017)
Building logical qubits in a superconducting quantum computing system. *Npj Quantum Information*, 3(1), 2

Girvin, S. M. (2011) Circuit QED: superconducting qubits coupled to microwave photons. *Quantum machines: measurement and control of engineered quantum systems*, 113, 2.

Clerk, A. A., Girvin, S. M., Marquardt, F., & Schoelkopf, R. J. (2010)
Introduction to quantum noise, measurement, and amplification
Reviews of Modern Physics, 82(2), 1155–1208

Clarke, J., & Wilhelm, F. K. (2008)
Superconducting quantum bits. *Nature*, 453(7198), 1031–1042

Devoret, M. H. (1997)
Quantum Fluctuations in Electrical Circuits.
In *Fluctuations Quantiques/Quantum Fluctuations* (p. 351)

...

Where are all my slides?



Slides!

zlatko-minev / zlatko-minev-quantum-repository

Code Issues Pull requests Actions Projects Security Insights Settings

Files

main

Go to file

- > _web
- > content - educational
- > content - research talks
- > content - tech notes

- .gitignore
- LICENSE.md
- README.md
- _config.yml
- index.html

zlatko-minev-quantum-repository / content - educational /

zlatko-minev Initial Commit

Name	Last commit message
..	
Career Talks	Initial Commit
Intro to Noise Mitigation - Near-T...	Initial Commit
Intro to Pauli Twirling (Tutorial)	Initial Commit
Intro to Quantum Error Mitigation ...	Initial Commit
Intro to Quantum Noise (Qiskit Gl...	Initial Commit
Intro to Quantum Noise (Qiskit Gl...	Initial Commit
Intro to Quantum Noise (Qiskit Gl...	Initial Commit
Superconducting Qubits 101 - Qu...	Initial Commit
A Prelude To Practical Quantum ...	Initial Commit

Zlatko Minev,
Ph.D.

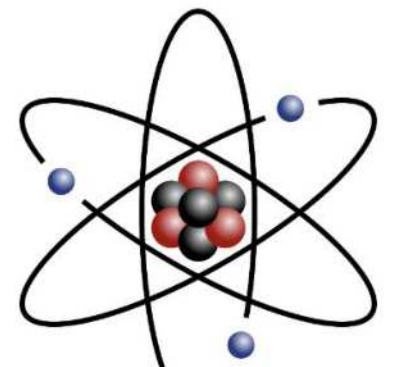
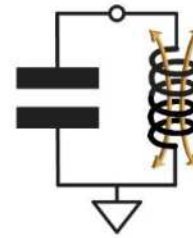
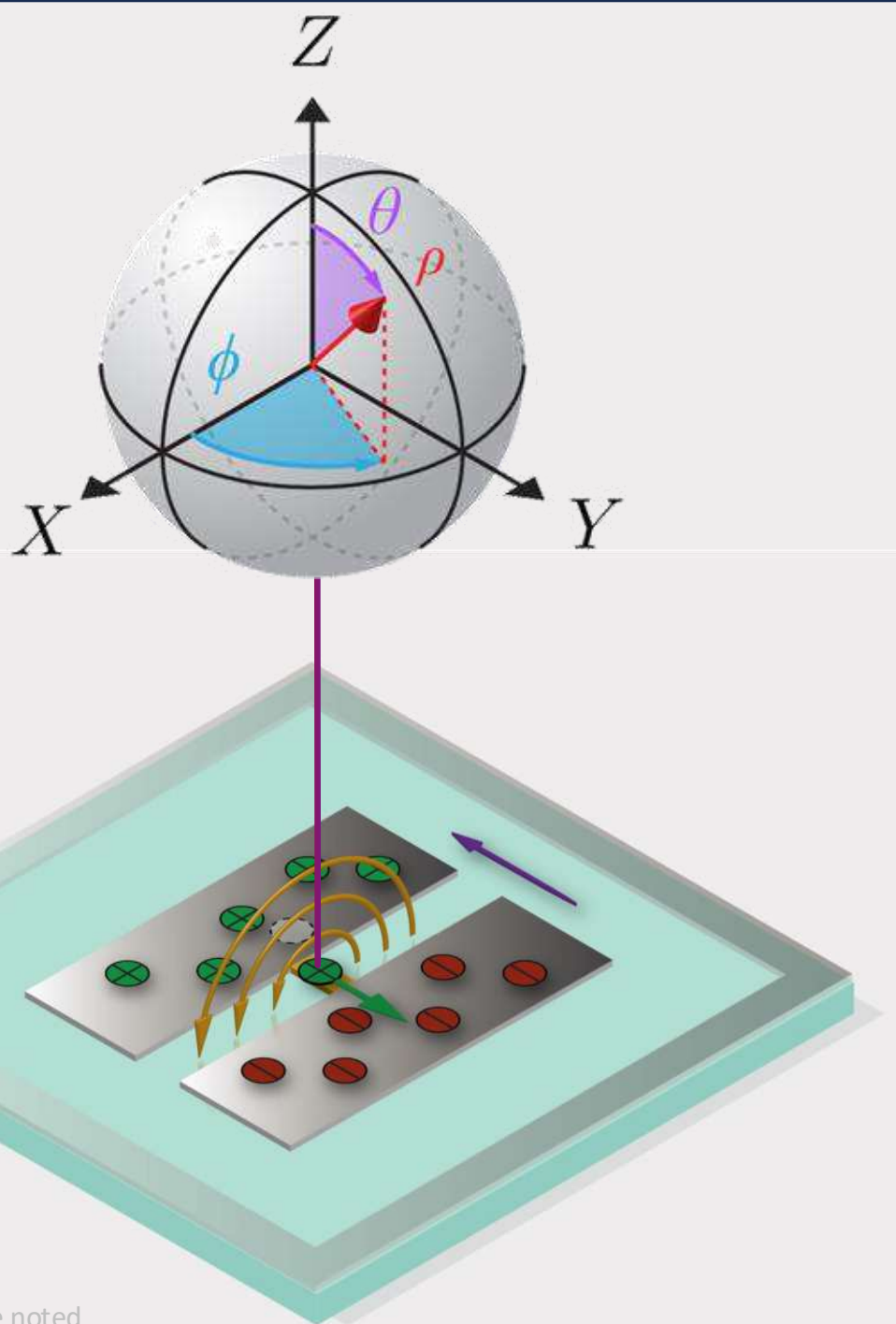
Zlatko's Blog

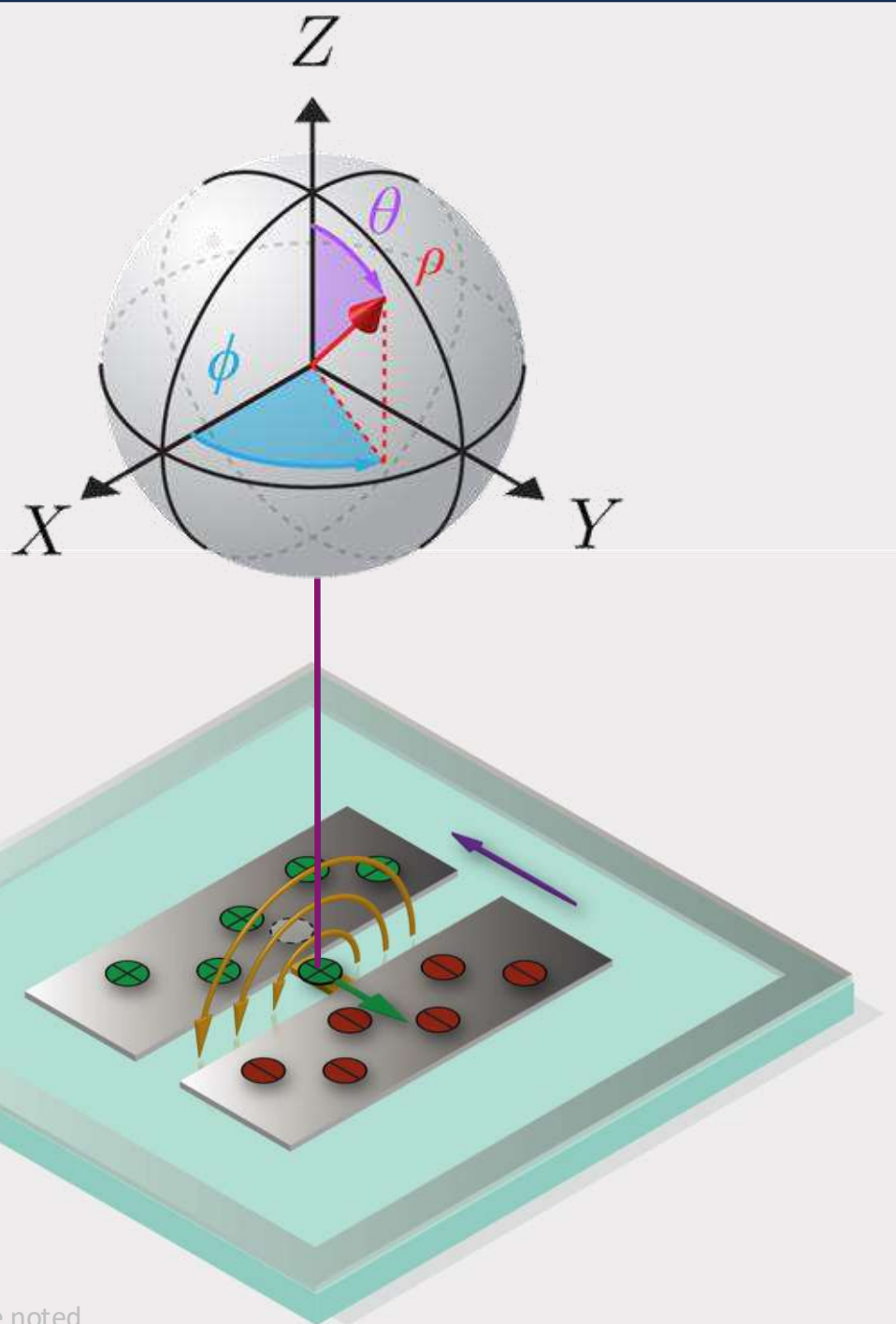
Casual notes, tutorials, and thoughts on quantum computing. Organized by categories:

- 🎓 **Tutorial + Lecture** - Summer school style lectures / tutorials
- 🧐 **Tech Note** - Useful derivations and technical notes
- 🗣️ **Talk Slides** - Research paper talks, slides, videos
- 📝 **Non-Tech** - Non technical blog

zlatko-minev.com/blog

Circuit Quantum Electrodynamics



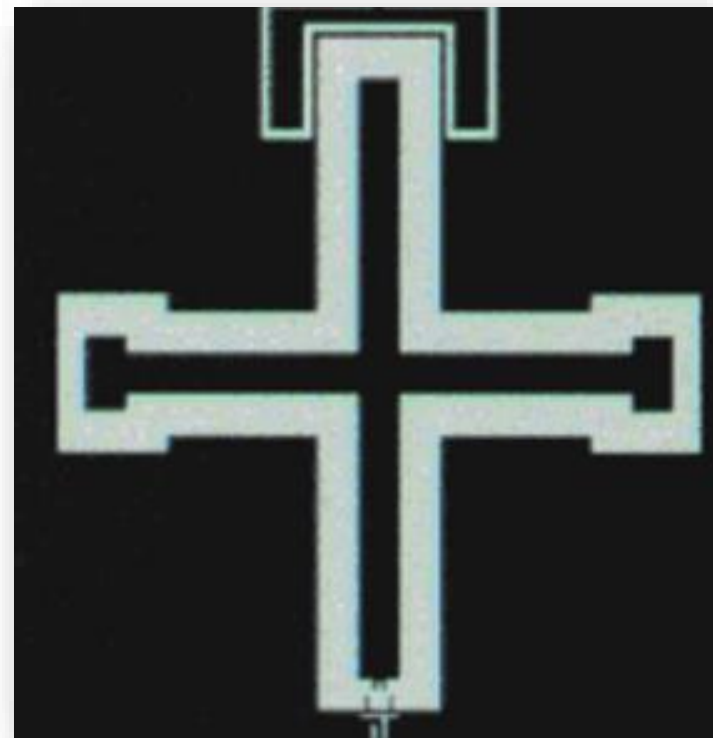
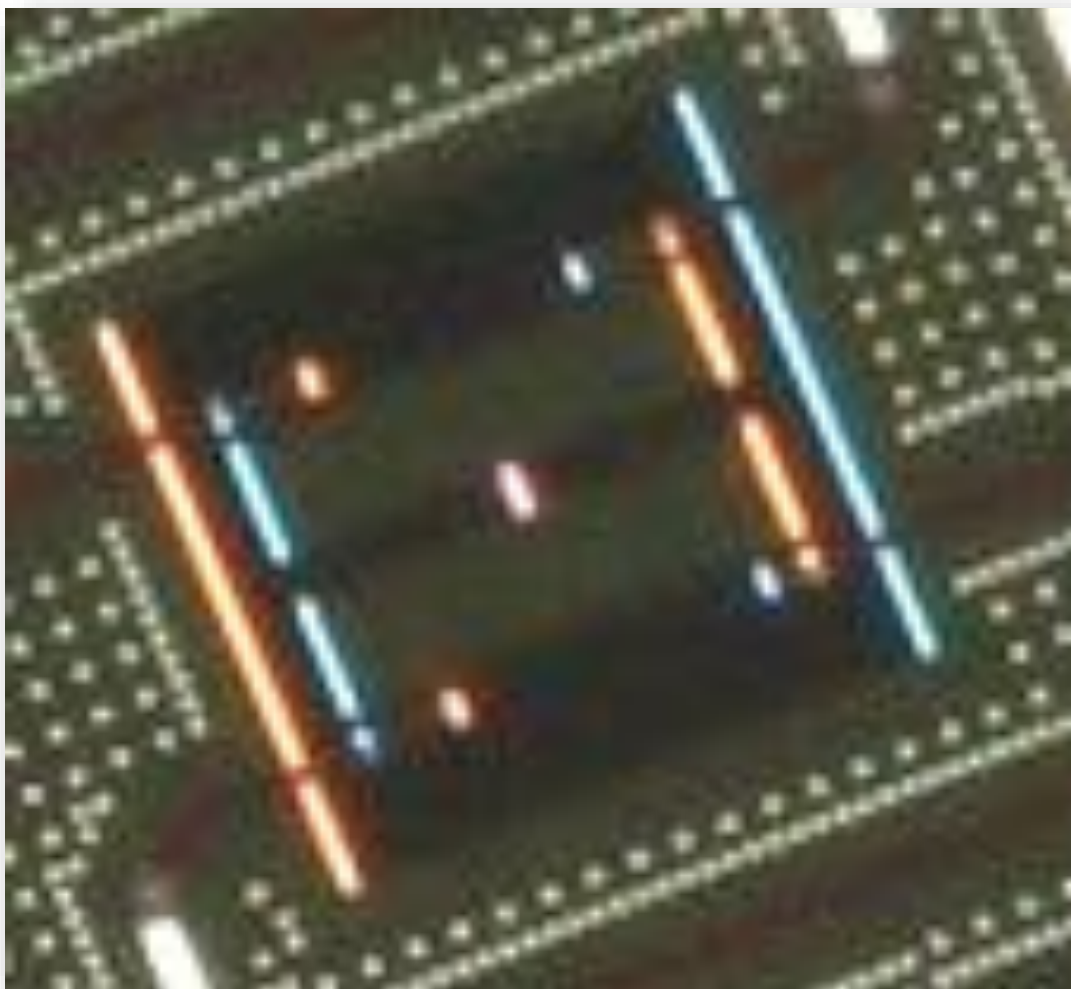


To quantum,

start classical

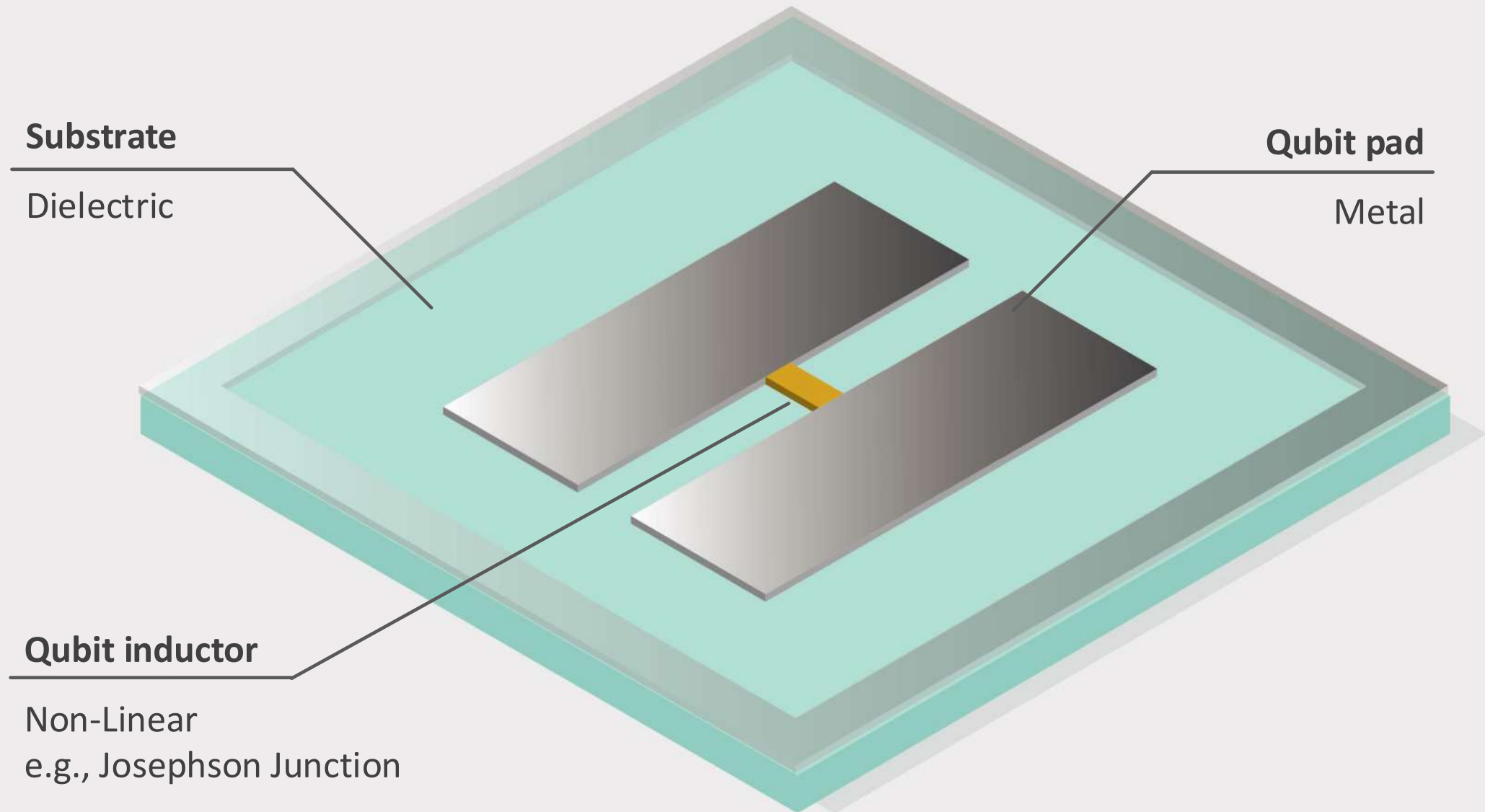
Classical oscillator

Transmon qubit



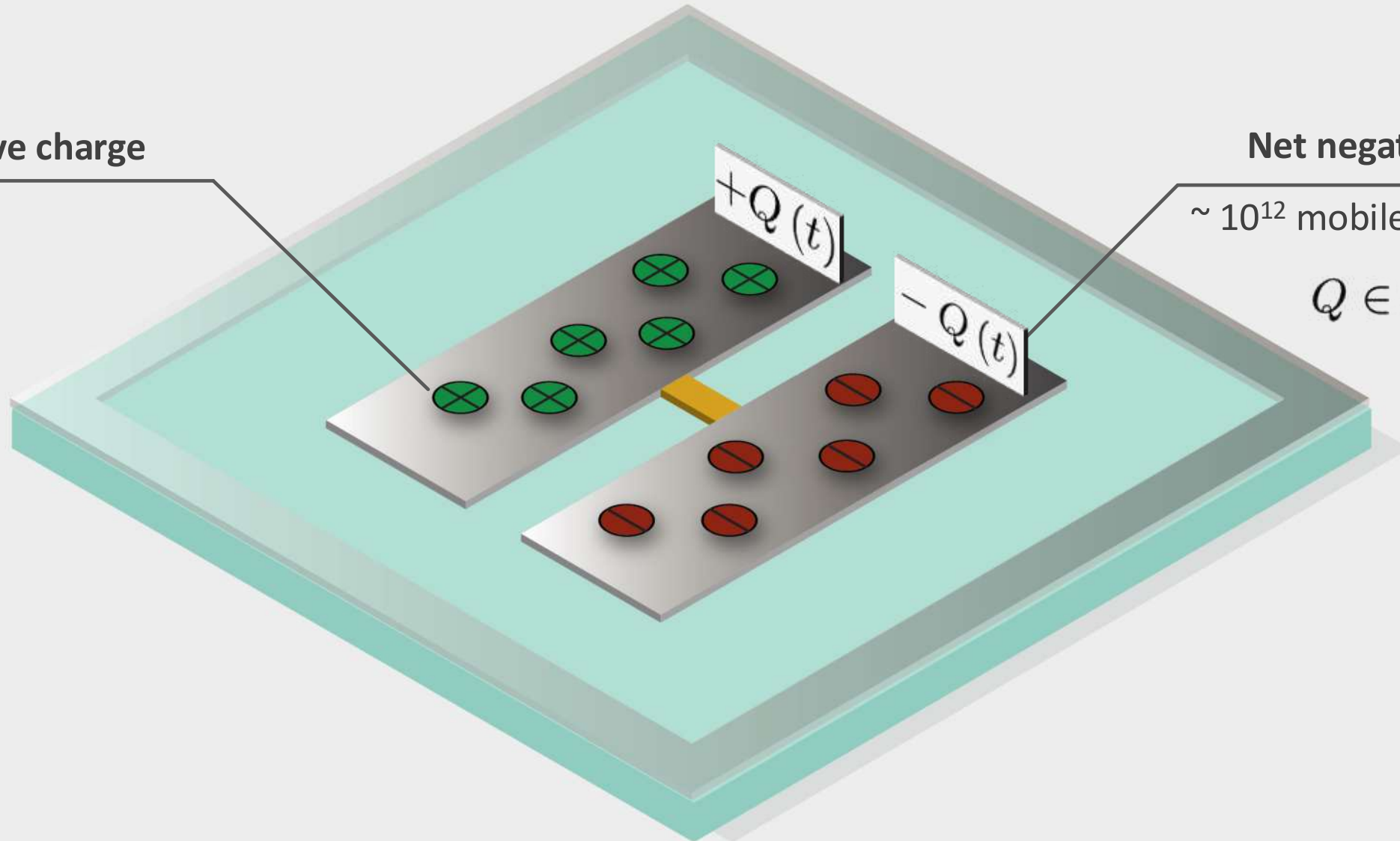
...

Transmon qubit



Transmon qubit: charge

Net positive charge

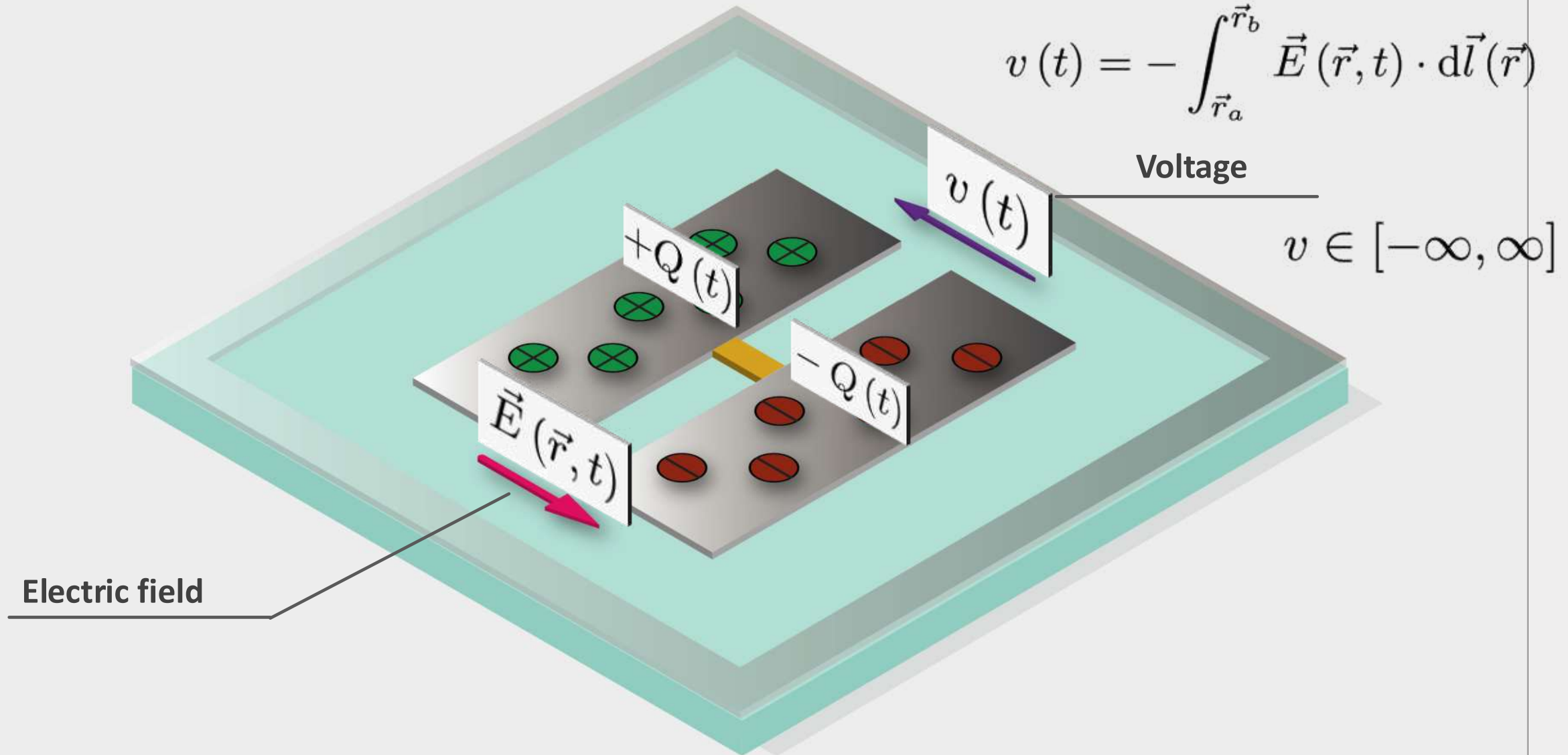


Net negative charge

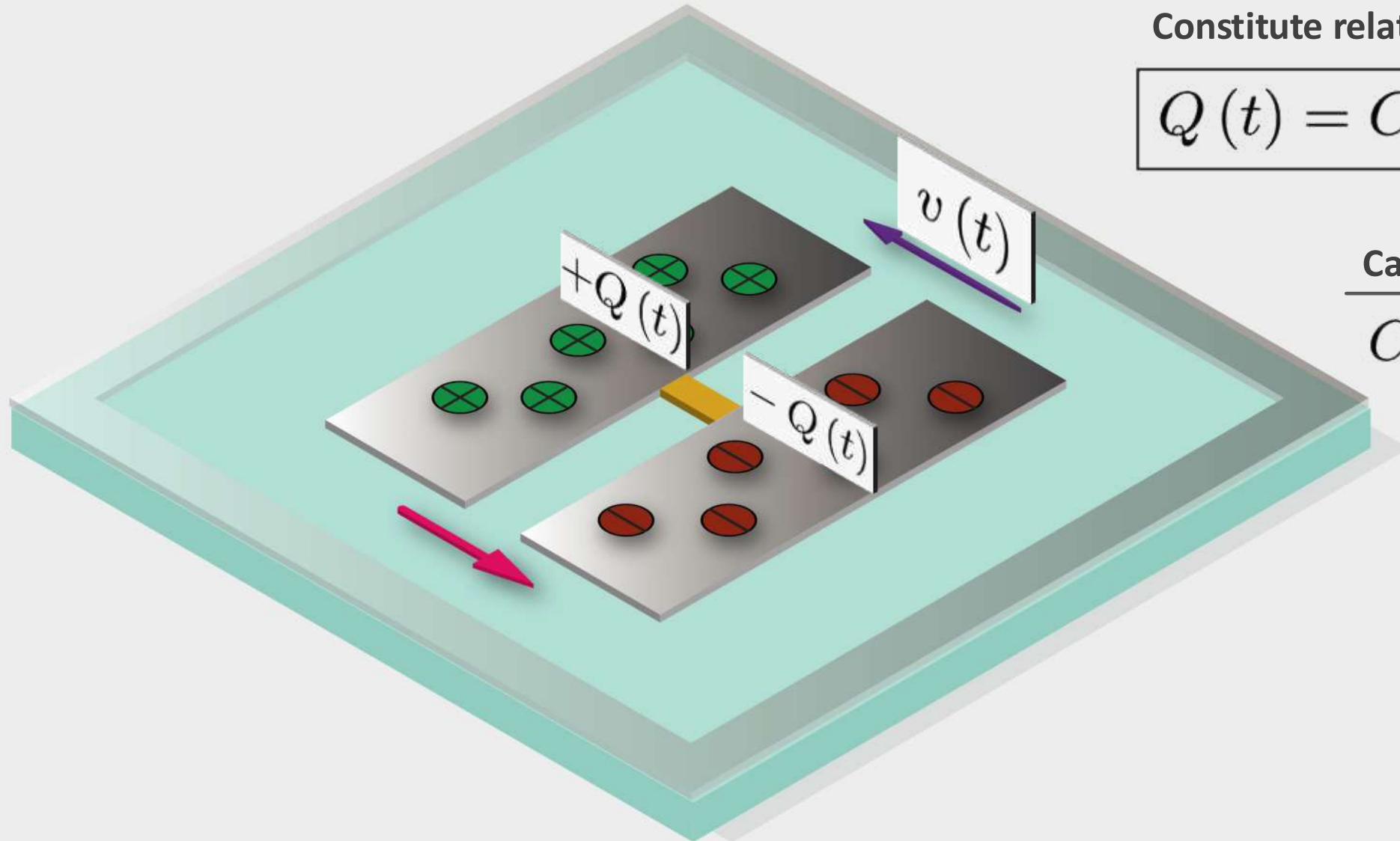
$\sim 10^{12}$ mobile e^-

$$Q \in [-\infty, \infty]$$

Electric field and voltage



Charge and capacitance



Constitute relationship

$$Q(t) = C v(t)$$

Capacitance

$$C \in (0, \infty)$$



Charge and current

Conservation of charge

Universal relationship

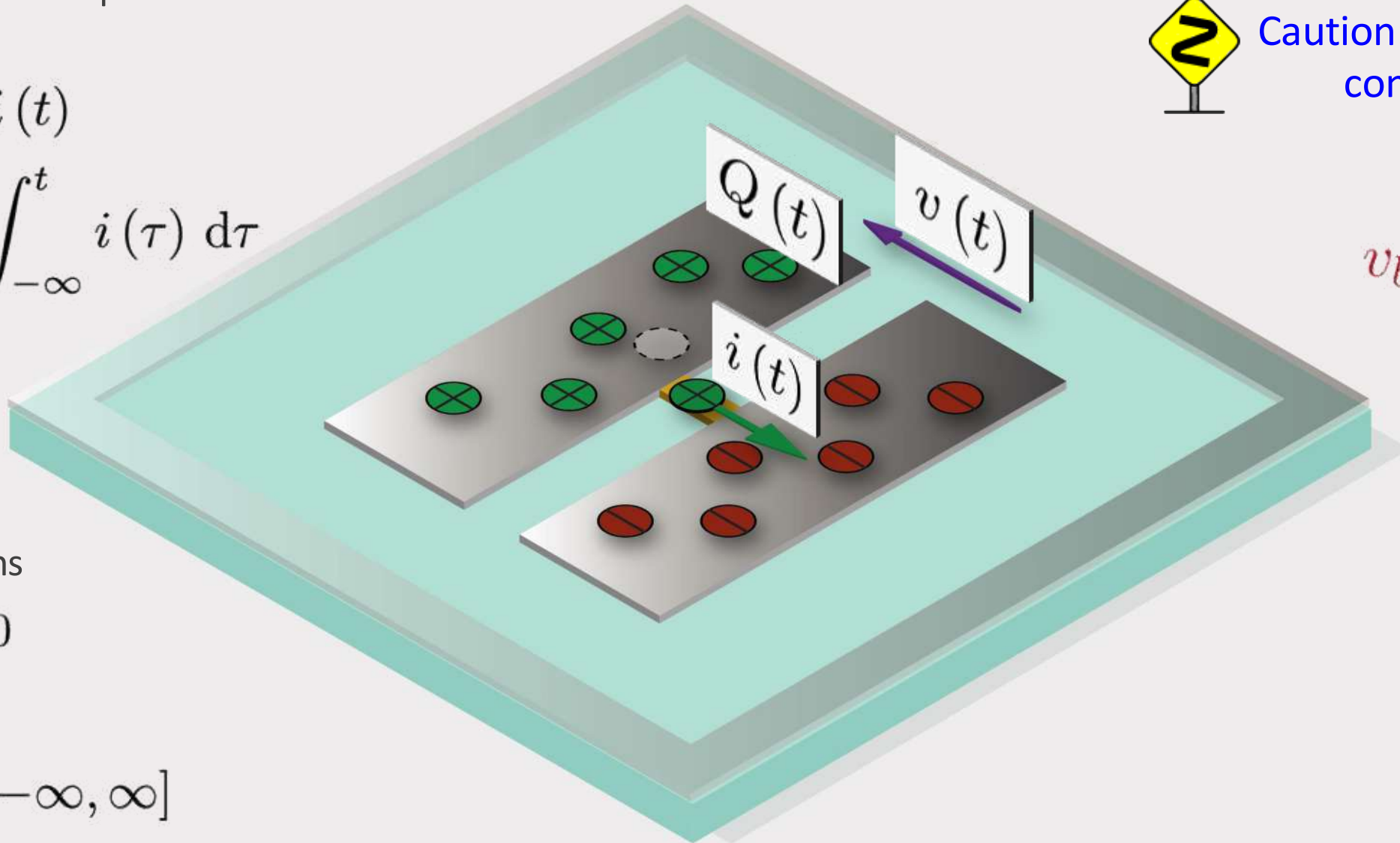
$$\frac{d}{dt}Q(t) = i(t)$$

$$Q(t) = \int_{-\infty}^t i(\tau) d\tau$$

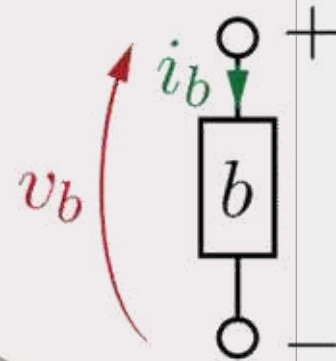
Initial conditions

$$Q(-\infty) = 0$$

$$i \in [-\infty, \infty]$$



Caution: Passive sign convention



Magnetic flux and inductance

Faraday's law of induction

Universal relationship

$$\Phi(t) = \int_{-\infty}^t v(\tau) d\tau$$

$$\frac{d}{dt}\Phi(t) = v(t)$$

Initial conditions

$$\Phi(-\infty) = 0$$

Magnetic flux

For kinetic inductors,
~98% of qubit inductive energy is not
in stored magnetic fields, but in
kinetic inductance

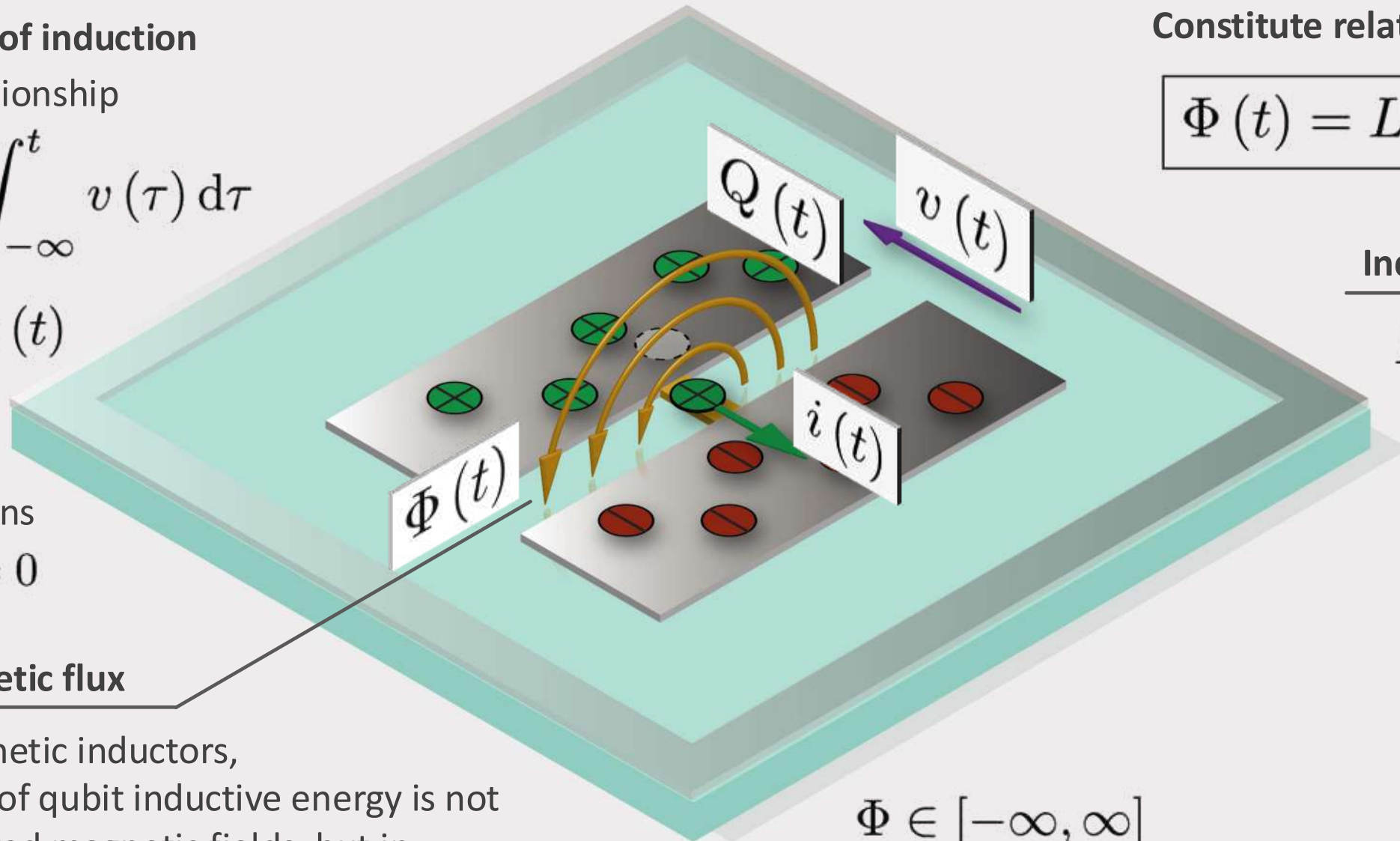
Constitute relationship

$$\Phi(t) = Li(t)$$

Inductance

$$L \in (0, \infty)$$

$$\Phi \in [-\infty, \infty]$$



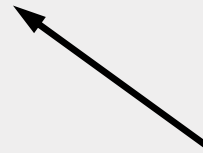
Power and energy

Universal

$$\frac{d}{dt} \mathcal{E}(t) = p(t) \equiv v(t) i(t)$$

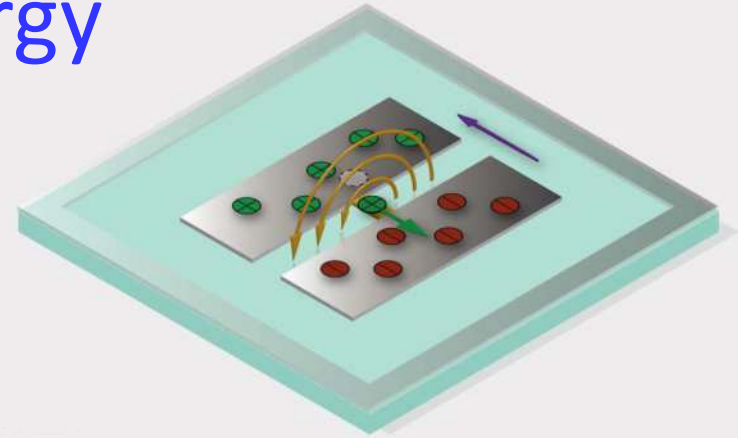


Energy stored *in*
(delivered *to*)
component



Instantaneous
power flowing *to*
component

$$\mathcal{E}(t) = \int_{-\infty}^t p(\tau) d\tau$$



$$\mathcal{E}_{\text{cap}}(\dot{\Phi}) = \frac{1}{2} C \dot{\Phi}^2$$

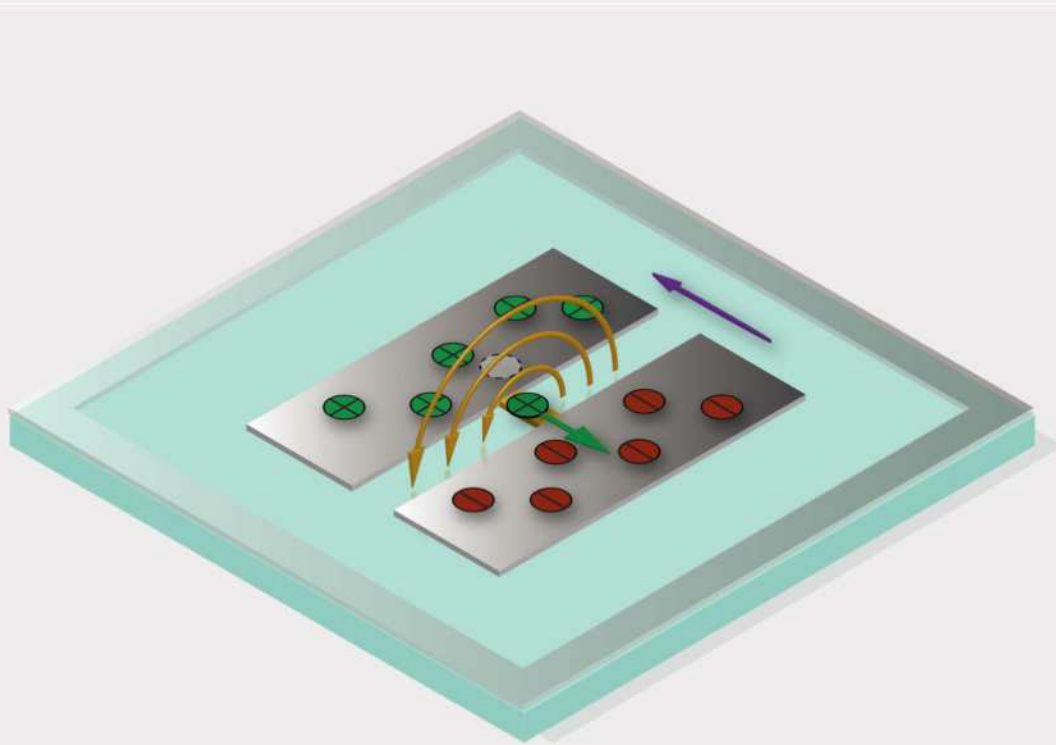
$$\mathcal{E}_{\text{cap}}(Q) = \frac{Q^2}{2C}$$



$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\mathcal{E}_{\text{ind}}(\dot{Q}) = \frac{1}{2} L \dot{Q}^2$$

Electromagnetic oscillator



capacitance value

node of the circuit

C

L

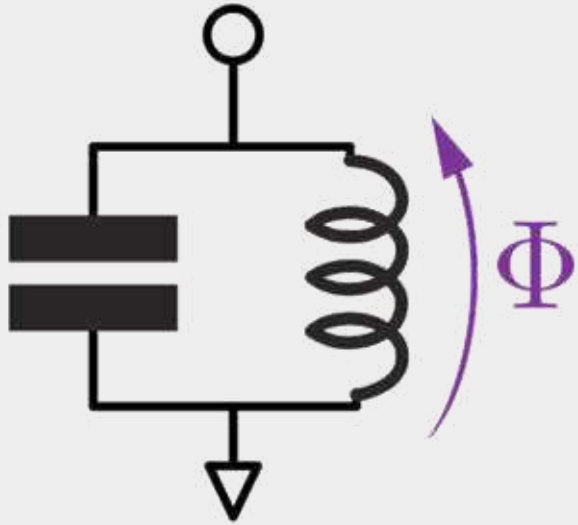
inductance
value

inductor

capacitor

ground

Oscillator analogy



$$C\ddot{\Phi} + L^{-1}\Phi = 0$$

$$\ddot{\Phi} = -\omega_0^2\Phi, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

Resonance
frequency

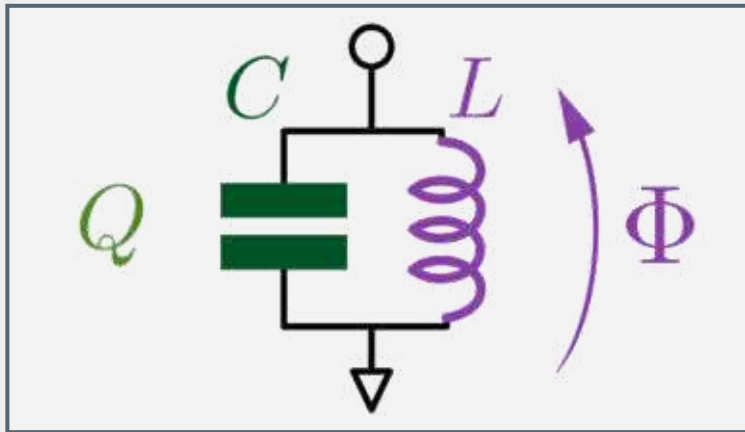


$\dot{\Phi}(t)$ Velocity

$\Phi(t)$ Deviation from equilibrium

Equilibrium position
 $x = 0, \quad \Phi = 0$

The LC classical harmonic oscillator



Position: $\Phi \mapsto x$

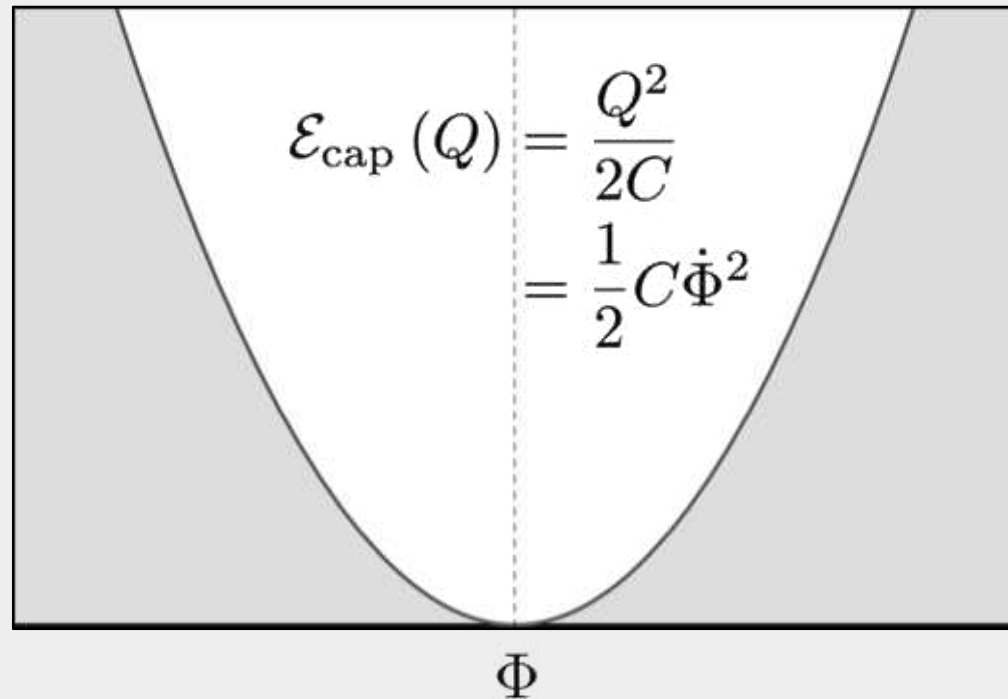
Mass: $C \mapsto m$

Momentum: $Q \mapsto p$

Spring constant: $L^{-1} \mapsto k$

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

Energy



$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

$$\omega_0^2 = \frac{1}{LC} \quad Z_0 = \frac{L}{C}$$

“It is by *logic* that we prove,
but by *intuition* that we discover.”

Henri Poincaré

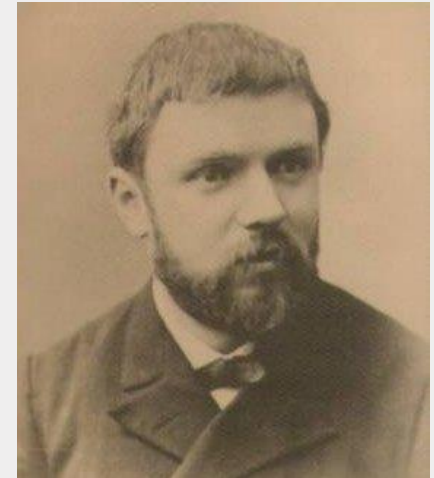


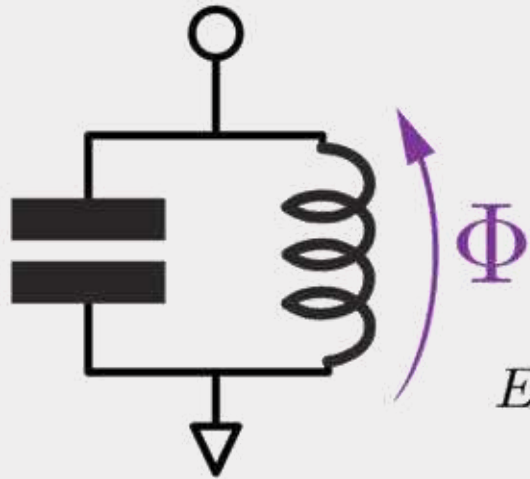
Photo by Eugène Pirou

Hamiltonian dynamics and phase space

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

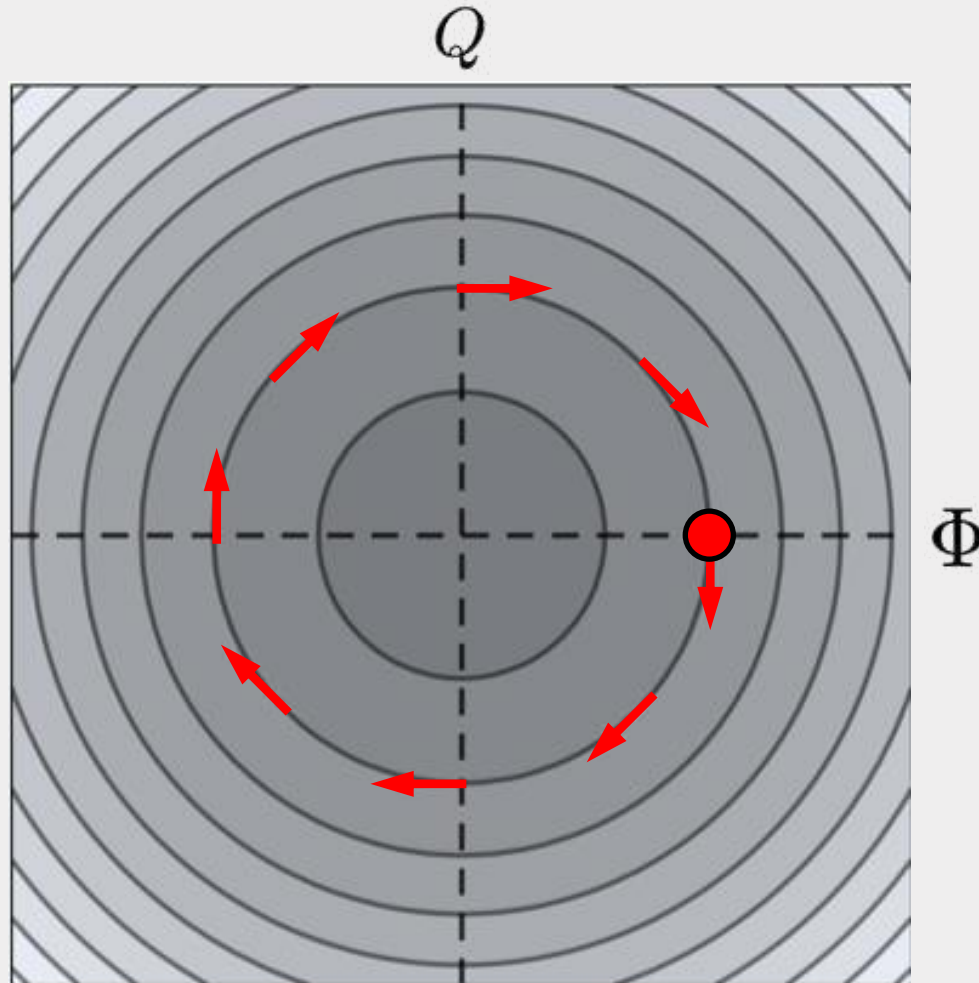
with $L = C = 1$

$$E = \hbar\omega_0 \left(n + \frac{1}{2} \right) = \frac{1}{2} (Q^2 + \Phi^2)$$



$$E = \frac{3\hbar\omega_0}{2}$$

$$E = \frac{\hbar\omega_0}{2}$$

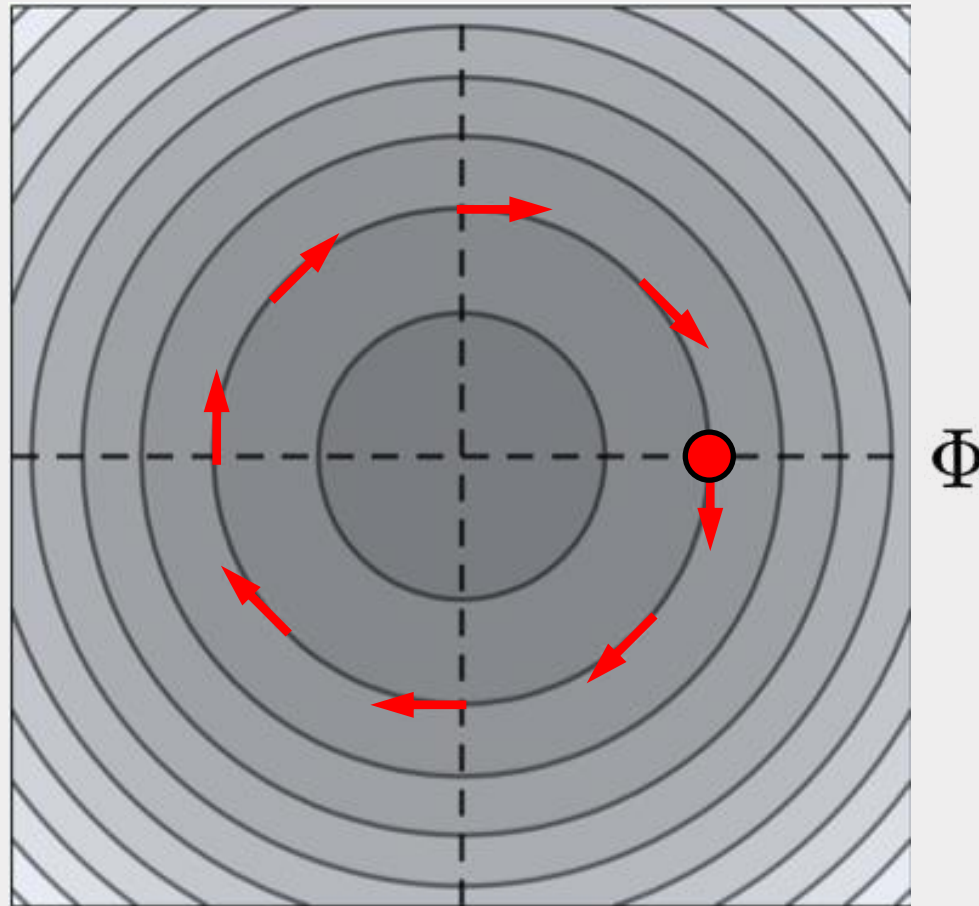
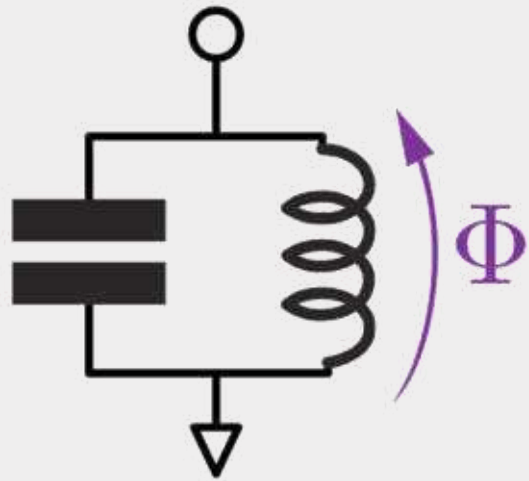


$$\dot{\Phi} = +\frac{\partial \mathcal{H}}{\partial Q} = +Q$$

$$\dot{Q} = -\frac{\partial \mathcal{H}}{\partial \Phi} = -\Phi$$

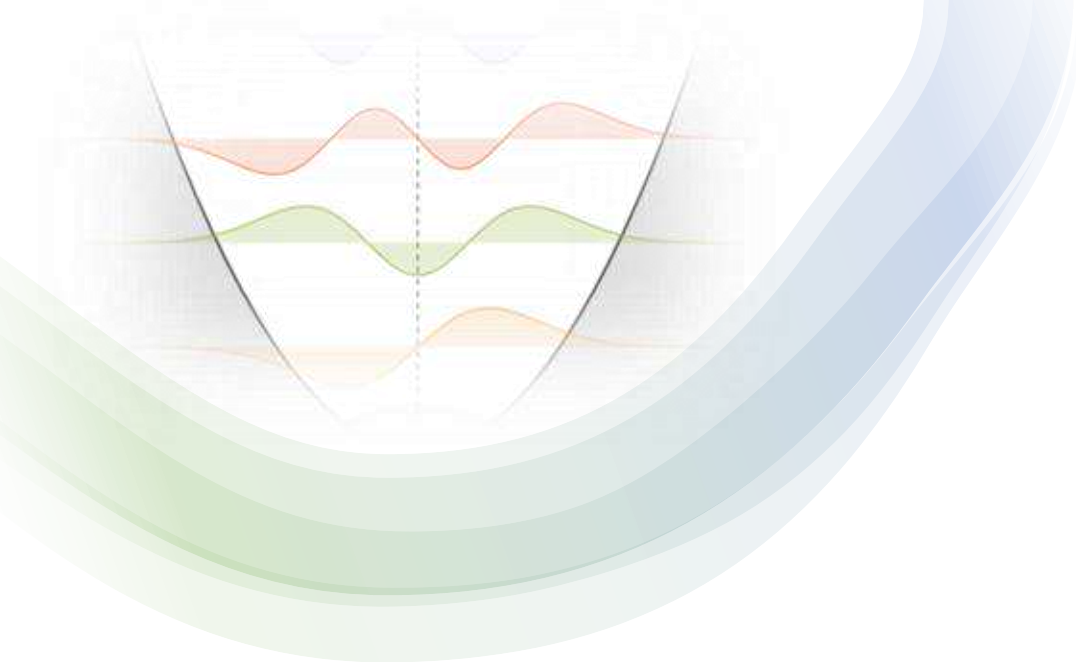
Complex action-angle variable

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L} \quad E = \hbar\omega_0 \left(n + \frac{1}{2} \right) \quad \alpha(t) = \sqrt{\frac{1}{2\hbar Z}} [\Phi(t) + iZQ(t)]$$



Classical analog of the bosonic ladder operator

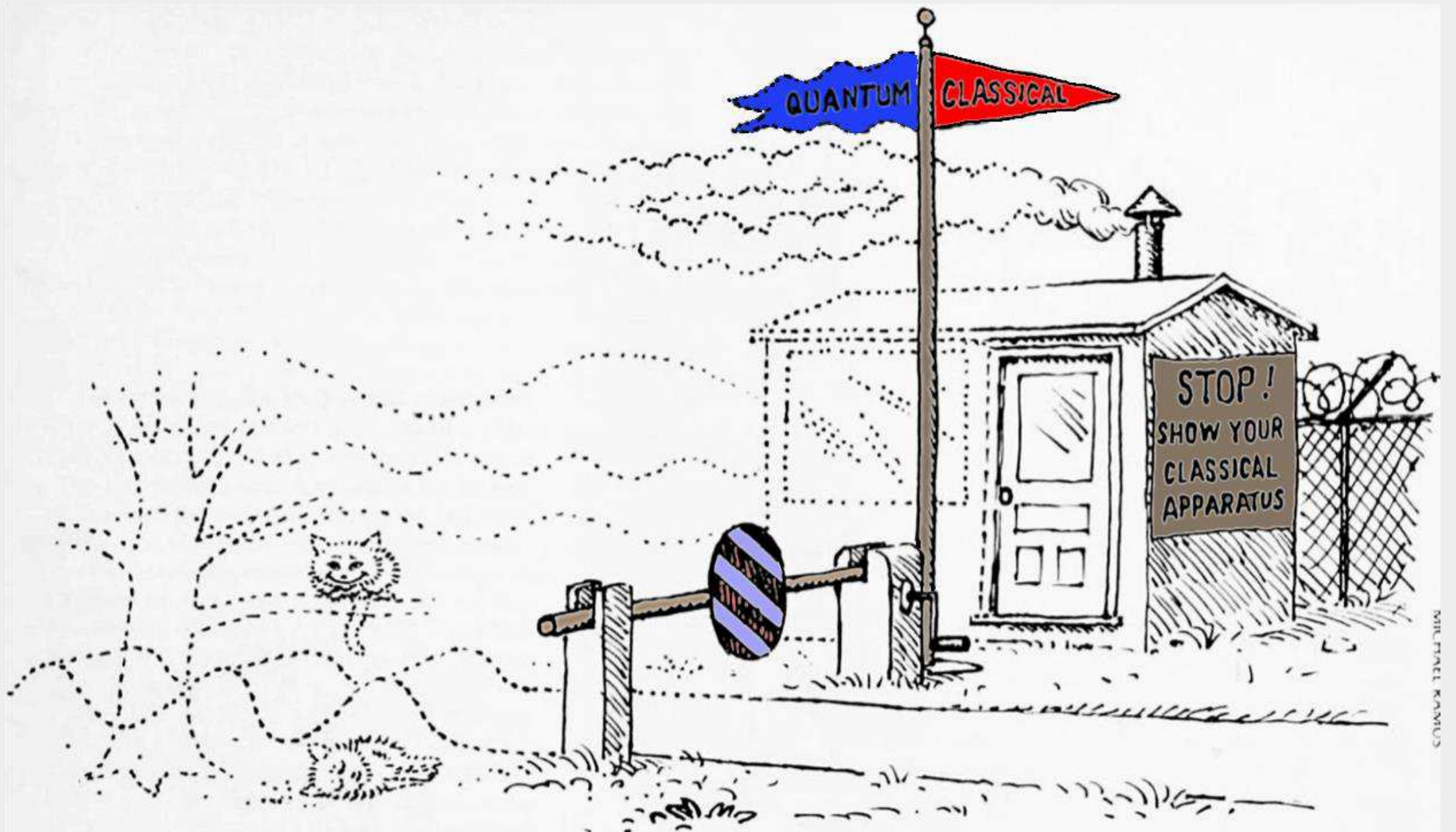
$$\alpha(t) = \alpha(0) e^{-i\omega_0 t}$$



Unveiling the quantum

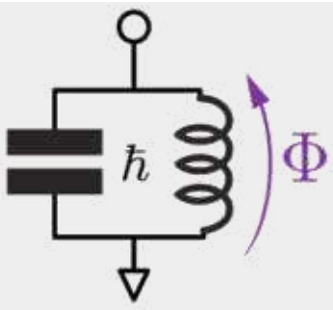
Quantum harmonic oscillator





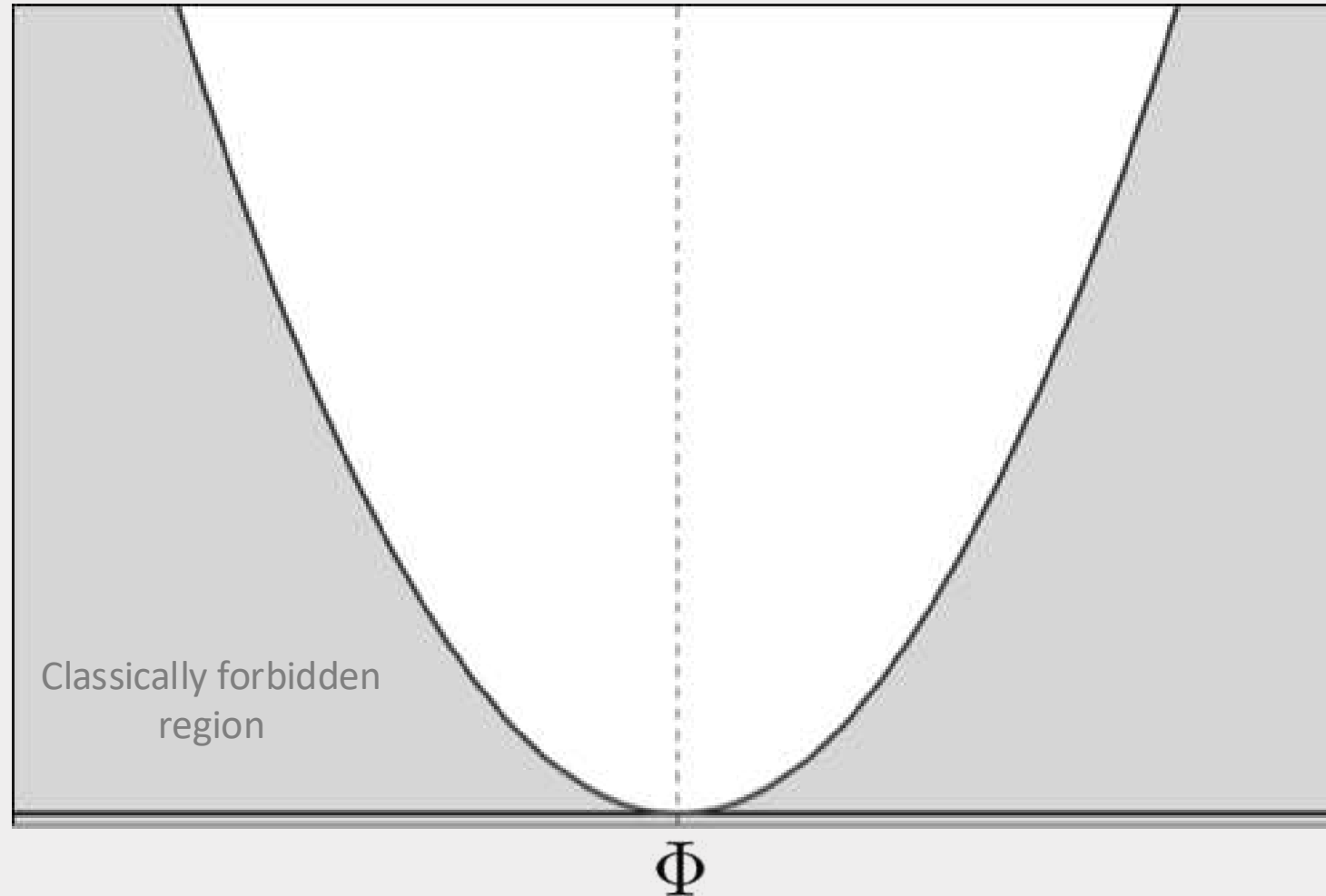
Drawing: Zurek, Physics Today (1991)

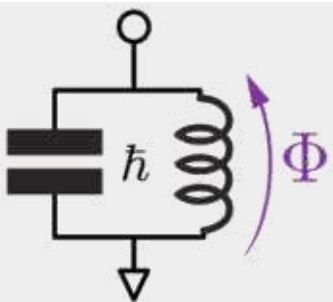
Wavefunctions of the quantum oscillator



$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

Energy



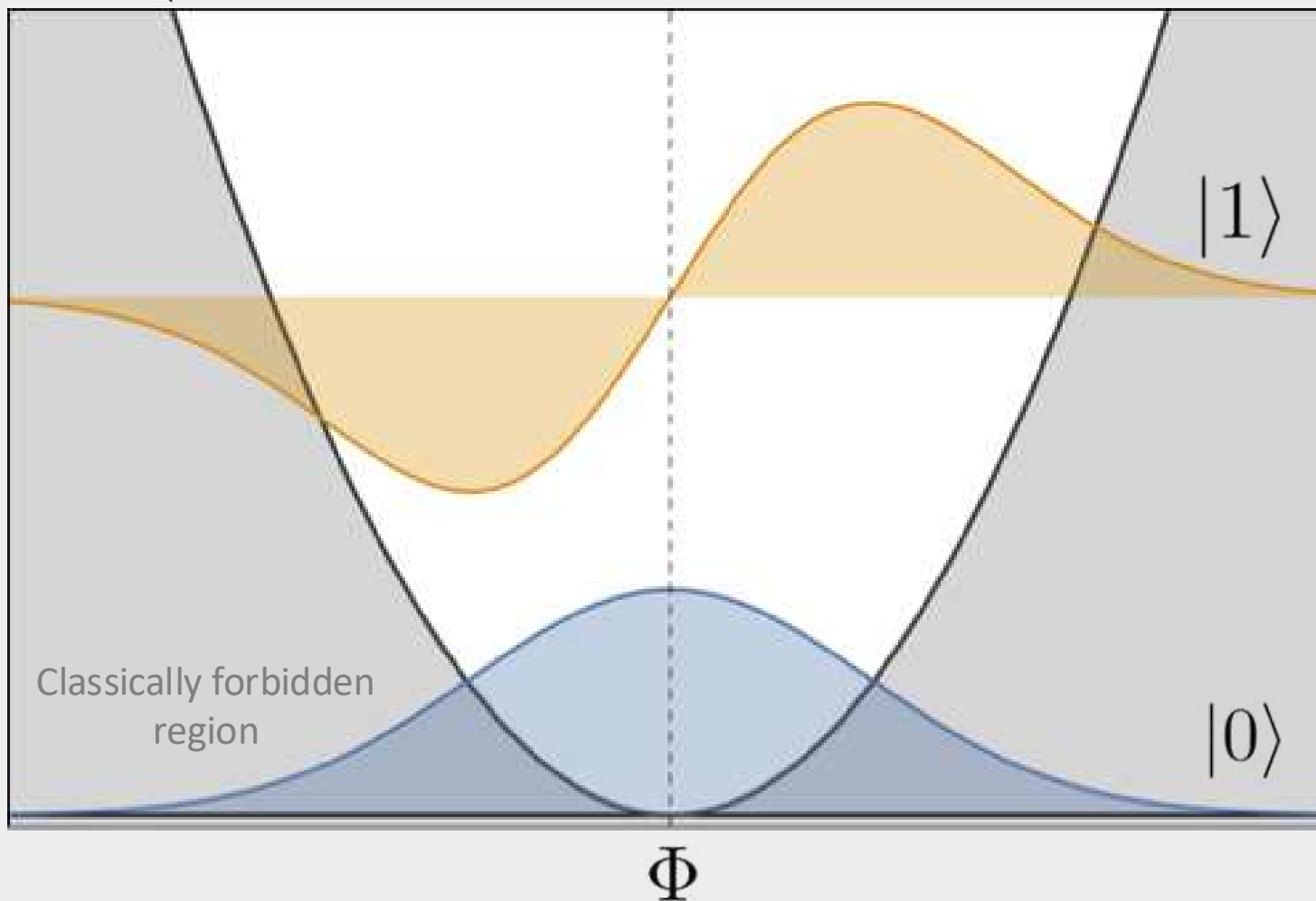


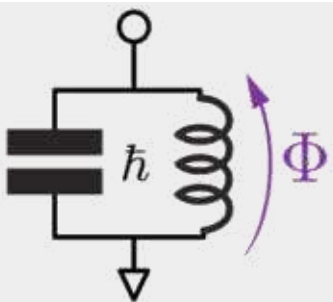
Wavefunctions of the quantum oscillator

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\psi_n(\Phi) \equiv \langle \Phi | n \rangle$$

Energy /
Scaled
wavefunction
amplitude



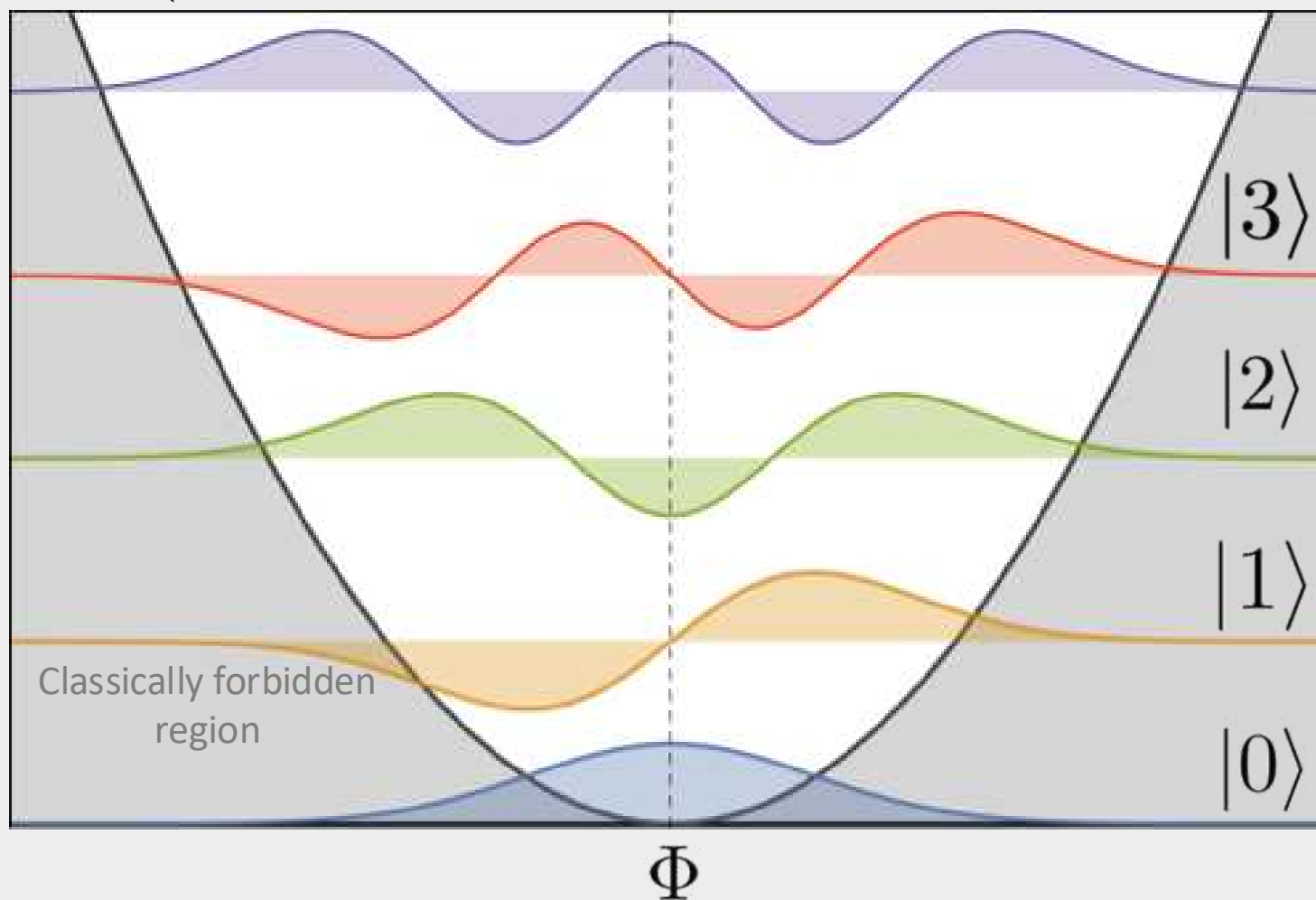


Wavefunctions of the quantum oscillator

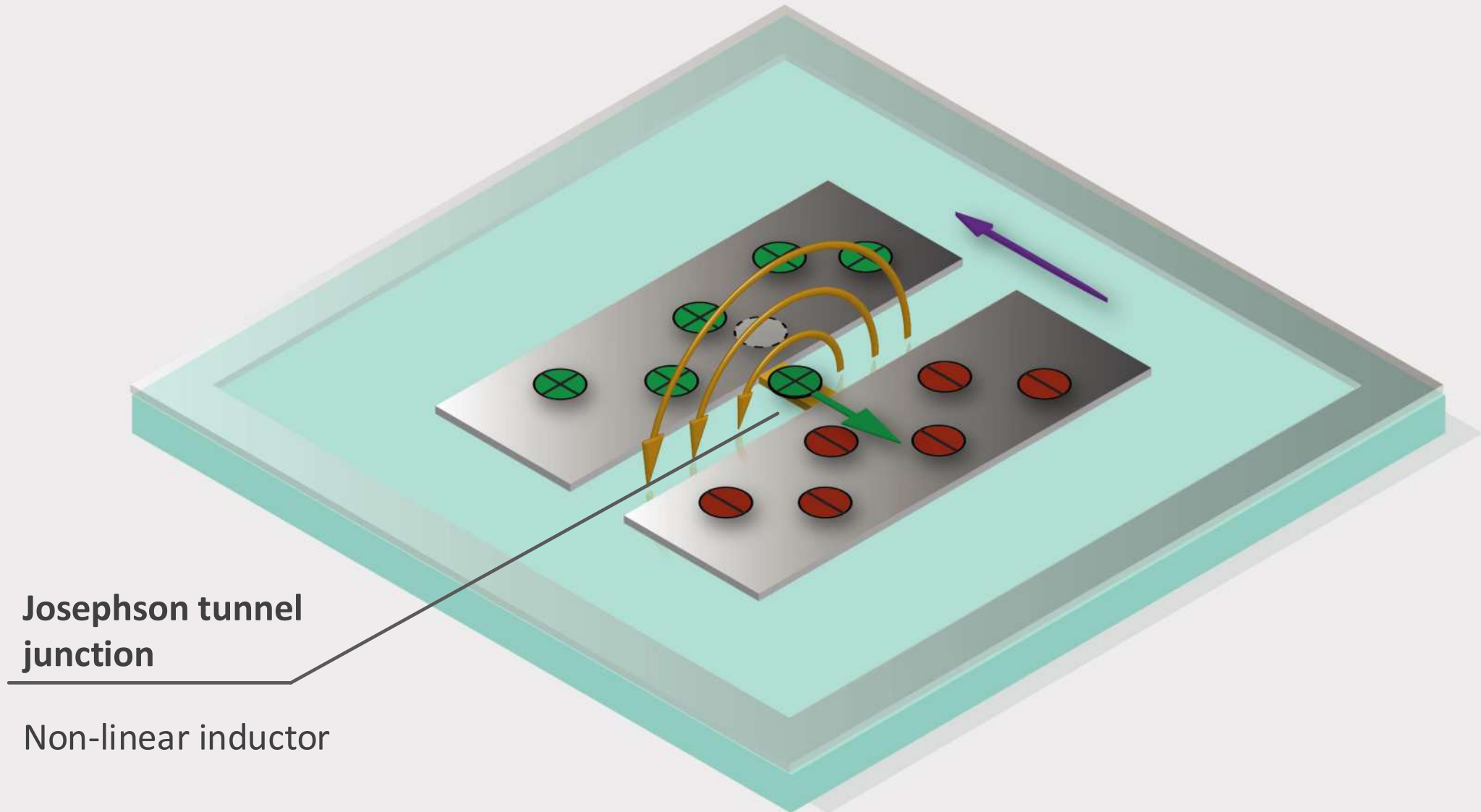
$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\psi_n(\Phi) \equiv \langle \Phi | n \rangle$$

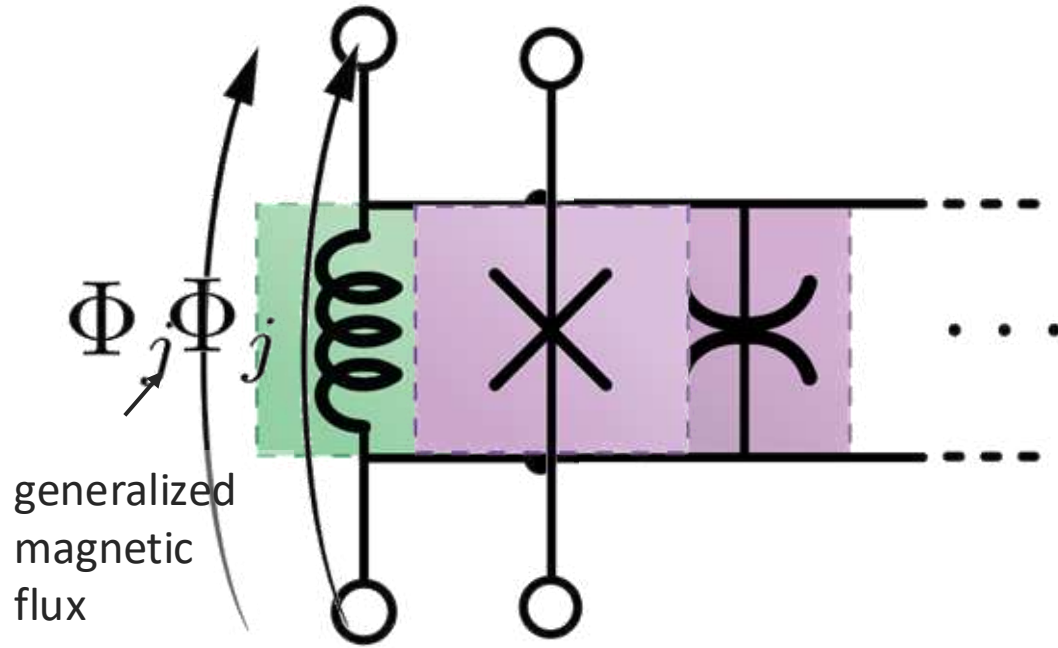
Energy /
Scaled
wavefunction
amplitude



The transmon as a non-linear oscillator



Josephson tunnel junction

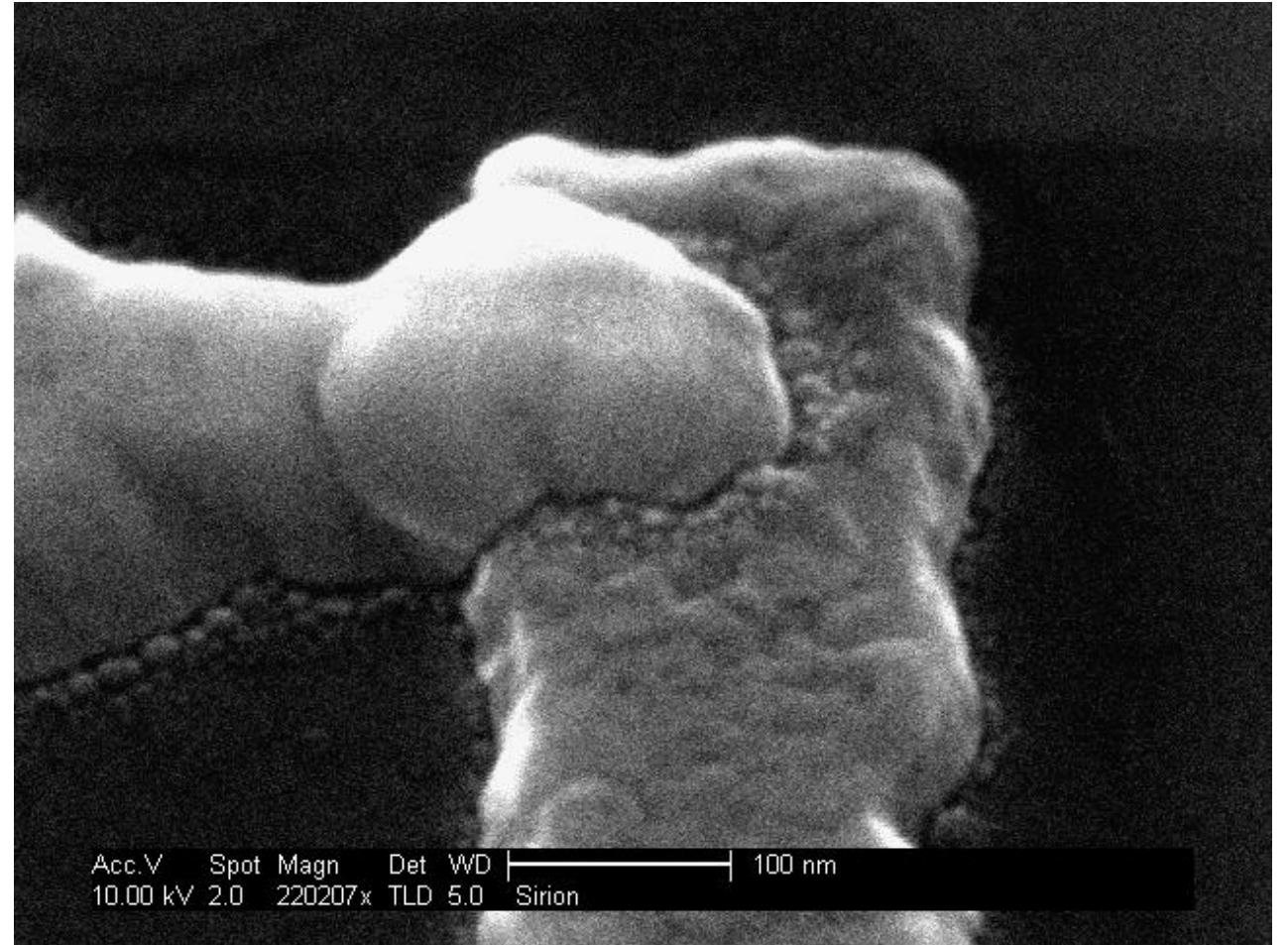


$$\mathcal{E}_j(\Phi_j) = E_j (1 - \cos(\Phi_j/\phi_0))$$

$$= \mathcal{E}_j^{\text{lin}}(\Phi_j) + \mathcal{E}_j^{\text{nl}}(\Phi_j)$$

$$\phi_0 \equiv \hbar/2e$$

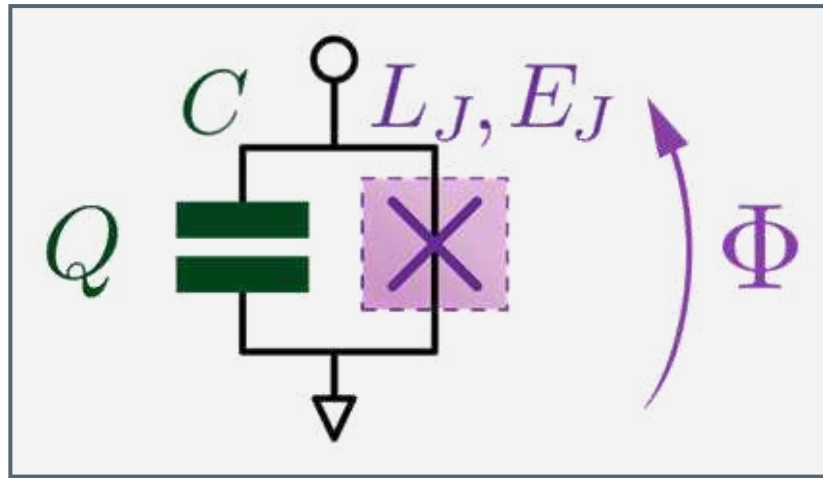
$$= \frac{E_j}{2} \left(\frac{\Phi_j}{\phi_0} \right)^2 - \frac{E_j}{4!} \left(\frac{\Phi_j}{\phi_0} \right)^4 + \mathcal{O}(\Phi_j^6)$$



SEM image: L. Frunzio

Circuit image: Minev et al., EPR to appear (2020)

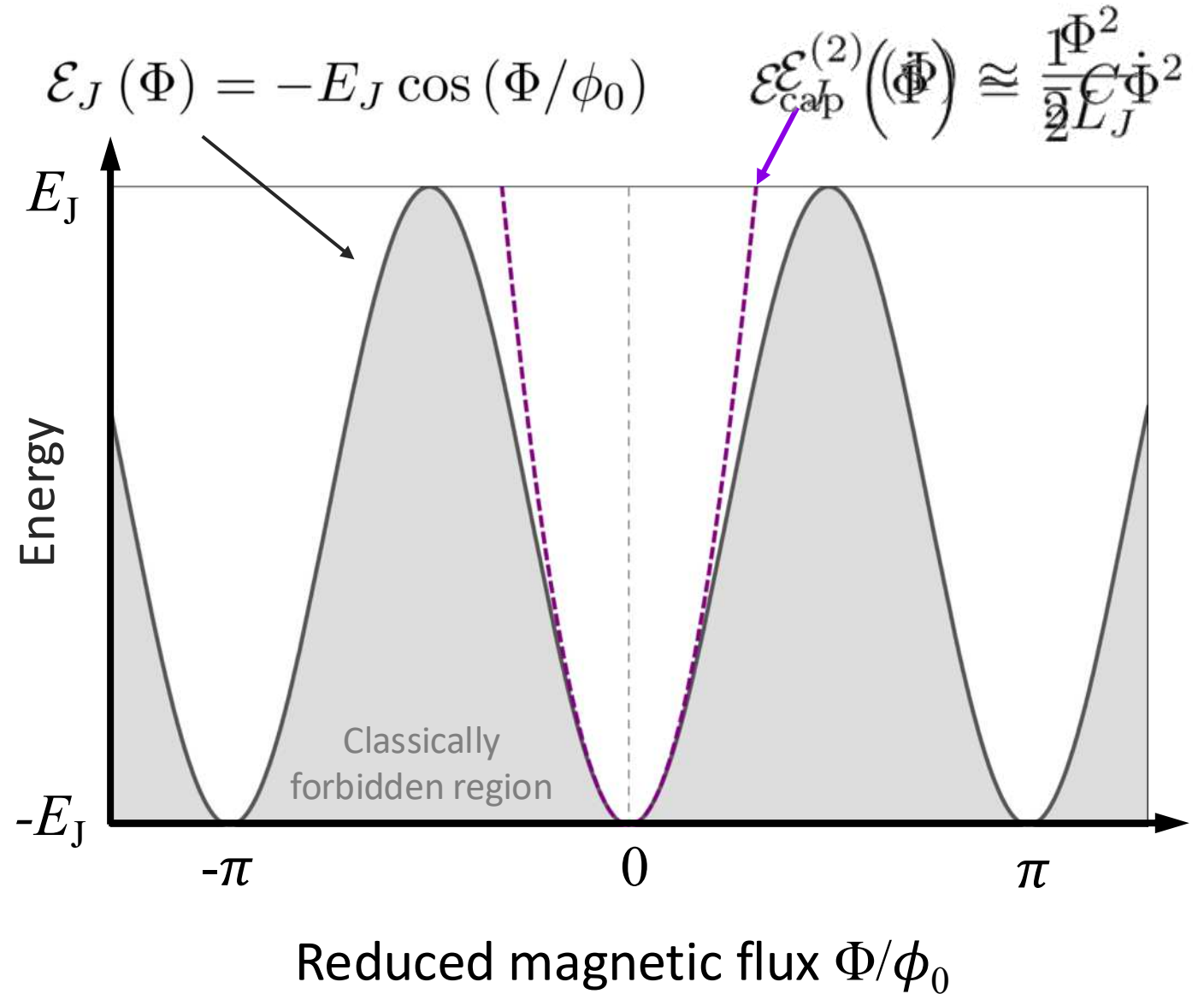
The Transmon qubit



$$E_J = \frac{\phi_0^2}{L_J}$$

$$\phi_0 = \frac{\hbar}{2e}$$

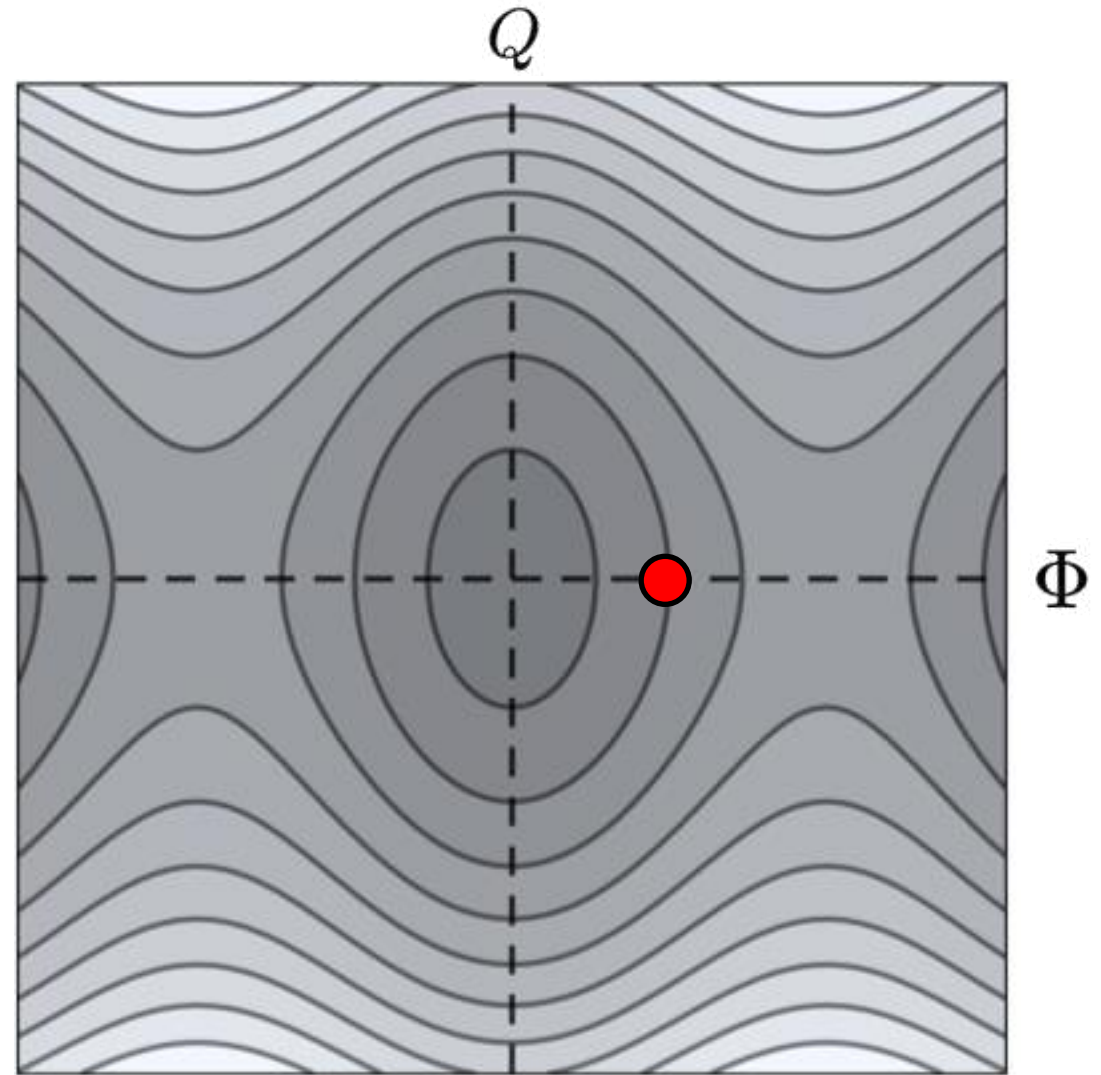
$$\approx 3.3 \times 10^{-16} \text{ Wb}$$

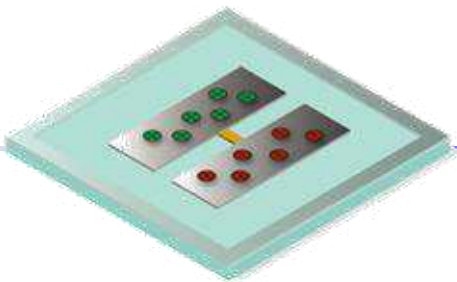


Semi-classical intuition: phase space picture

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} - E_J \cos(\Phi/\phi_0)$$

$$\begin{aligned} E &= \hbar\omega_0 \left(n + \frac{1}{2} \right) \\ &= \frac{Q^2}{2C} - E_J \cos(\Phi/\phi_0) \end{aligned}$$

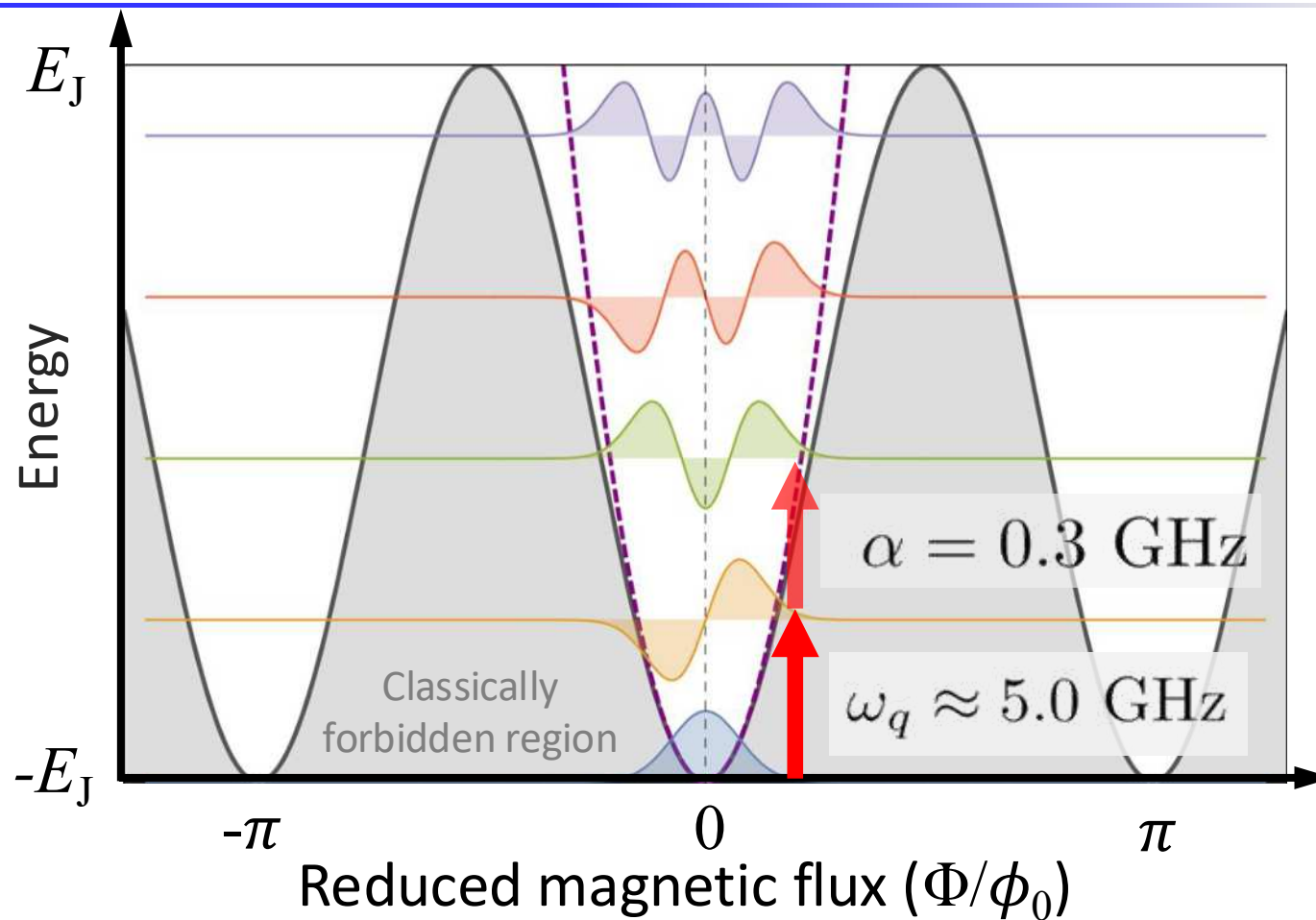




Exploring a real transmon qubit

$$\hat{H}_4^{\text{RWA}} \approx \hbar\omega_q \hat{N} - \frac{\hbar\alpha}{2} \hat{N} (\hat{N} - 1)$$

Dispersive, states didn't change to 1st order
 "Lamb shift" due to ZPF



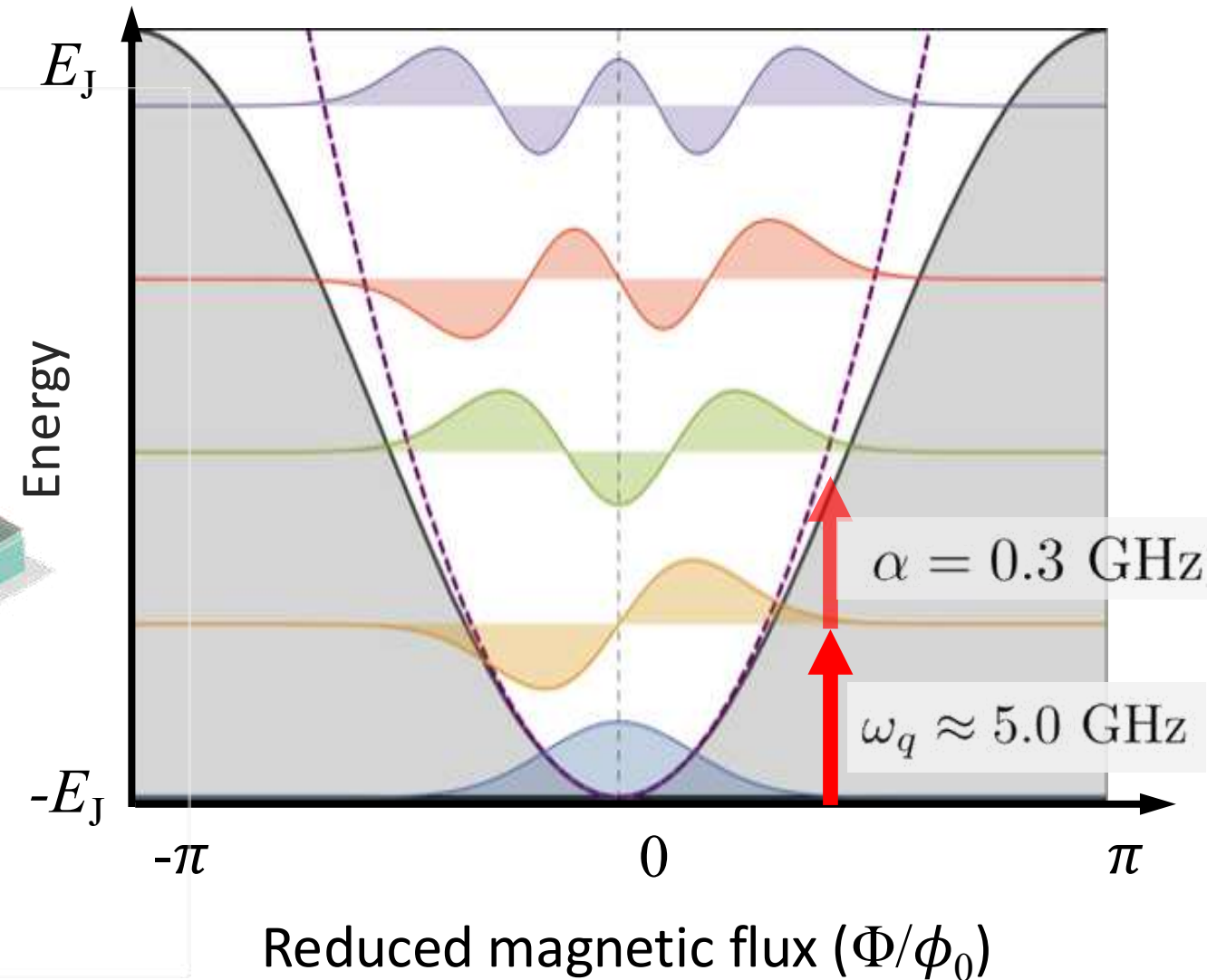
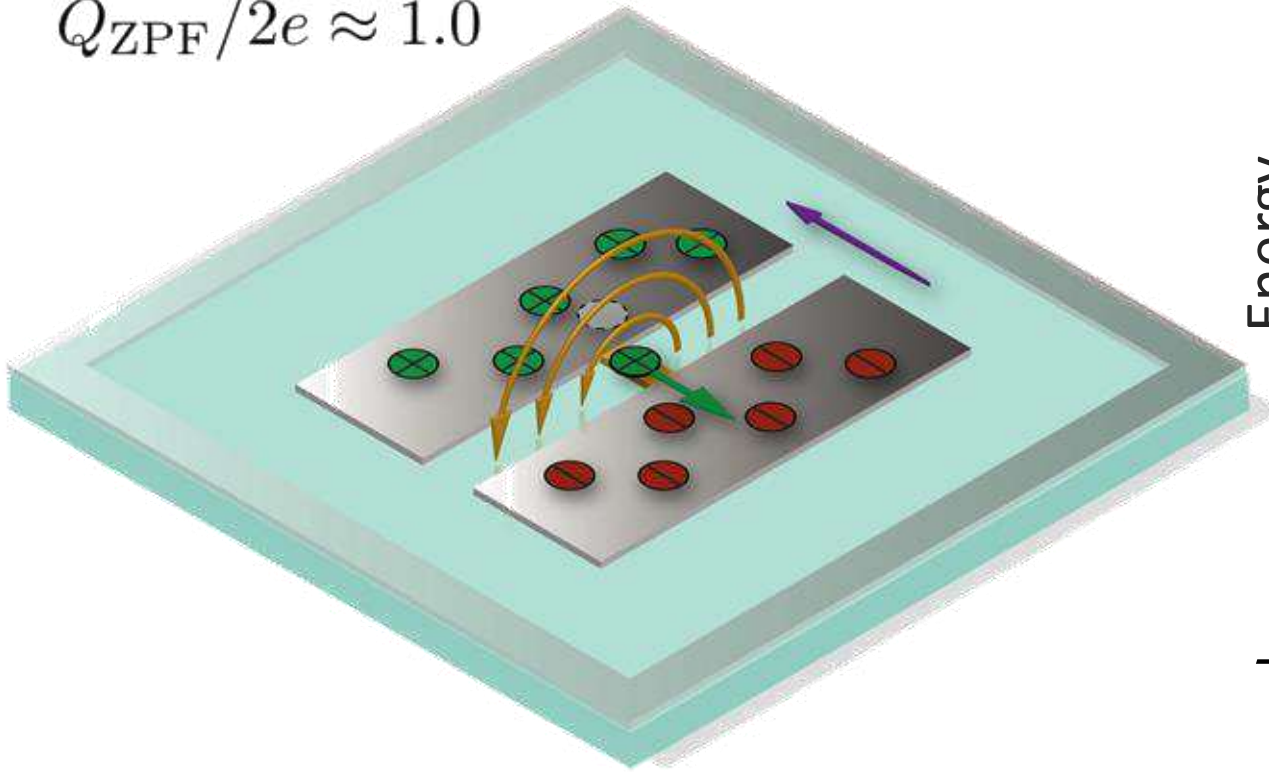
Parameters used in figure (of a measured qubit)

$$\begin{aligned} L_J &= 14 \text{ nH} & E_J &= \frac{\phi_0^2}{L_J} = 12 \text{ GHz} & \omega_0 &= \sqrt{\frac{1}{LC}} = 5.3 \text{ GHz} \\ C_J &= 65 \text{ fF} & E_C &= \frac{e^2}{2C} = 0.3 \text{ GHz} & Z &= \sqrt{\frac{L}{C}} \approx 450 \text{ } \Omega \end{aligned}$$

Quantum fluctuations of the transmon qubit

$$\Phi_{\text{ZPF}}/\phi_0 \approx 0.5$$

$$Q_{\text{ZPF}}/2e \approx 1.0$$



FIREHOSE MODE: ON

RAPID OVERVIEW OF ADVANCED QUANTUM METHODS
& COMMUNITY CHIP DESIGNS

HOLD ONTO
YOUR QUBITS!



QUANTUM ANALYSIS



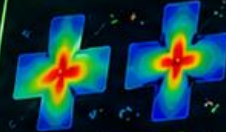
Understand the physics
behind your device

BLACK-BOX QUANTIZATION



Extract modes & energies
without assumptions

EIGENMODE SIMULATIONS



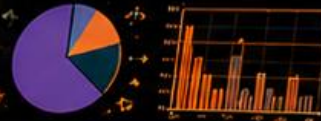
See the invisible.
Design with confidence.

CAPACITIVE SIMULATIONS



Accurate capacitances.
Reliable models.

ENERGY PARTICIPATION RATIO



Know where energy lives.
Design better. Debug faster.

EXAMPLES OF COMMUNITY CHIP DESIGNS



Real designs. Real impact. Real inspiration.

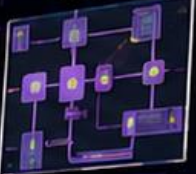
QUBIT XY



FLUX TUNABLE



QUBIT ZZ



TRANSMON



RESONATOR



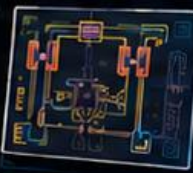
COUPLER



3D CAVITY



MULTI-QUBIT



READOUT



PARAMETRIC



TUNABLE BUS



SNAIL



QUBIT ARRAY



LOGICAL QUBIT



AND MORE!



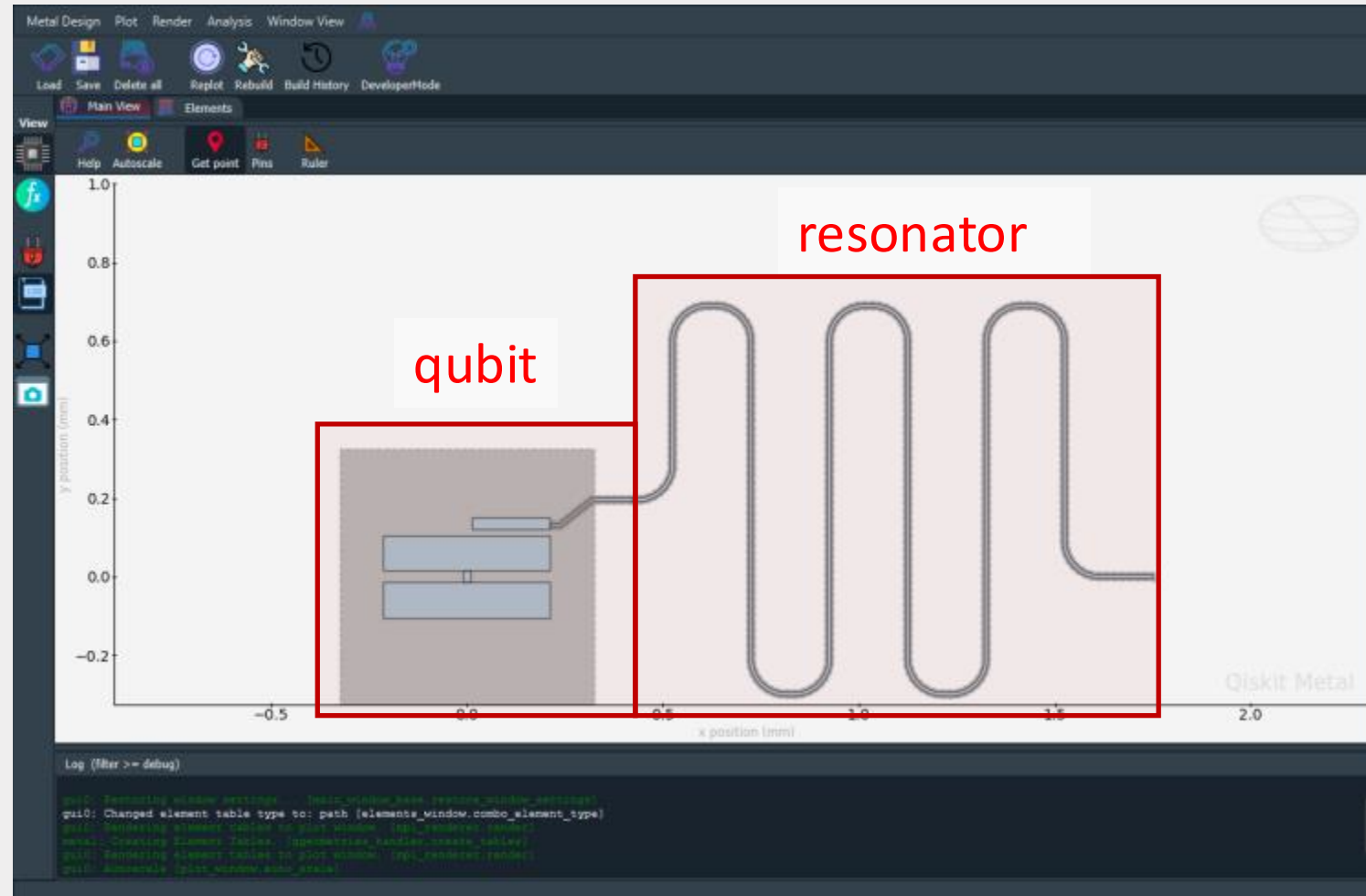
THE COMMUNITY IS AMAZING! ❤️



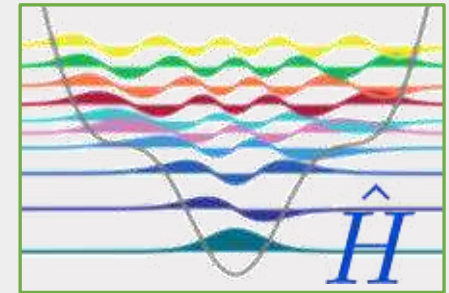
HIGH INFORMATION VELOCITY AHEAD BUCKLE UP!



How do we quantize?

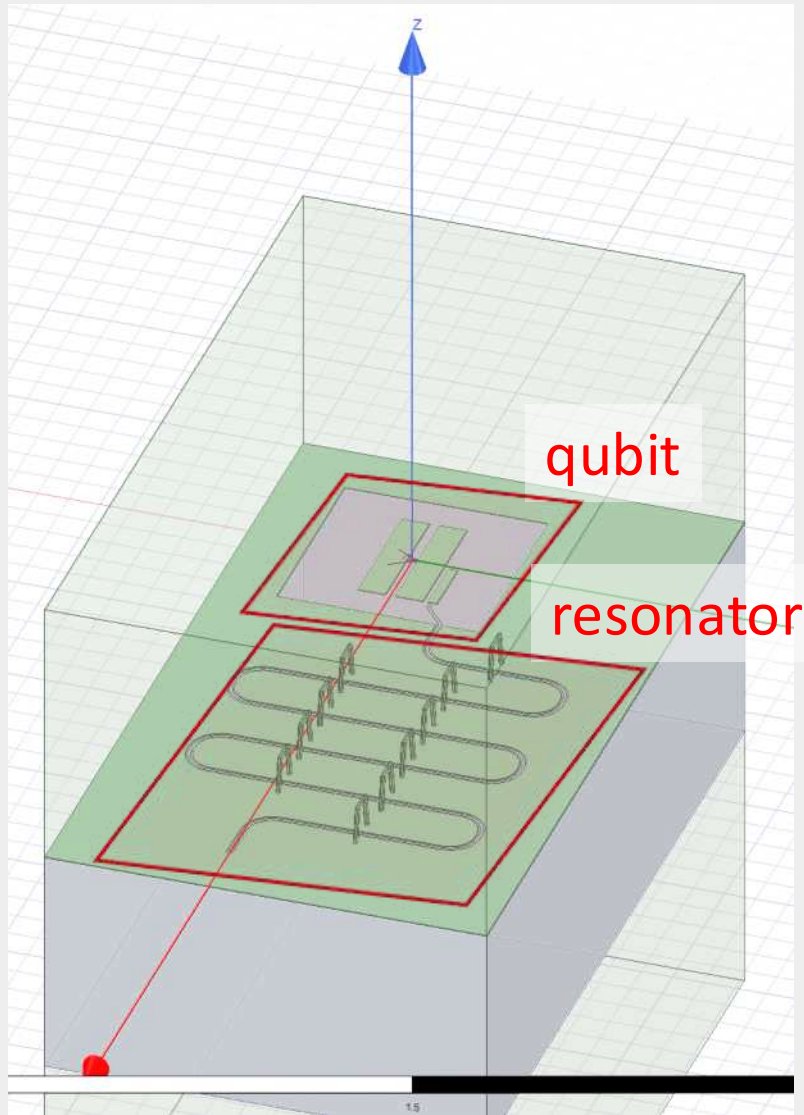


?

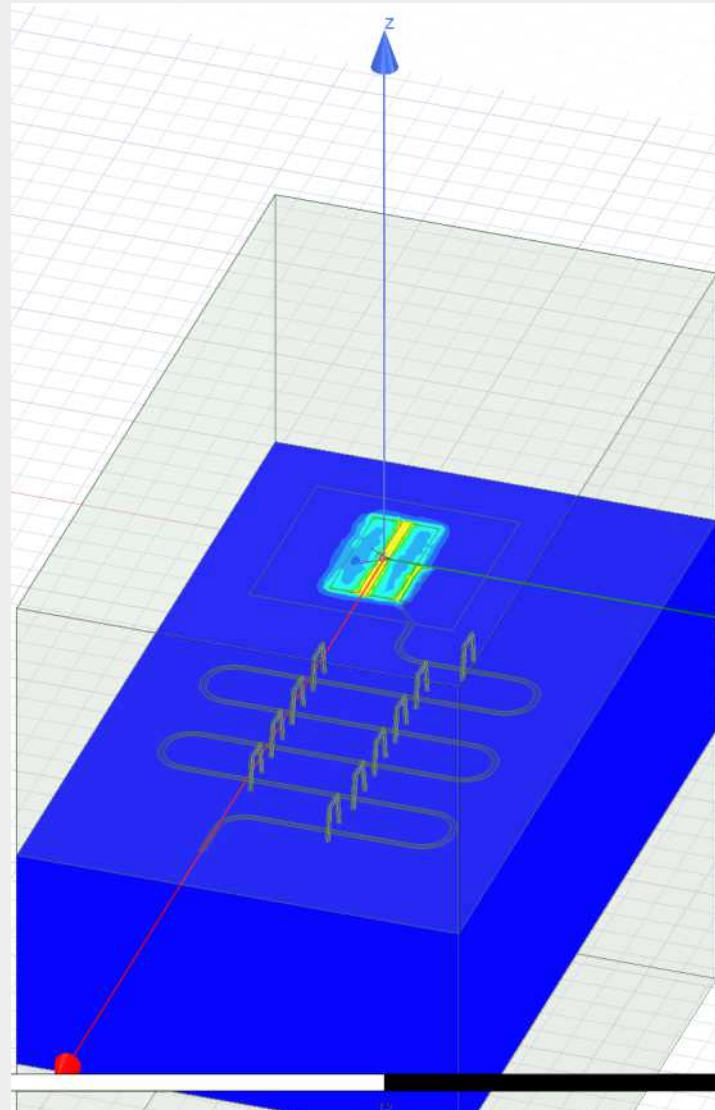


Modeling step 1: Classical E&M approximations

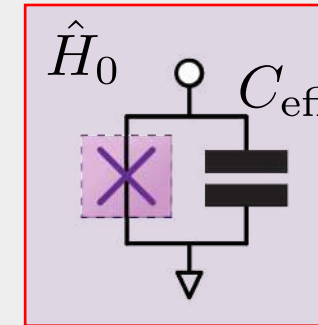
Physical (linear) model



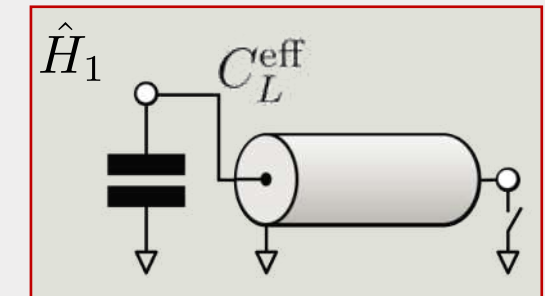
Eigenmodes



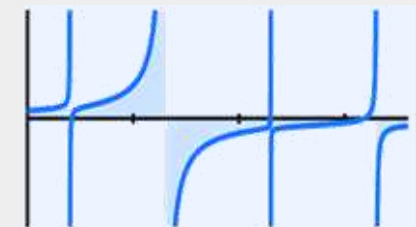
Quasi-lumped models



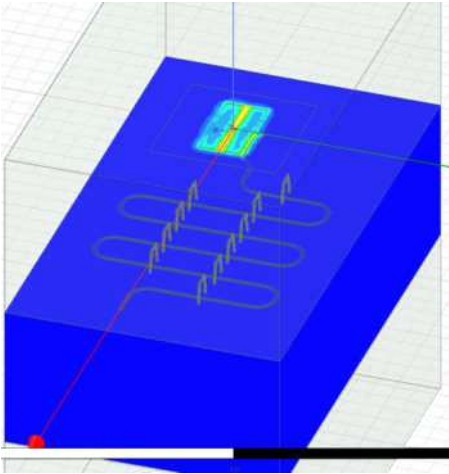
$$\begin{matrix} n_0 & n_1 & n_2 & n_3 & n_4 & n_5 \\ \begin{matrix} n_0 \\ n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \end{matrix} & \begin{pmatrix} C_{0\Sigma} & -C_{01} & -C_{02} & -C_{03} & -C_{04} & -C_{05} \\ & C_{1\Sigma} & -C_{12} & -C_{13} & -C_{14} & -C_{15} \\ & & C_{2\Sigma} & -C_{23} & -C_{24} & -C_{25} \\ & & & C_{3\Sigma} & -C_{34} & -C_{35} \\ & & & & C_{4\Sigma} & -C_{45} \\ & & & & & C_{5\Sigma} \end{pmatrix} \end{matrix}$$



Impedance



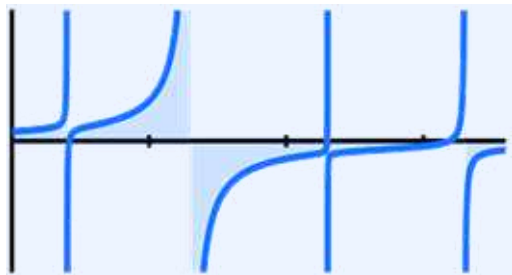
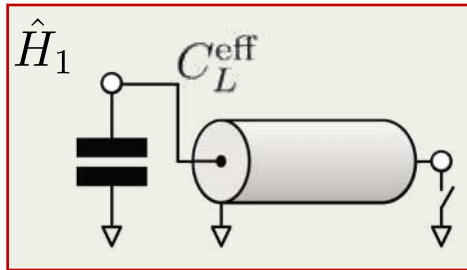
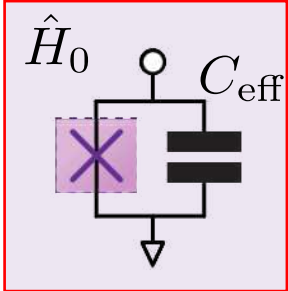
Modeling step 2: Quantize



$$\hat{H}_{\text{tot}}$$

$$\hat{H}_{\text{tot}} = \hat{H}_{\text{sys}} + \hat{H}_{\text{int}}$$

$$\hat{H}_{\text{tot}} = \hat{H}_{\text{lin}} + \hat{H}_{\text{nl}}$$



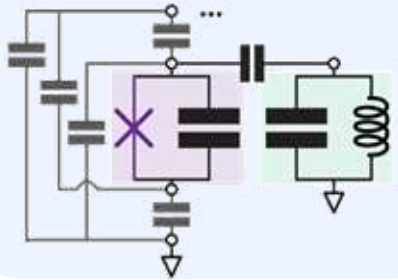
There's a description for every job!

Landscape of quantization methods

quasi-static

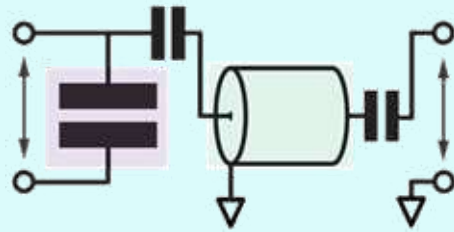
full-wave

lumped



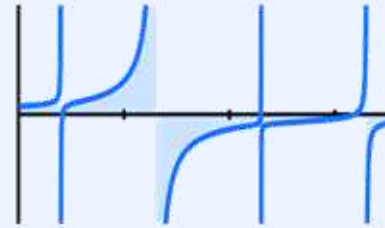
Yurke & Denker (1984), Devoret (1997),
Burkard *et al.* (2004), Koch *et al.*...

quasi-lumped



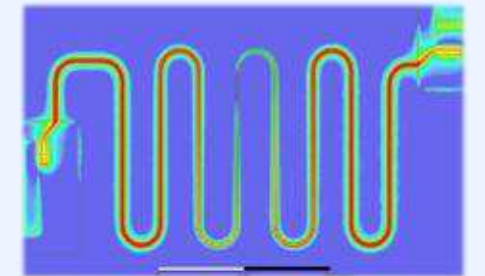
Malekakhlagh *et al.* (2017, 2019),
Gely *et al.* (2019), Parra-Rodriguez *et al.* (2019),
Minev *et al.* (2021), ...

impedance



Nigg *et al.* (2012), Bourassa *et al.* (2012),
Solgun *et al.* (2014, 2015, 2017) ...

energy



Minev (2018)
Minev *et al.* (2020)

more speed



more information, accuracy



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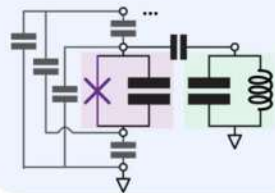


Quantization methods library

quasi-static

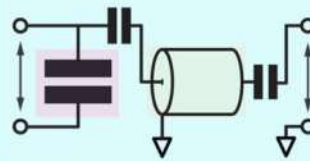
full-wave

lumped



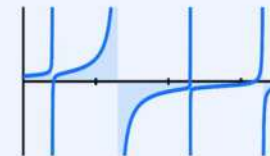
Yurke & Denker (1984), Devoret (1997), Burkard et al. (2004), Koch et al. ...

quasi-lumped



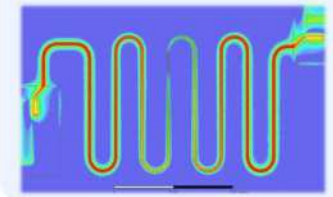
Malekakhlagh et al. (2017, 2019), Gely et al. (2019), Parra-Rodriguez et al. (2019), Minev et al. (2021), ...

impedance



Nigg et al. (2012), Bourassa et al. (2012), Solgun et al. (2014, 2015, 2017) ...

energy



Minev (2018)
Minev et al. (2020)

more speed



more information, accuracy



Zlatko Minev
IBM Quantum

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Quantization methods library: Tutorials

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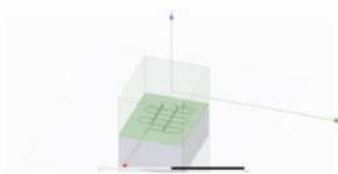
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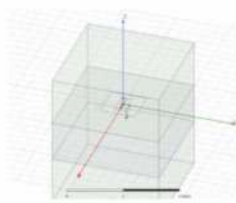
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Tutorials: Quantum analysis library



Eigenmode and EPR



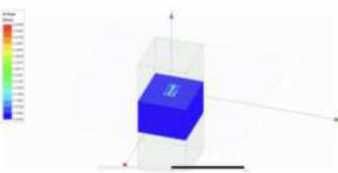
Impedance Analysis



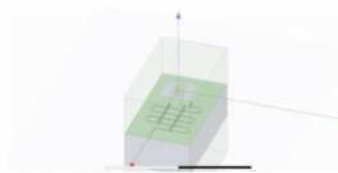
Capacitance Matrix and LOM Analysis



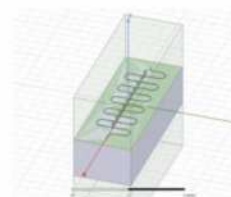
Sweep a option in QComponent, get Ansys HFSS driven-modal scatter matrix, impedance matrix, and admittance matrix.



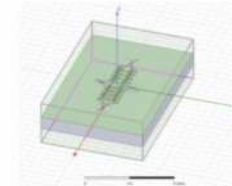
Analyzing and Tuning a Transmon Qubit



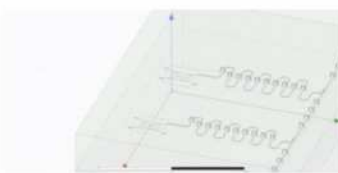
Analyzing a transmon and a resonator



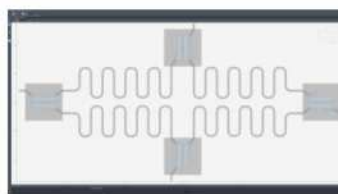
Analyzing a resonator



How to Render a Metal Design into Ansys



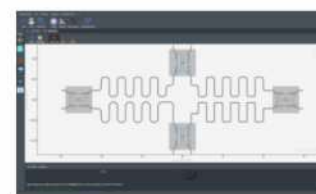
Driven Modal Simulation and S-Parameters



Sweep an option in QComponent, get capacitance matrix from Ansys Q3D and pyEPR



Sweep an option in QComponent, get Ansys HFSS eigenmode solution matrix.



How to Render a Metal Design into GDS

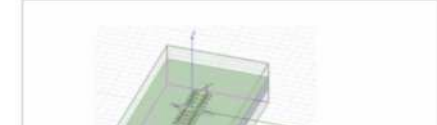
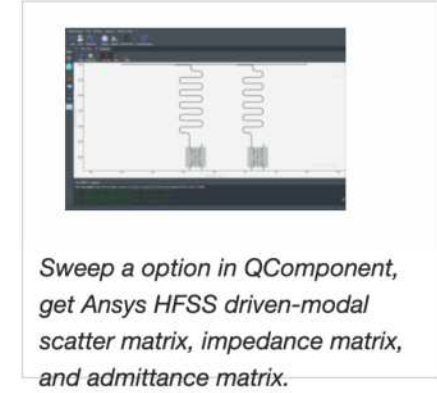
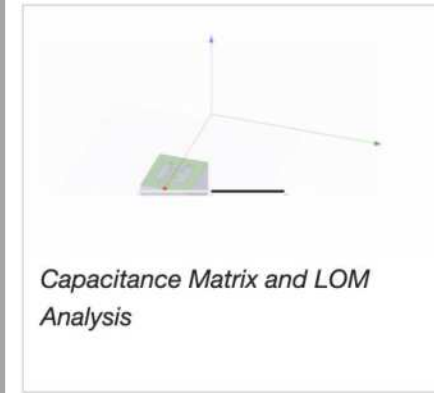
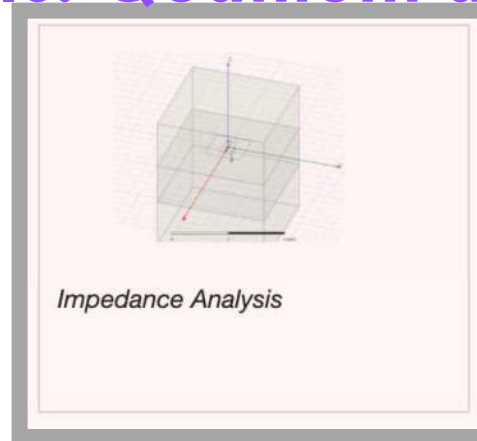
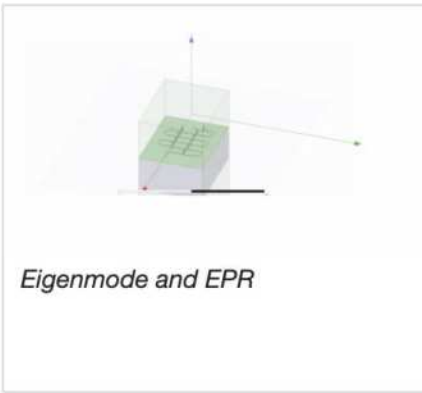
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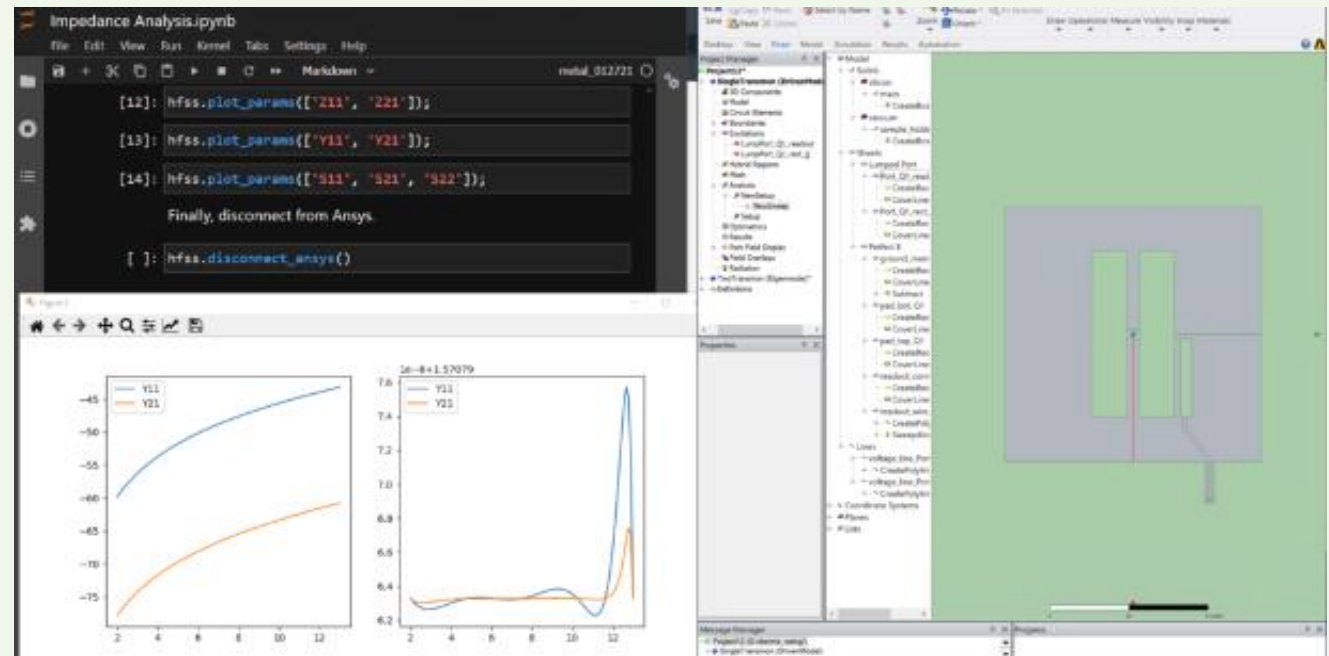
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S, Z, Y Impedance Scattering

arXiv:1204.0587 ...



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Sweep a option in QComponent,
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and admittance matrix.

Eigenmode

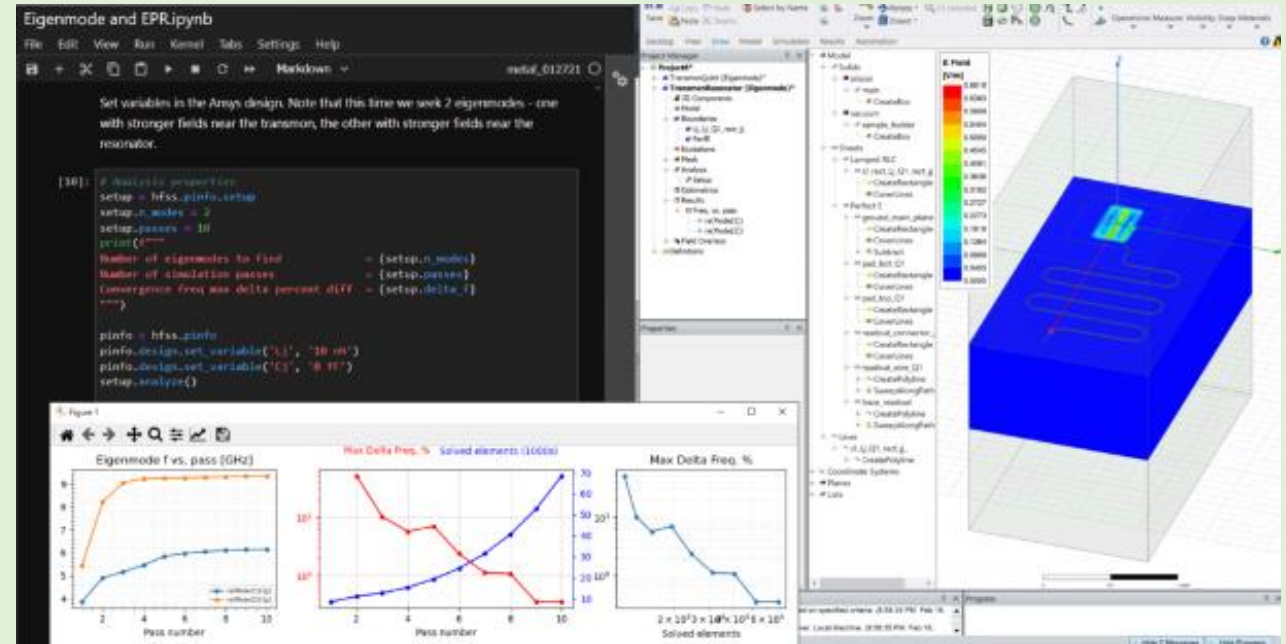
Energy-participation quantization of Josephson circuits

[Zlatko K. Minev](#) , [Zaki Leghtas](#), [Shantanu O. Mundhada](#), [Lysander Christakis](#), [Ioan M. Pop](#)
& [Michel H. Devoret](#)

[npj Quantum Information](#) **7**, Article number: 131 (2021) | [Cite this article](#)

[Minev arXiv: 1902.10355](#)

[Minev arXiv: 2010.00620](#)



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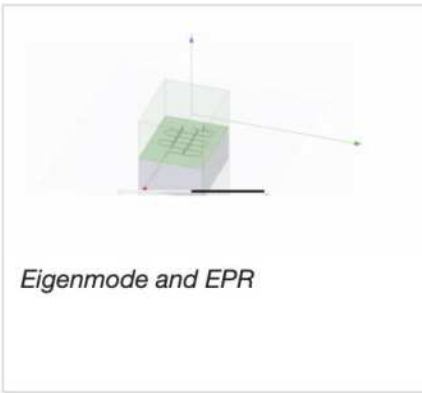
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QGeometry
GUI

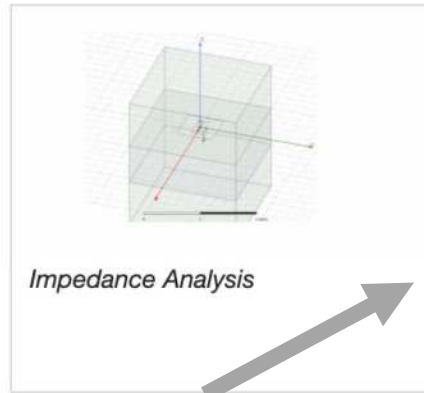
Code of Conduct



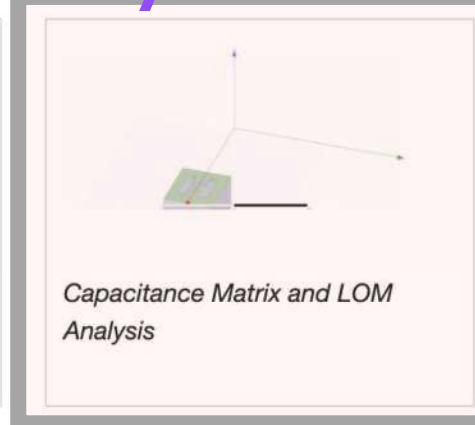
Tutorials: Quantum analysis library



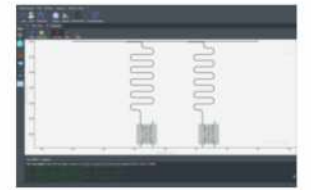
Eigenmode and EPR



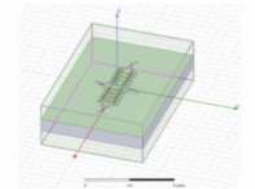
Impedance Analysis



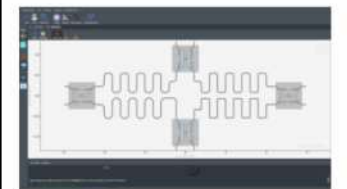
Capacitance Matrix and LOM Analysis



Sweep a option in QComponent, get Ansys HFSS driven-modal scatter matrix, impedance matrix, and admittance matrix.



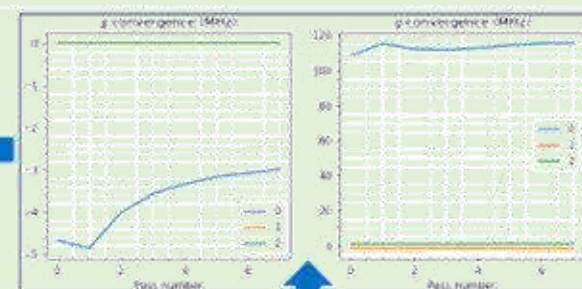
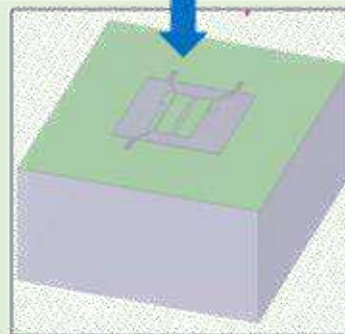
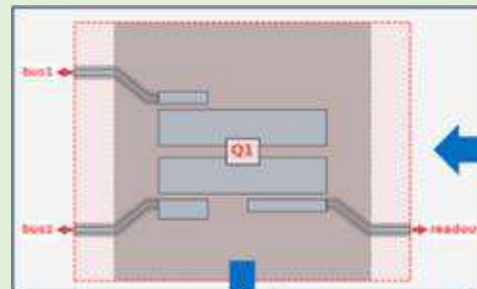
How to Render a Metal Design into Ansys



How to Render a Metal Design into GDS

Capacitive

arXiv:2103.10344 ...



```
# Using the analysis results, get capacitance matrix as a dataframe
q3d.get_capacitance_matrix()
```

	bus1_connector_pad_Q1	bus2_connector_pad_Q1	ground
bus1_connector_pad_Q1	47.71247	-0.38203	
bus2_connector_pad_Q1	-0.38203	51.80766	
ground_main_plane	-33.11632	-35.62303	
pad_bot_Q1	-1.34246	-12.58801	
pad_top_Q1	-12.12145	-1.59640	
readout_connector_pad_Q1	-0.17280	-0.96276	

Resonators
Composite Bi-Partite Systems
Qubit Couplers
Input-Output Coupling
Small Quantum Chips
Design Flow

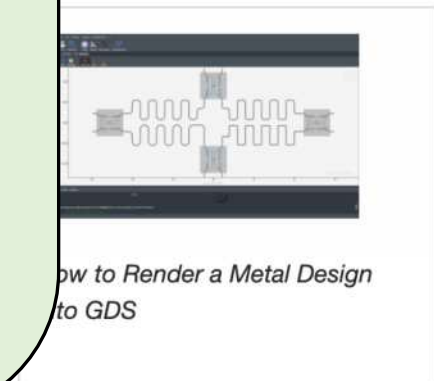
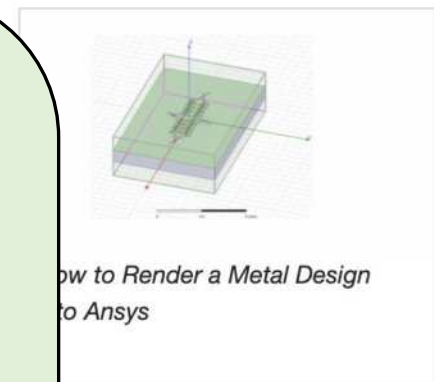
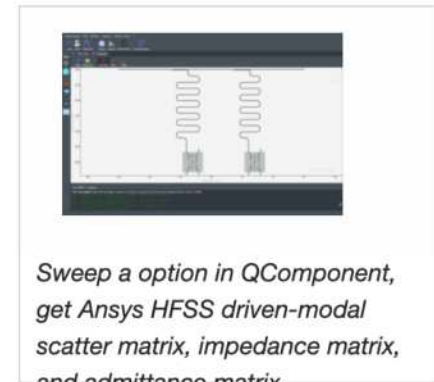
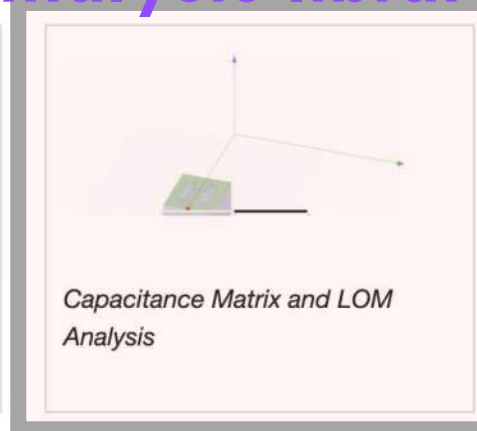
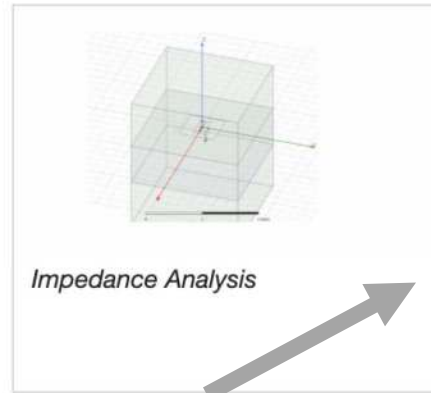
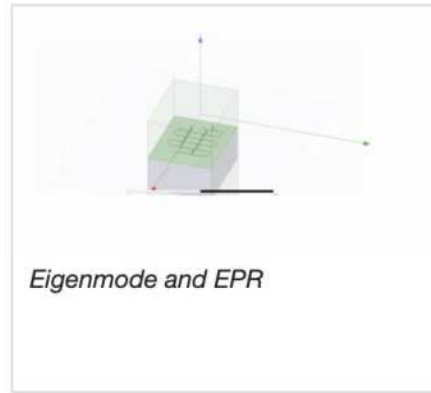
Libraries
All Quantum Devices

API References

Overview
QDesigns
QComponents
Analyses
QRenderes
Toolbox
QGeometry
GUI

Code of Conduct

Tutorials: Quantum analysis library



Cornell University

arXiv.org > quant-ph > arXiv:2103.10344

Quantum Physics

Circuit quantum electrodynamics (cQED) with modular quasi-lumped models

Zlatko K. Minev,* Thomas G. McConkey, Maika Takita, Antonio Corcoles, and Jay M. Gambetta
IBM Quantum, IBM T.J. Watson Research Center, Yorktown Heights, US

WIP: General capacitive analysis code
map to known building blocks: e.g.,
transmon, fluxonium, zero-pi, etc.

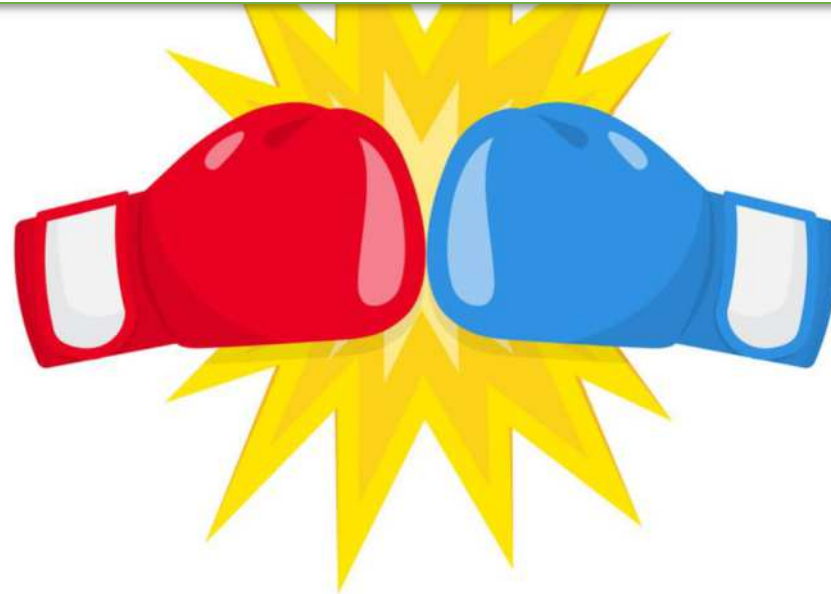
Planar devices & comparison to other methods

Quantum Physics

[Submitted on 2 Feb 2021]

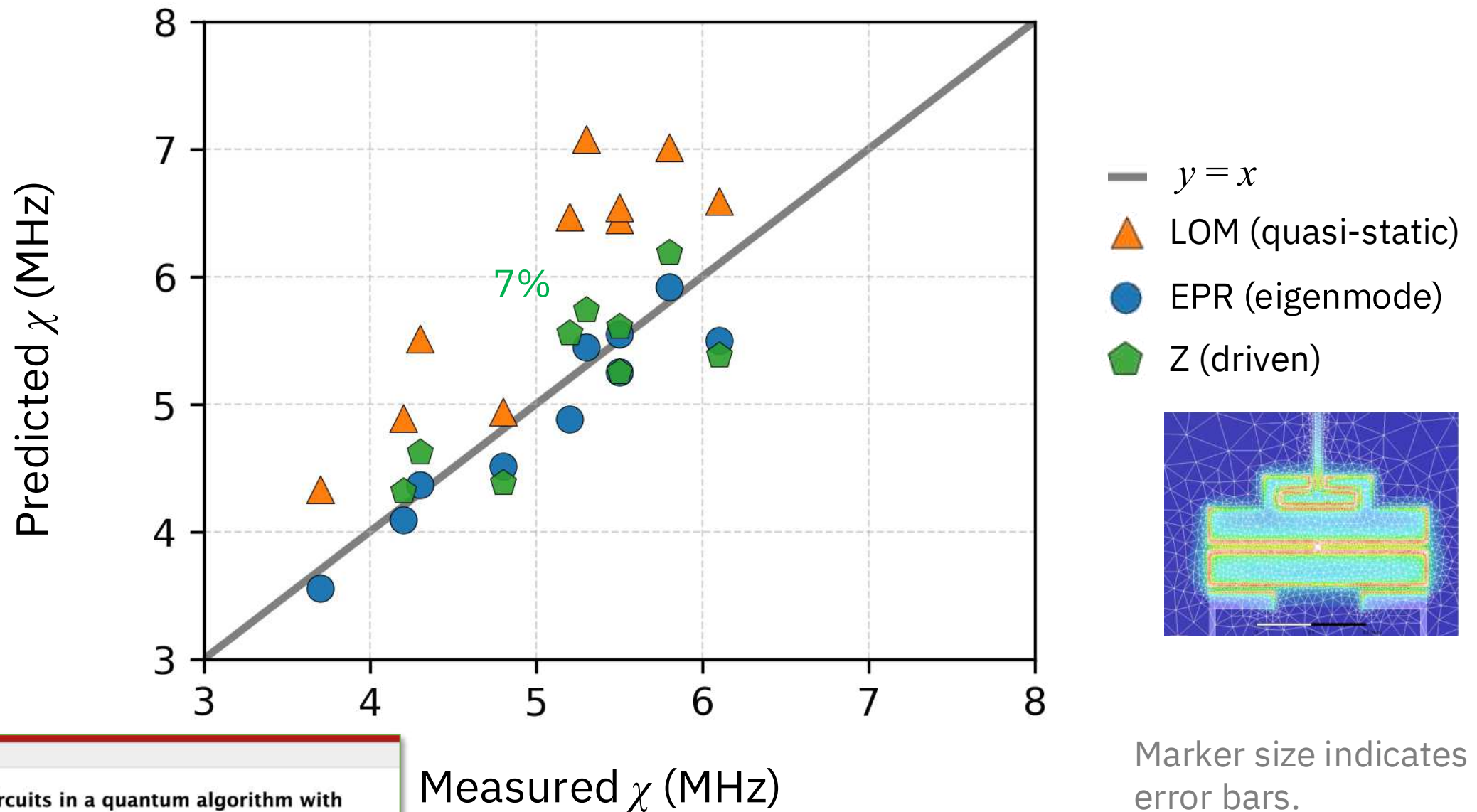
Exploiting dynamic quantum circuits in a quantum algorithm with superconducting qubits

Antonio D. Corcoles, Maika Takita, Ken Inoue, Scott Lekuch, Zlatko K. Mineev, Jerry M. Chow, Jay M. Gambetta

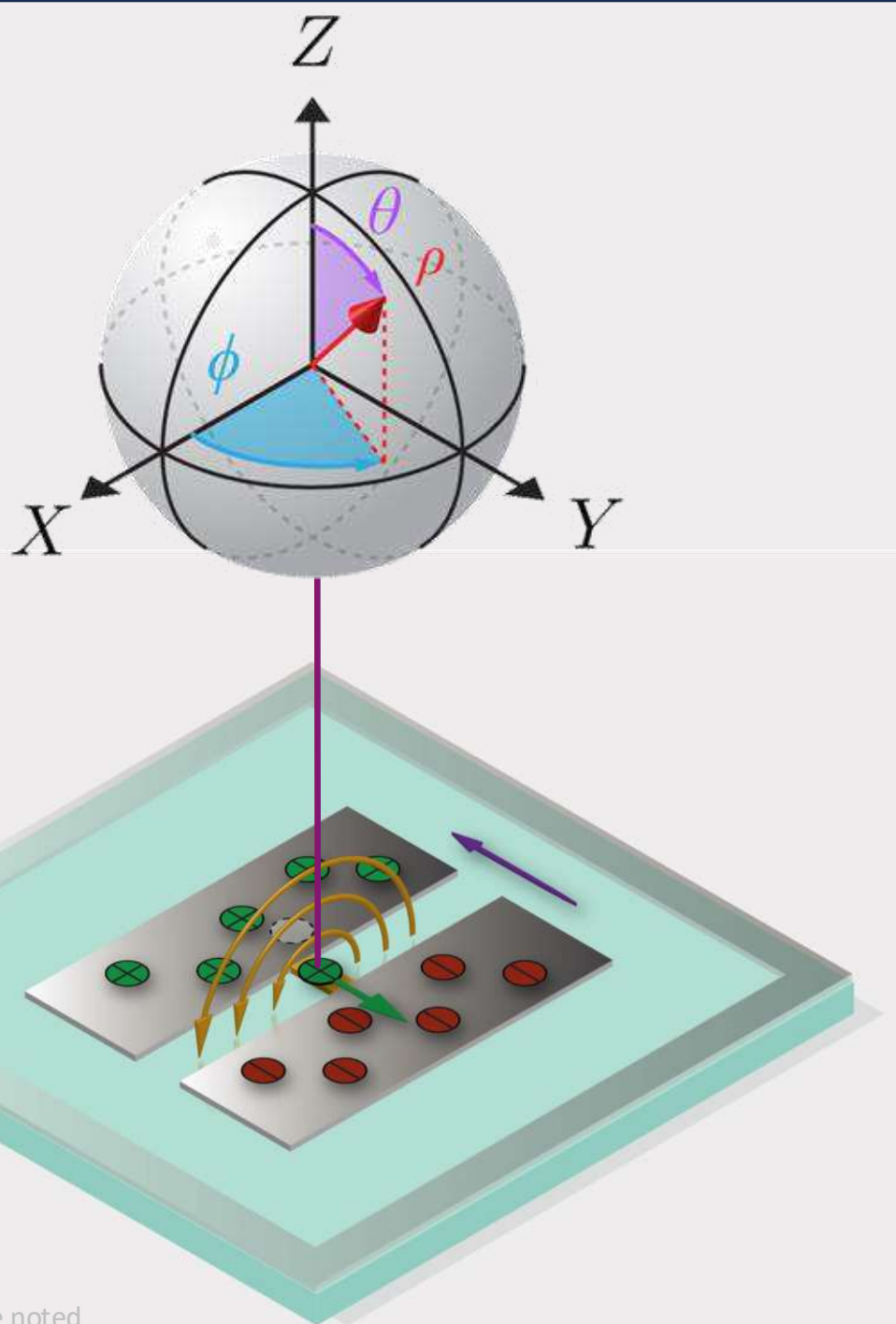


Automated with
Qiskit | quantum device
design

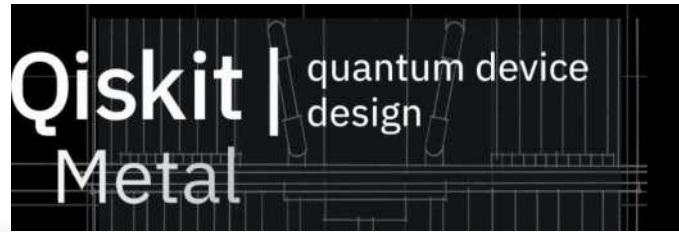
Measured vs. predicted: qubit-readout cross-Kerr



Examples from the Community

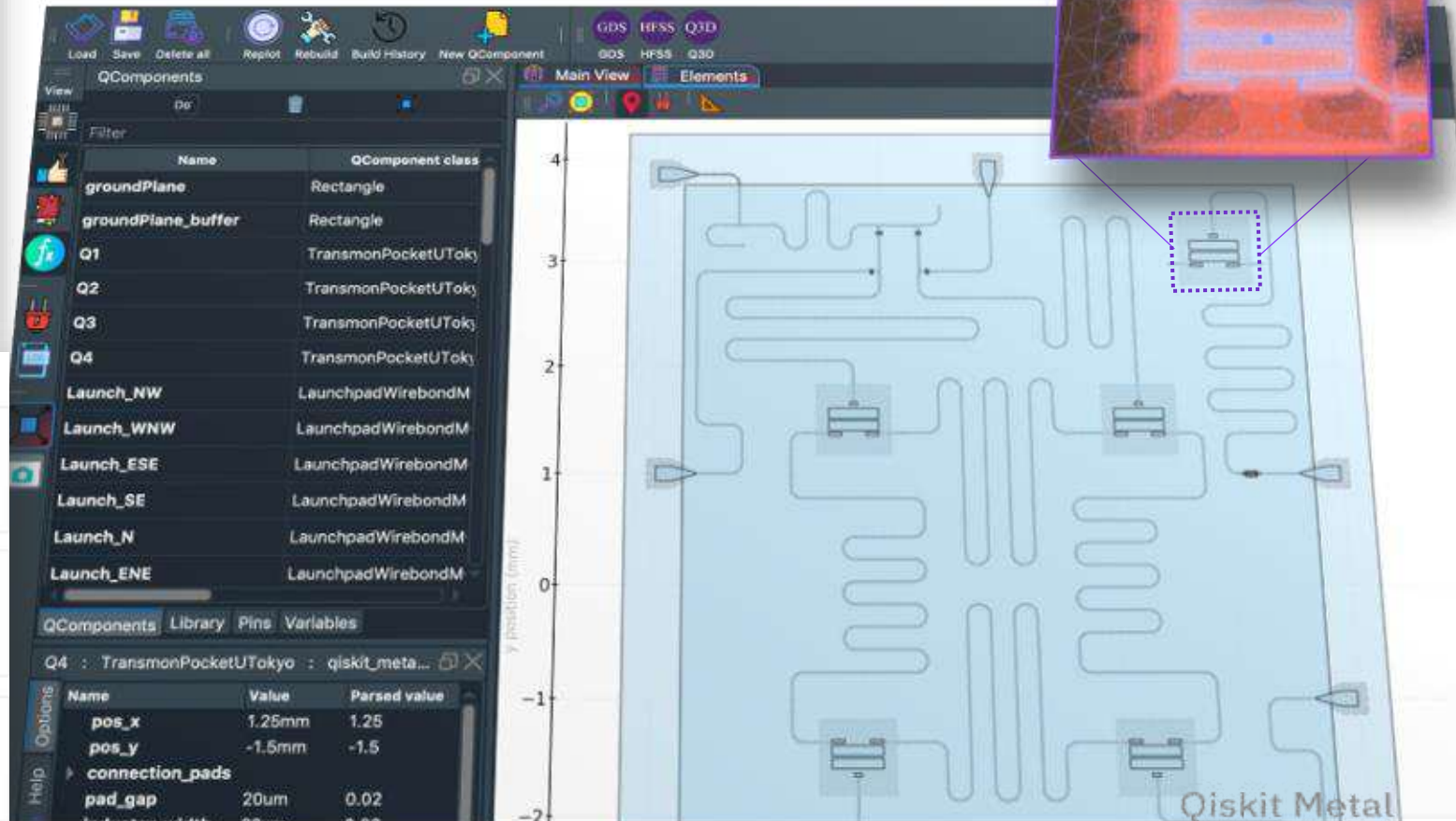
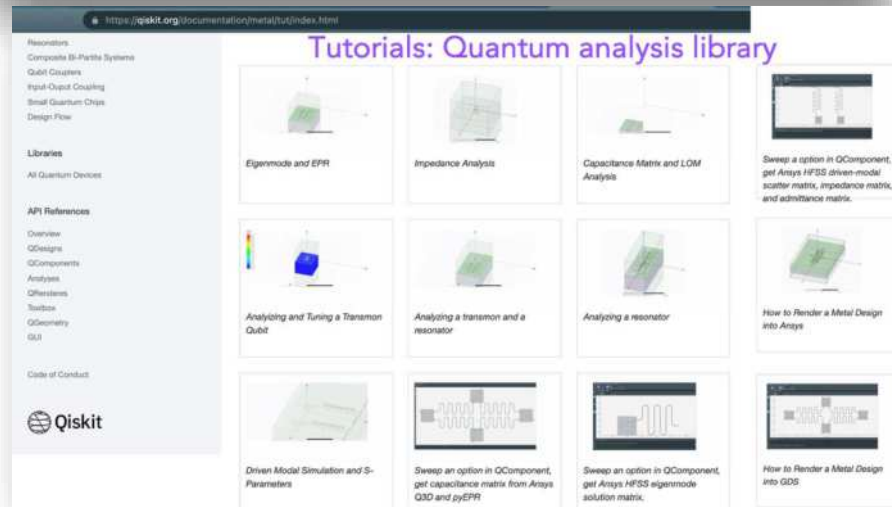
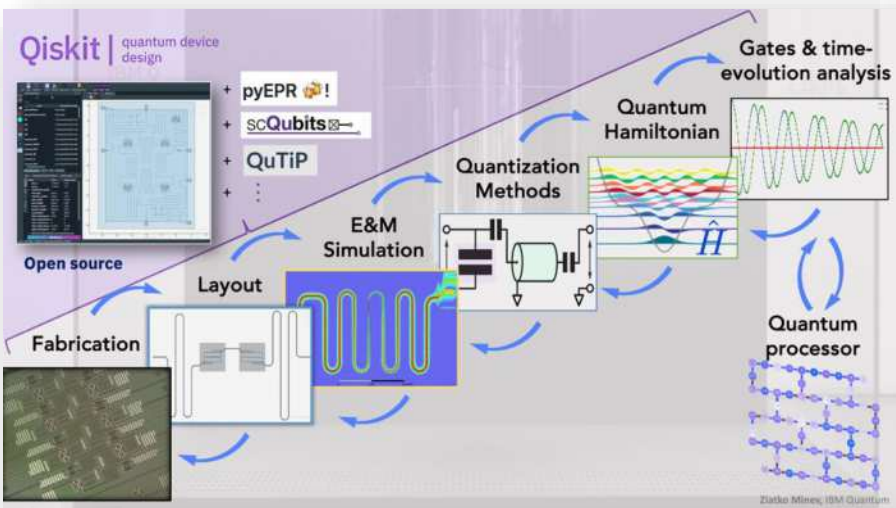


Automation in Practice: Qiskit Metal + Community

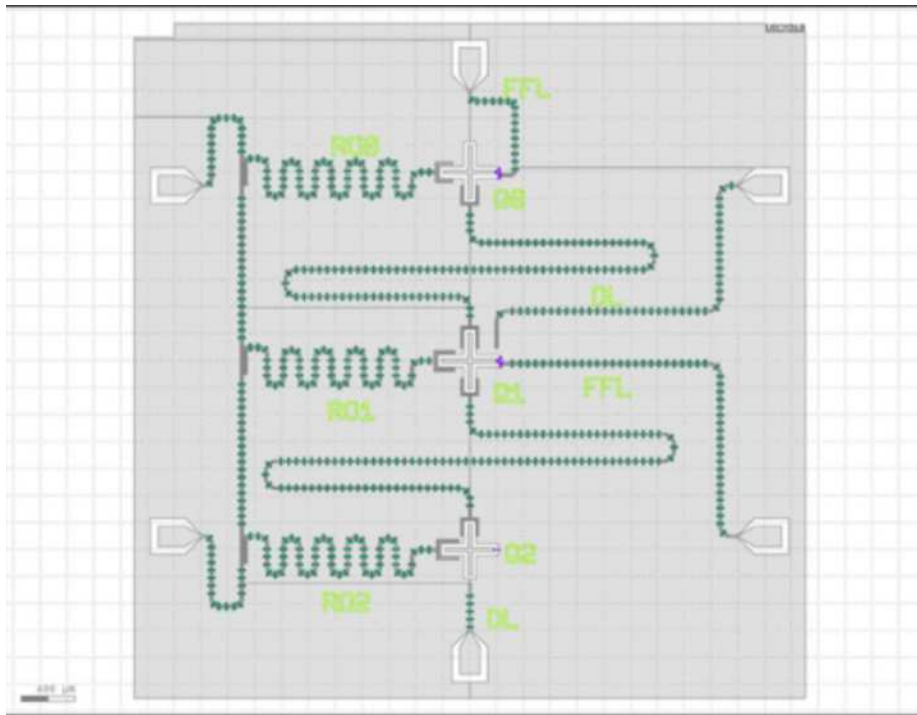


- + advanced talks
- + workshops
- + lunch meeting discussion maintainers

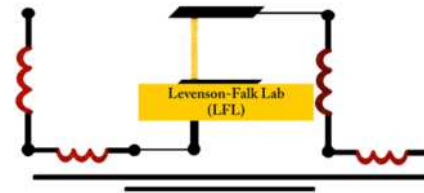
Community ecosystem driven (Apache-2.0)
github.com/qiskit-community/qiskit-metal



Example designs from the community, folks like you!



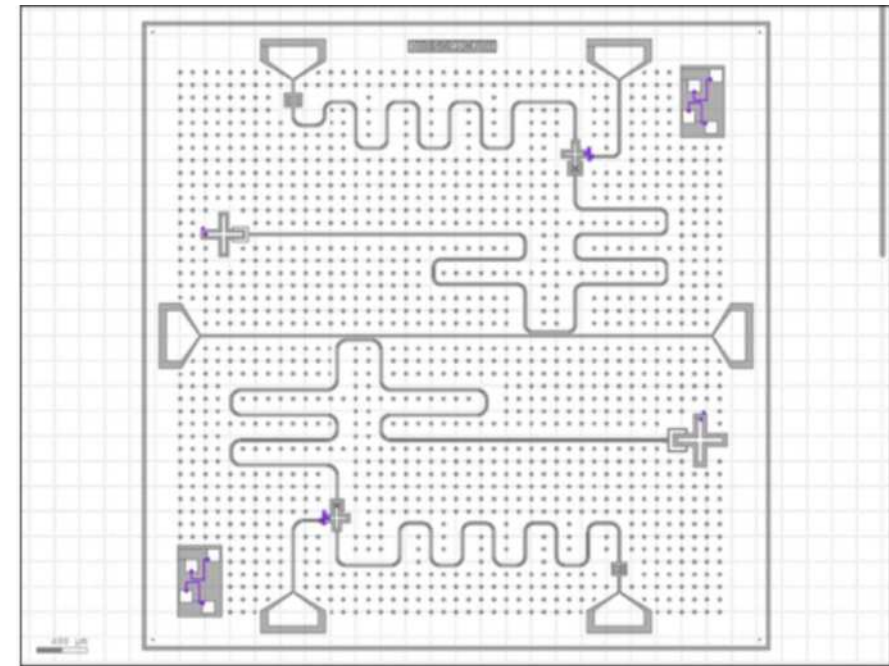
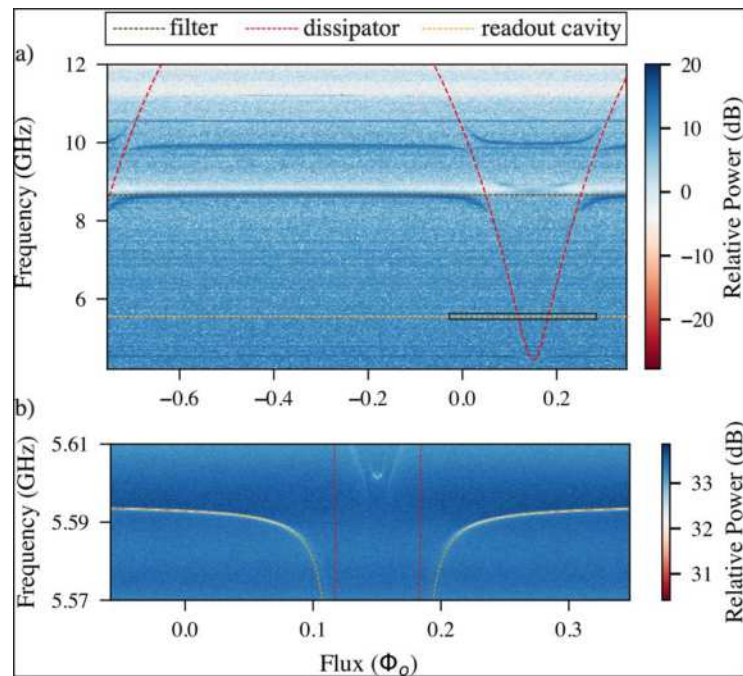
USC Viterbi
School of Engineering



Sadman Ahmed Shanto



Eli Levenson-Falk

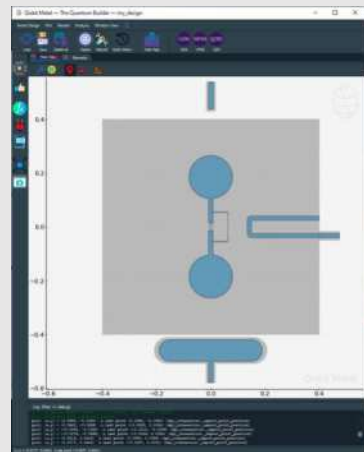


Zlatko Minev

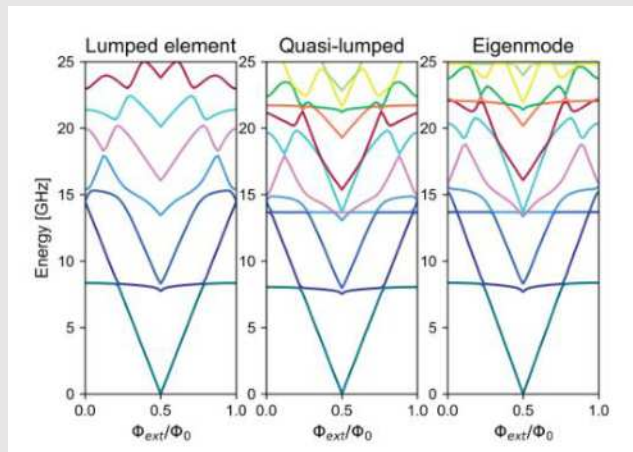
Example designs from the community, folks like you!

QuTech / Technische Universiteit Delft

Figen Yilmaz, C. Andersen Lab: Fluxonium chip design
Roald van den Boogaart-Andersen Lab: Fluxonium spectrum (eigenmode = pyEPR) to show how they extended the analysis methods to be able to handle fluxonium qubits
Sara Buhktari-Andersen Lab: Fabricated resonator design being measured

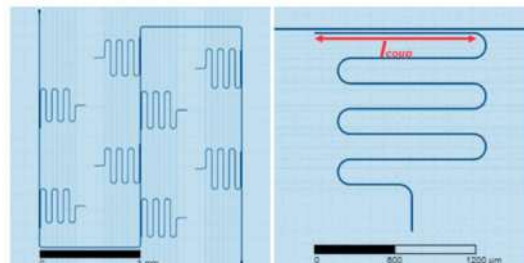


“Qiskit-Metal is the best tool if one studies superconducting qubits” -- Figen Yilmaz



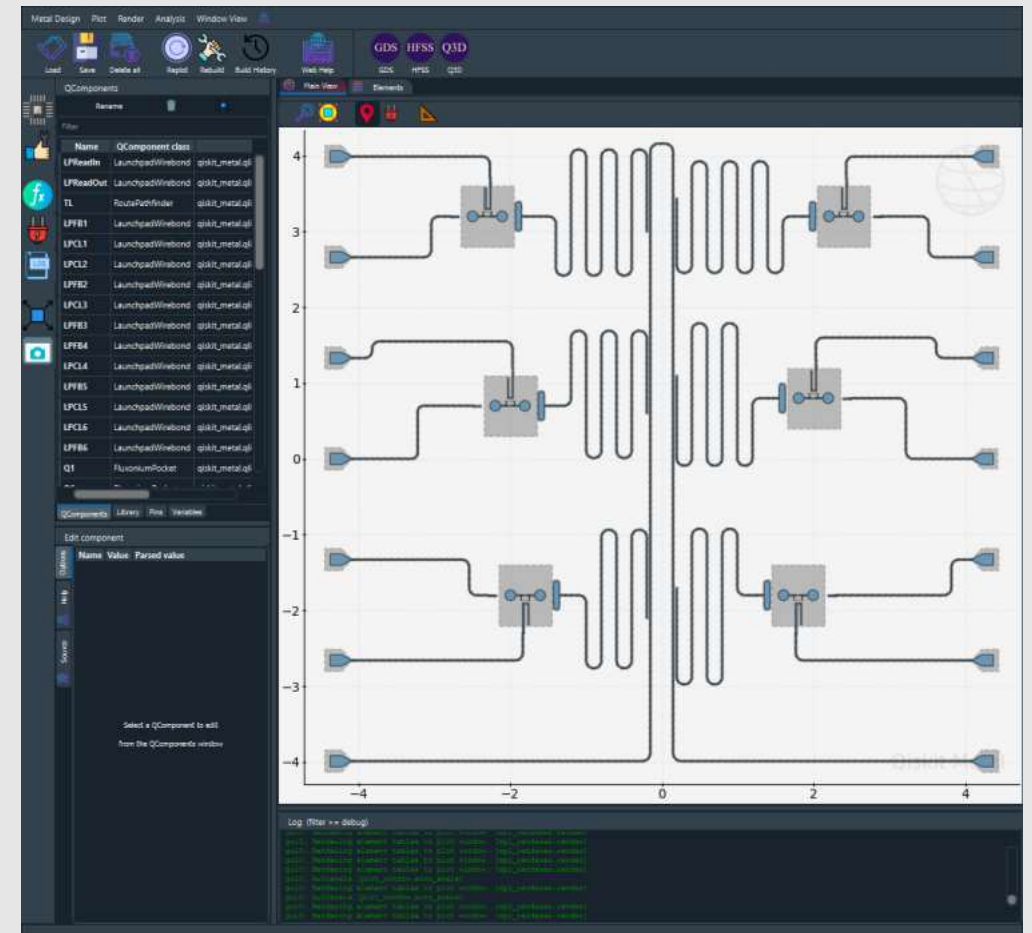
3.1. Final design

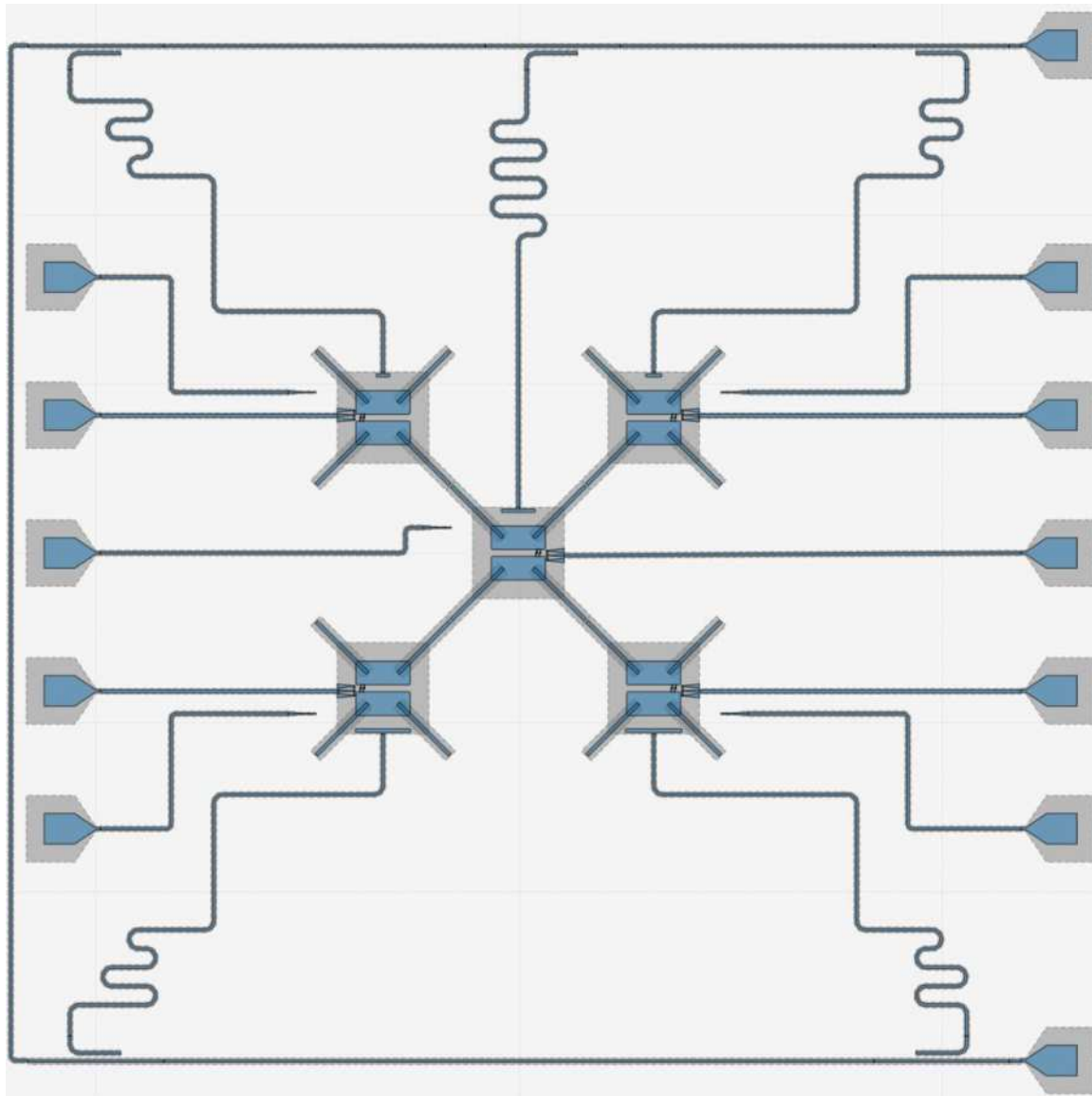
The final design consists of an S-shaped feedline routed between two launchpads. Eight resonators are capacitively coupled to this feedline. Every component on this 9 x 9 mm device is implemented by a coplanar waveguide transmission line. The schematic representations of the full device and single resonator geometry are depicted in Figure 6.



QuTech / Technische Universiteit Delft

Intermediate chip design.





Example designs from the community, folks like you!



5 tunable transmon weight-4 devices
(direct coupling and Delft/ETH style)



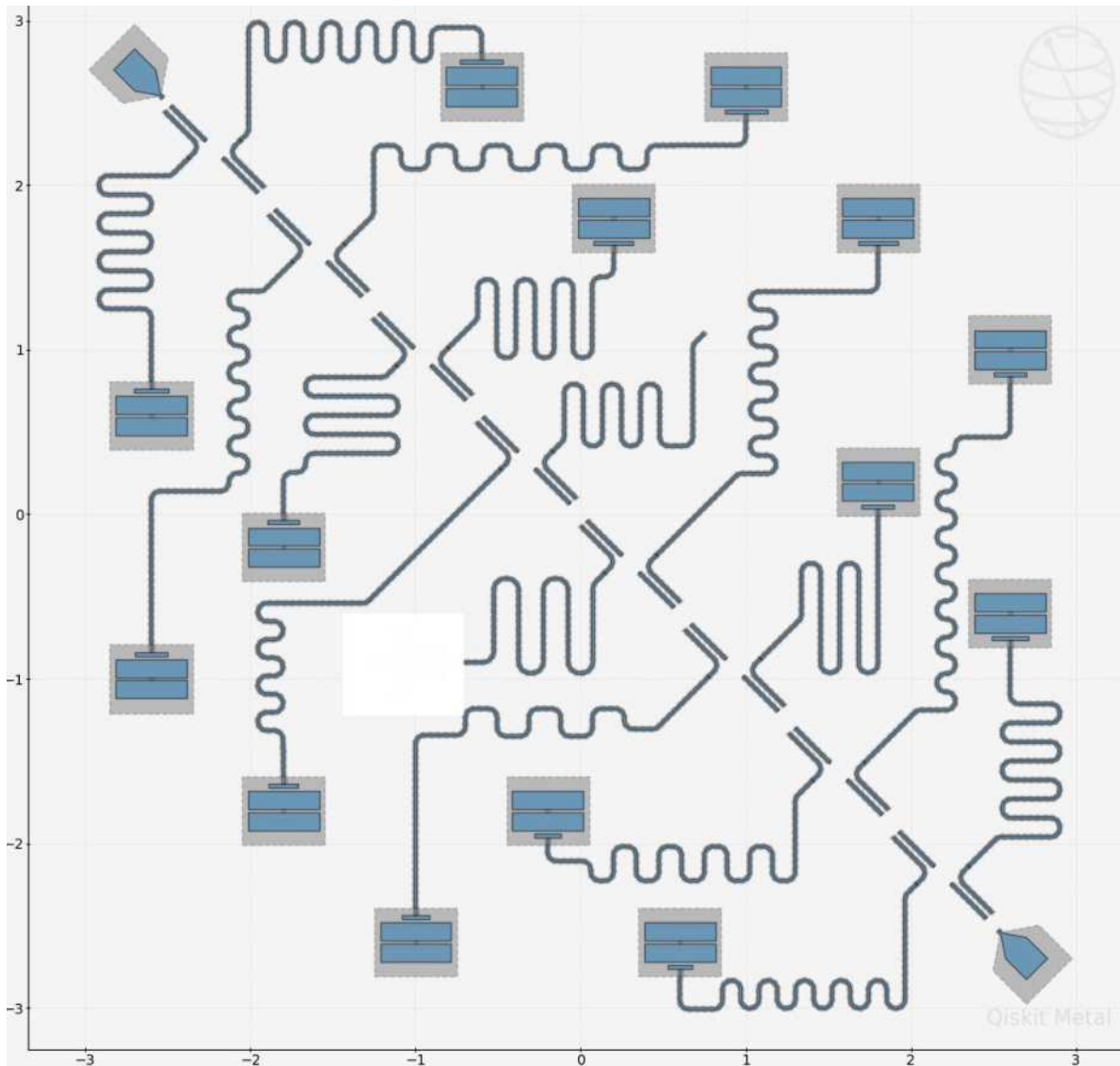
Designs:

- Benjamin Joecker, PhD
- Thue Christian Thann, PhD Candidate



Affiliations:

- Niels Bohr Institute, University of Copenhagen
- Novo Nordisk Quantum Computing Programme (NQCP)



Example designs from the community, folks like you!

NQCP
NIELS BOHR INSTITUTE



Standard candle devices for tracing reproducibility

Designs:

-Thue Christian Thann, PhD Candidate

Affiliations:

- Niels Bohr Institute, University of Copenhagen
- Novo Nordisk Quantum Computing Programme (NQCP)



Example designs from the community, folks like you!

NQCP
NIELS BOHR INSTITUTE



Christopher. W. Warren

Niels Bohr Institute
Verified email at nbi.ku.dk

quantum computing quantum simulation
quantum information

12 Tunable qubit chip; directly coupled (dubbed “Adamite”)

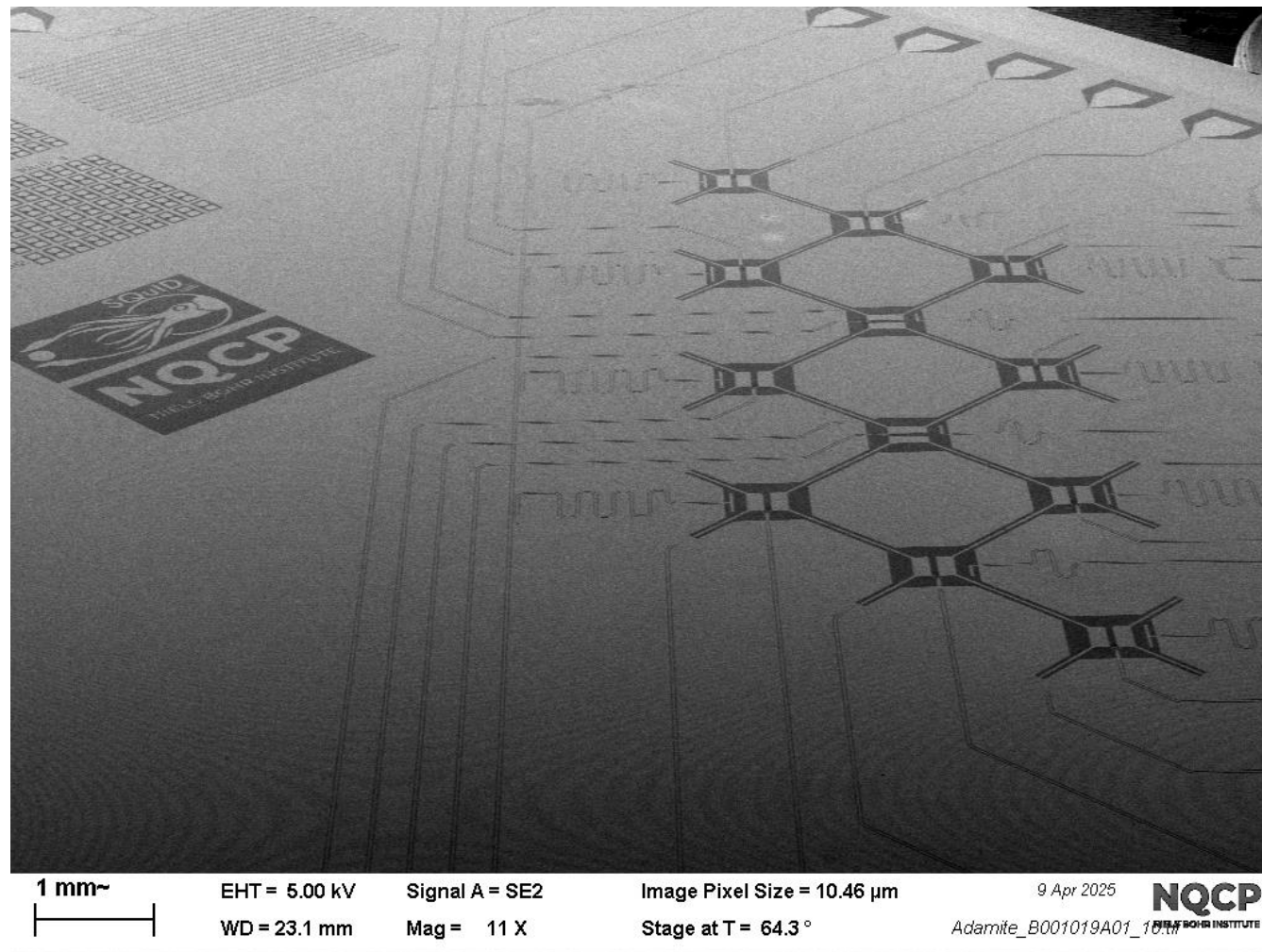
Design:

- Christopher W. Warren, PhD

Affiliations:

- SQulD Lab, Niels Bohr Institute, University of Copenhagen
- Novo Nordisk Quantum Computing Programme (NQCP)

Zlatko Minev



Example designs from the community, folks like you!

NQCP
NIELS BOHR INSTITUTE



Christopher. W. Warren

Niels Bohr Institute
Verified email at nbi.ku.dk

quantum computing quantum simulation
quantum information

12 Tunable qubit chip; directly coupled (dubbed “Adamite”)

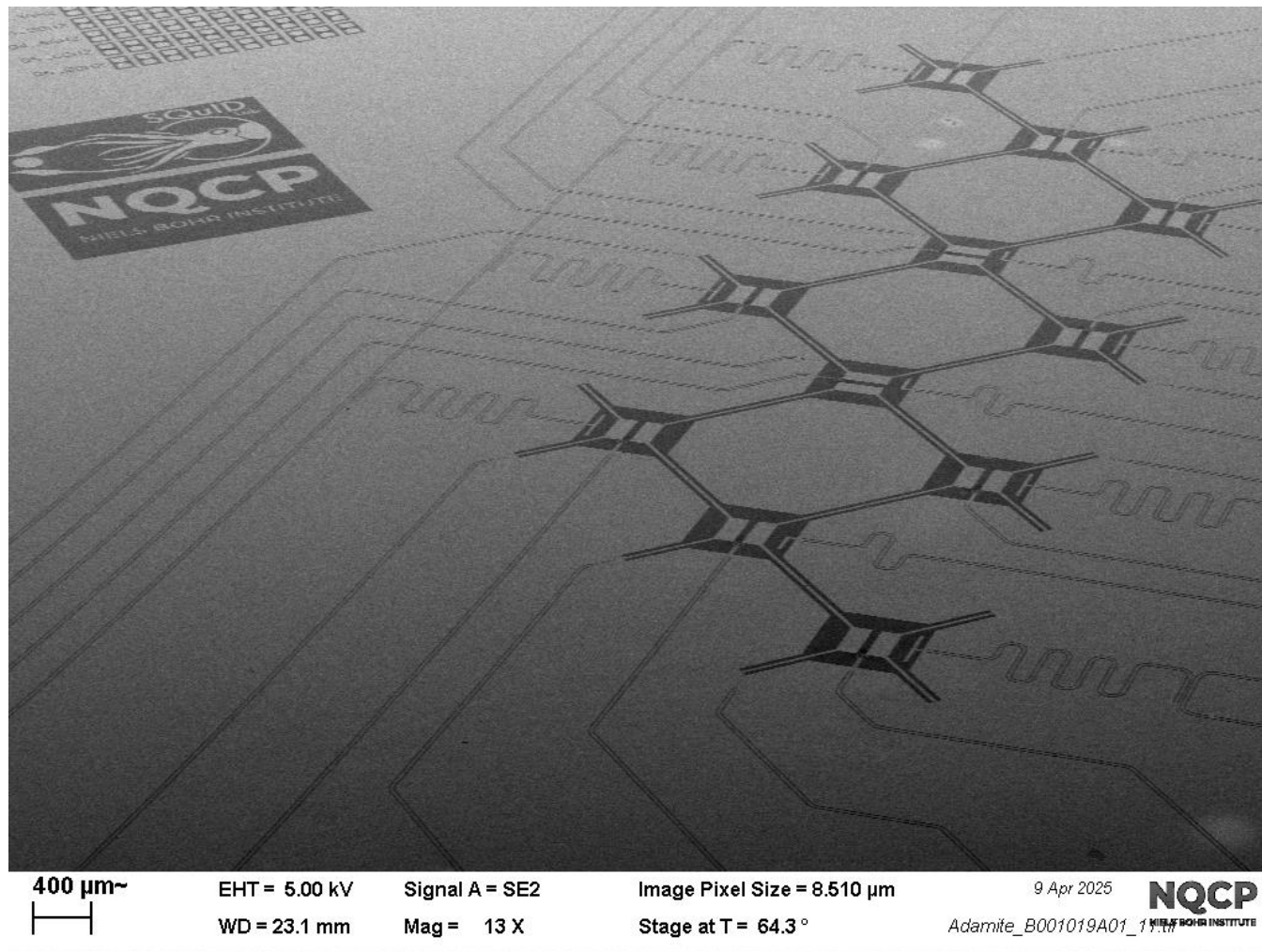
Design:

- Christopher W. Warren, PhD

Affiliations:

- SQuID Lab, Niels Bohr Institute, University of Copenhagen
- Novo Nordisk Quantum Computing Programme (NQCP)

Zlatko Minev



Example designs from the community, folks like you!

NQCP
NIELS BOHR INSTITUTE



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quantum computing quantum simulation
quantum information

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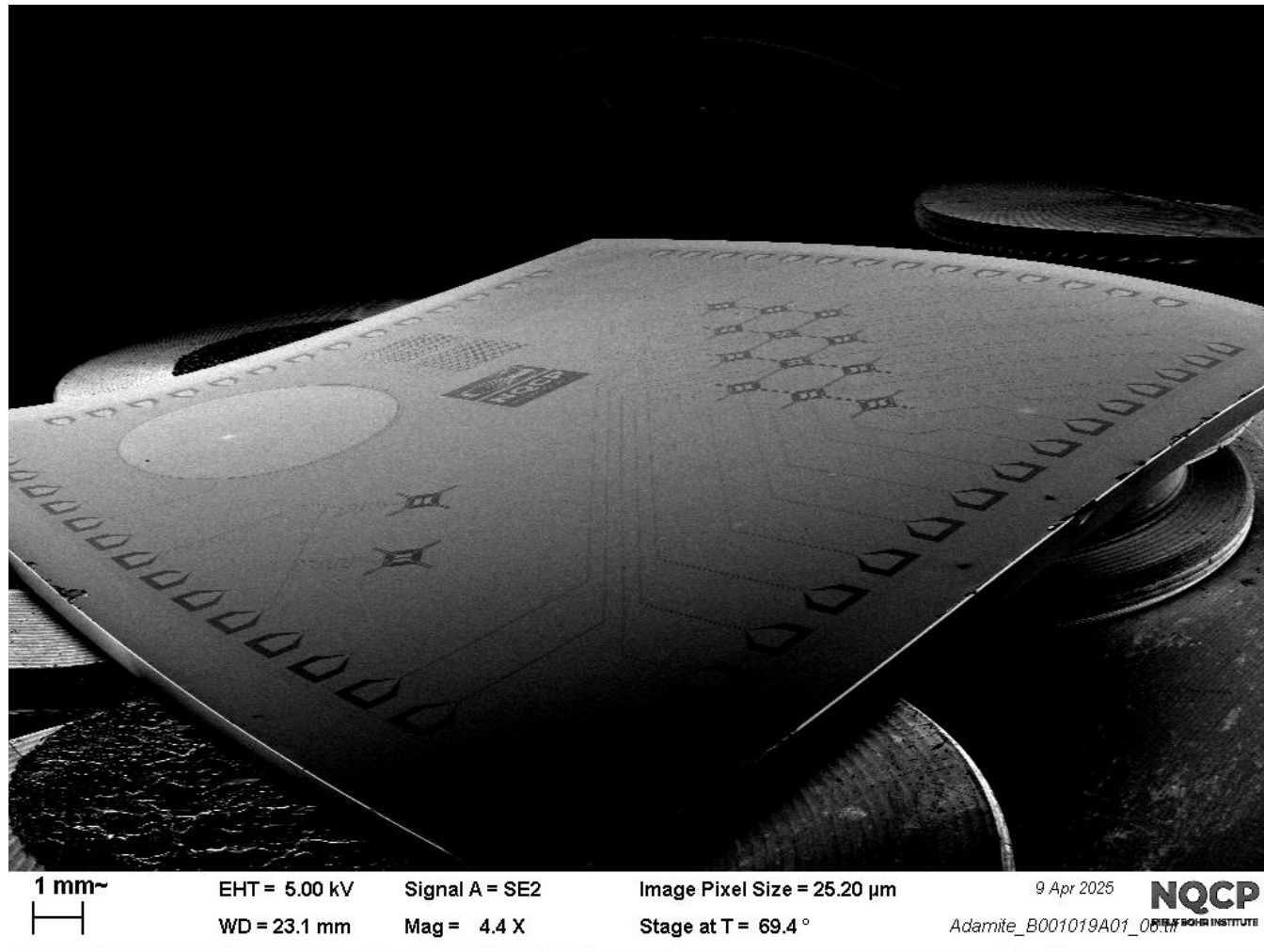
Design:

- Christopher W. Warren, PhD

Affiliations:

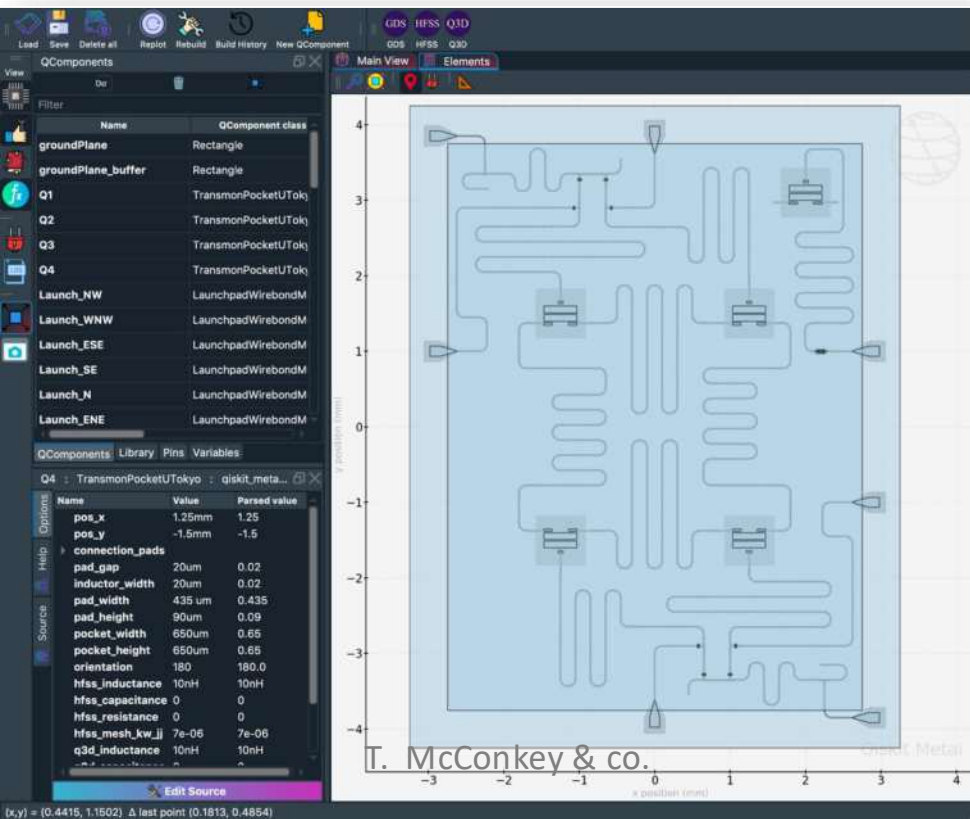
- SQulD Lab, Niels Bohr Institute, University of Copenhagen
- Novo Nordisk Quantum Computing Programme (NQCP)

Zlatko Minev

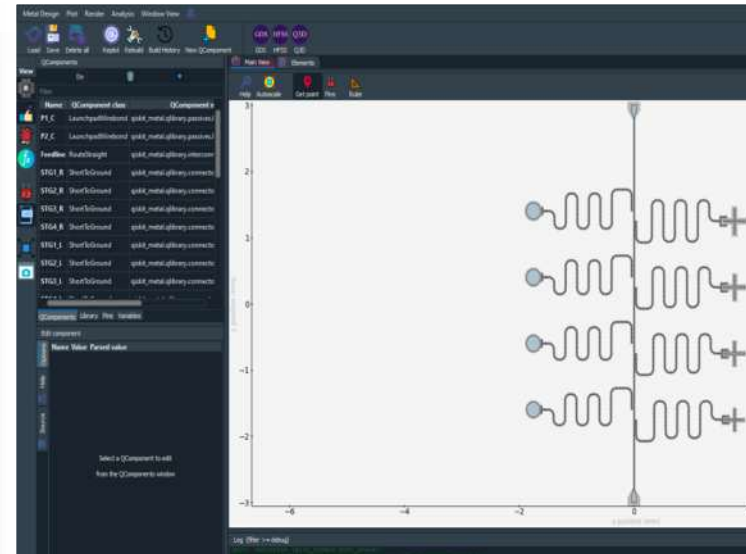


Example designs from the community, folks like you!

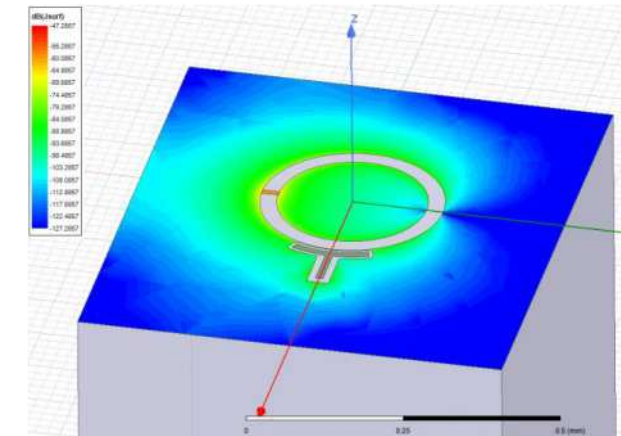
IBM 5Q Tsuru U Tokyo



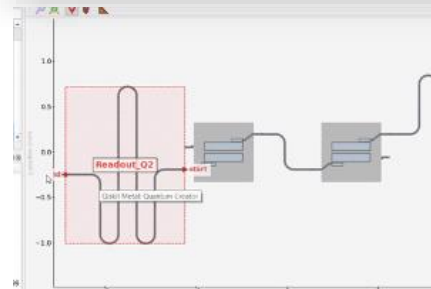
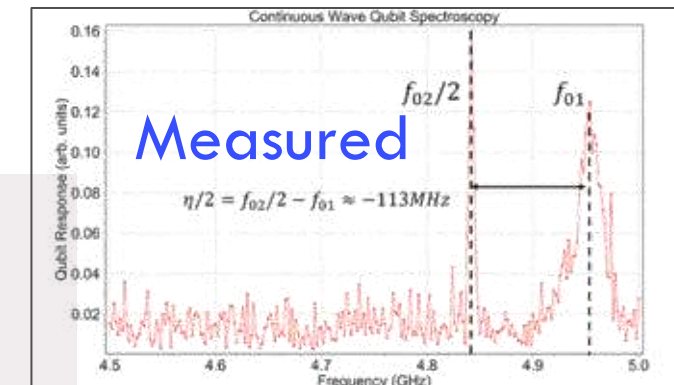
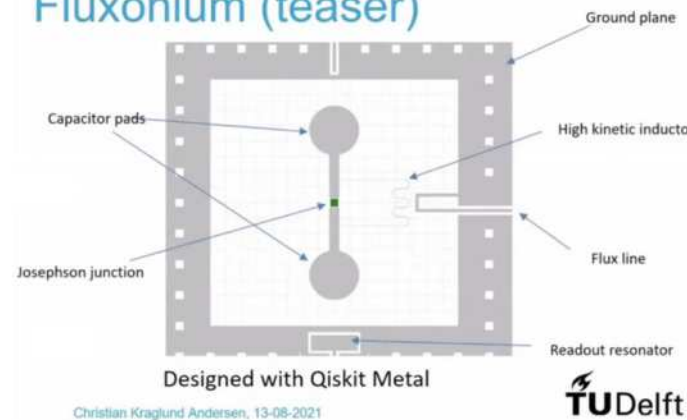
Chalmers 8Q



Simulated



TU Delft Fluxonium
Fluxonium (teaser)



APS physics **MARCH MEETING 2021**
MARCH 15-19 ONLINE

CEC/ICMC **21**
VIRTUAL CONFERENCE JULY 19-23
Cryogenics • Through • Collaboration



IEEE QUANTUM WEEK IEEE International Conference on Quantum Computing and Engineering — QCE21

Zlatko Minev

Get Started

Installing Quantum Metal

Quick Start

Tutorials

Tutorials

QComponent Gallery

Example Designs

Videos & Education

Ecosystem

Built on / with Quantum Metal

Concepts

Roadmap

Quantum Metal Workflow

Quantization Methods Overview

Frequently Asked Questions

API Reference

Overview

QDesigns

QComponents

Analyses

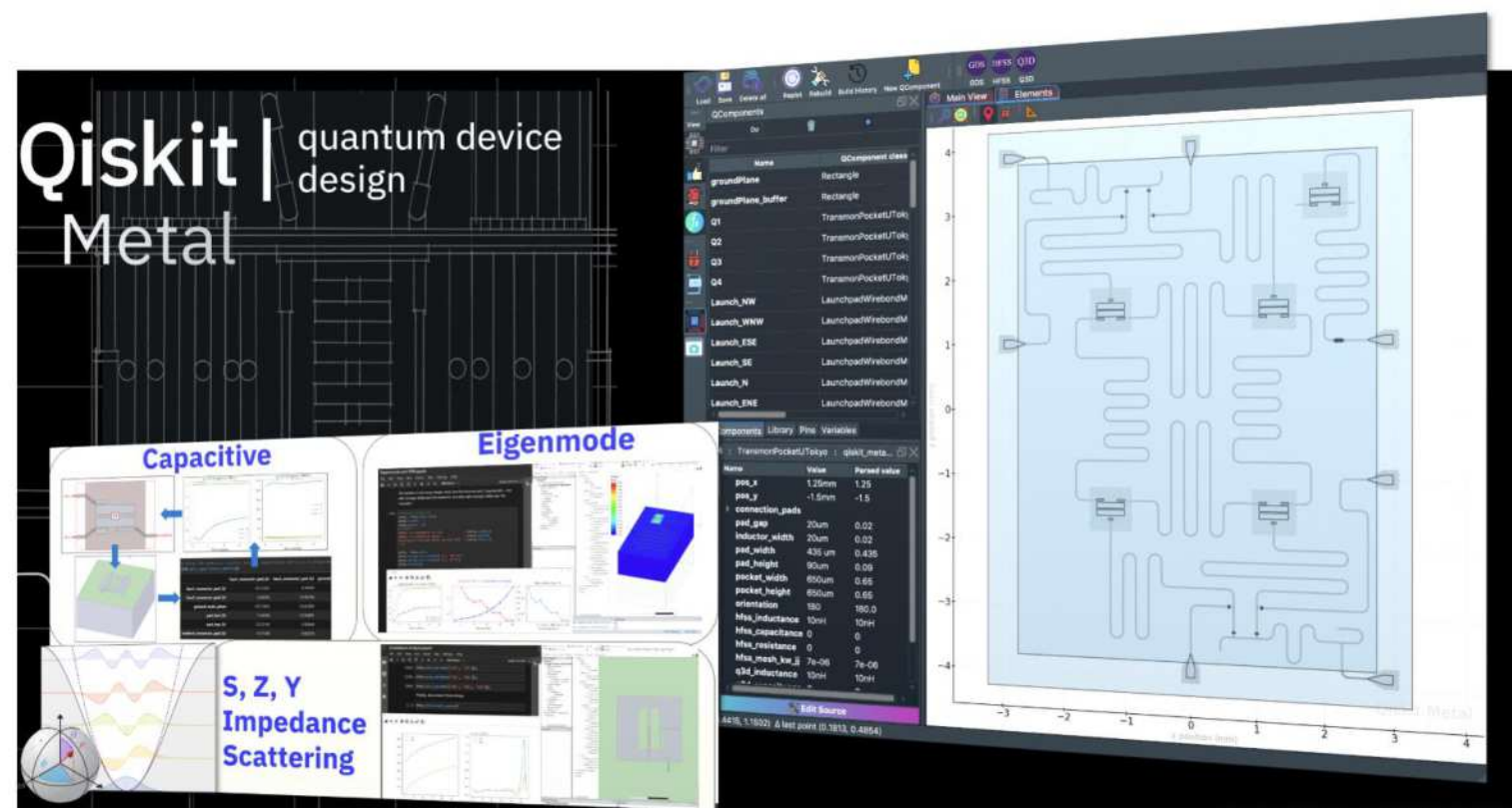
QRenderers

Toolbox

Quantum Metal (formerly Qiskit Metal)



Quantum Device Design & Analysis (Q-EDA)



Quantum Metal | Open source · Community-maintained quantum EDA

Design and analyze superconducting quantum chips with a Python API + optional GUI that plugs into your favourite EDA tools. No Qt required — works headlessly in any Jupyter notebook or plain Python script.



Install

Get started in 60
seconds



Tutorials

Hands-on notebooks



Component Gallery

Browse every
QComponent visually



Community

Join the Discord



Star History

Try it now — zero install



Open in Colab

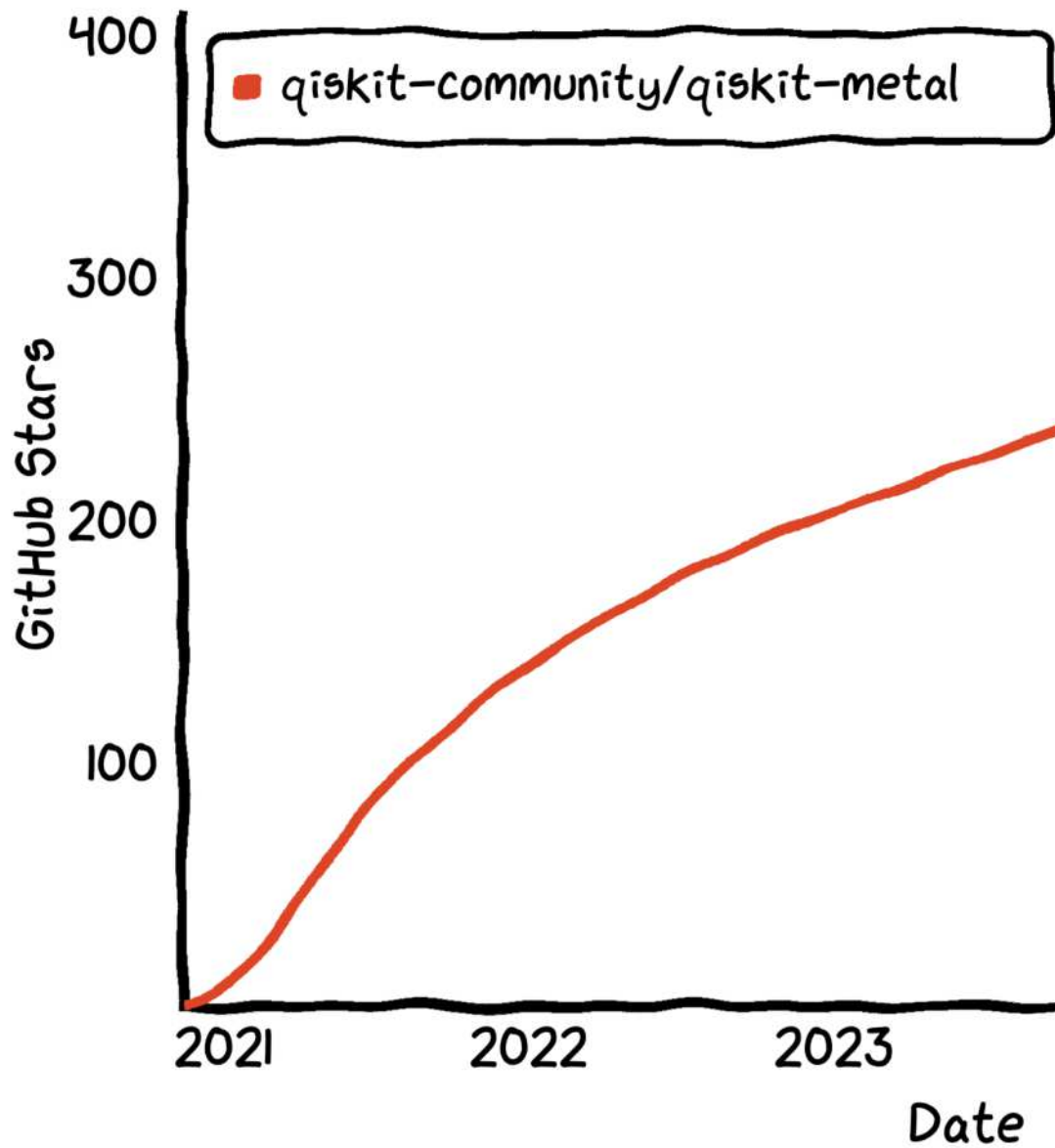


launch

binder



Open in Codespaces



Feature matrix — what each install gives you, at a glance:

Feature	lite	[gui]	[ansys]	[mesh]	[full]
import qiskit_metal	✓	✓	✓	✓	✓
qm.view(design) (headless matplotlib)	✓	✓	✓	✓	✓
Build designs + components from qlibrary	✓	✓	✓	✓	✓
GDS export	✓	✓	✓	✓	✓
LOAnalysis, LOM math, capacitance reductions	✓	✓	✓	✓	✓
MetalGUI desktop app	—	✓	—	—	✓
HFSS / Q3D renderers	—	—	✓	—	✓
EPR analyses (EigenmodeSim, LumpedElementsSim)	—	—	✓	—	✓
gmsh mesher (foundation for Elmer / Palace)	—	—	—	✓	✓



star-history.com

Zlatko Minev

Feature matrix — what each install gives you, at a glance:

 Tip

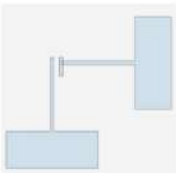
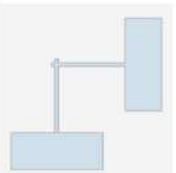
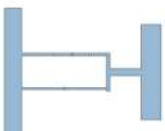
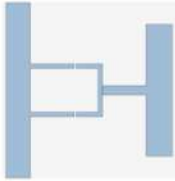
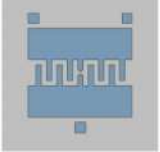
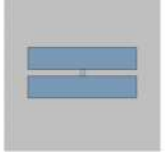
Extras **compose** — combine them with comma-separated names:

```
pip install "quantum-metal[gui,ansys]" # MetalGUI + HFSS/Q3D
pip install "quantum-metal[gui,mesh]"  # MetalGUI + gmsh
```

Feature	lite	[gui]	[ansys]	[mesh]	[full]
<code>import qiskit_metal</code>	✓	✓	✓	✓	✓
<code>qm.view(design)</code> (headless matplotlib)	✓	✓	✓	✓	✓
Build designs + components from <code>qlibrary</code>	✓	✓	✓	✓	✓
GDS export	✓	✓	✓	✓	✓
<code>LOAnalysis</code> , LOM math, capacitance reductions	✓	✓	✓	✓	✓
<code>MetalGUI</code> desktop app	—	✓	—	—	✓
HFSS / Q3D renderers	—	—	✓	—	✓
EPR analyses (<code>EigenmodeSim</code> , <code>LumpedElementsSim</code>)	—	—	✓	—	✓
gmsh mesher (foundation for Elmer / Palace)	—	—	—	✓	✓

Built on Quantum Metal

These projects use Quantum Metal directly — they consume `QDesign` objects, extend the `QComponent` library, or build workflows on top of Metal as their foundation:

Josephson Junction Dolan  The base "JJ_Dolan" inherits the "QComponent" class. jj_dolan	Josephson Junction Manhattan  The base "JJ_Manhattan" inherits the "QComponent" class. jj_manhattan	SNAIL (Superconducting Nonlinear Asymmetric Inductive eLement)  A SNAIL (Superconducting Nonlinear Asymmetric Inductive eLement). SNAIL	Squid Loop  The base "SQUID_LOOP" inherits the "QComponent" class. SQUID_LOOP
Star Qubit  A single configurable circle with multiple pads. StarQubit	Transmon Concentric  The base <code>TransmonConcentric</code> class. TransmonConcentric	Transmon Concentric Type 2  The base <code>TransmonConcentric</code> class. TransmonConcentricType2	Transmon Cross  The base <code>TransmonCross</code> class. TransmonCross
Transmon Cross Flux Line  The base <code>TransmonCrossFL</code> class. TransmonCrossFL	Transmon Interdigitated  The base "TransmonInterdigitated" inherits the "QComponent" class. TransmonInterdigitated	Transmon Pocket  The base <code>TransmonPocket</code> class. TransmonPocket	Transmon Pocket 6  Transmon pocket with 6 connection pads. TransmonPocket6



LFL-Lab @ USC · MIT

Validated qubit-design database + physics-based parameter interpolation. Search the DB, interpolate to your target Hamiltonian, generate the design in Quantum Metal.

Published in Quantum journal (Sept 2024). Includes an MCP server for AI agents.



Fermilab + Northwestern

ML-based inverse design — predicts Quantum Metal design parameters from target qubit, resonator, coupler, and Hamiltonian properties using multi-layer perceptrons.



Northwestern

Python toolkit for AWS Palace — Quantum Metal gmsh export, JSON config builders, mesh utilities, LOM analysis.



SQD Lab @ UQ · Apache 2.0

Simulation wrapper for `QDesign` → AWS Palace / COMSOL workflows. Eigenmode, capacitance, driven, inductance. Accepts Quantum Metal `QDesign` objects directly.



Onri Jay Benally · CC-BY-4.0

End-to-end workflow from Quantum Metal design through electron-beam lithography. Quantum Metal + PHIDL + GDSTK + CuPy + KLayout + Blender. Headless Colab install scripts. Fab-facing process docs (EBPG / BEAMER). Real fabricated chip examples.

This list grows with the community. If you're building on Quantum Metal in a research project, an internal tool, an educational notebook, or a startup, we'd genuinely love to know — [open an issue](#) and we'll add you.

The important thing is not to stop questioning. Curiosity has its own reason for existence.

One cannot help but be in awe when he contemplates the mysteries of eternity, of life, of the marvelous structure of reality.

It is enough if one tries merely to comprehend a little of this mystery each day.

Albert Einstein

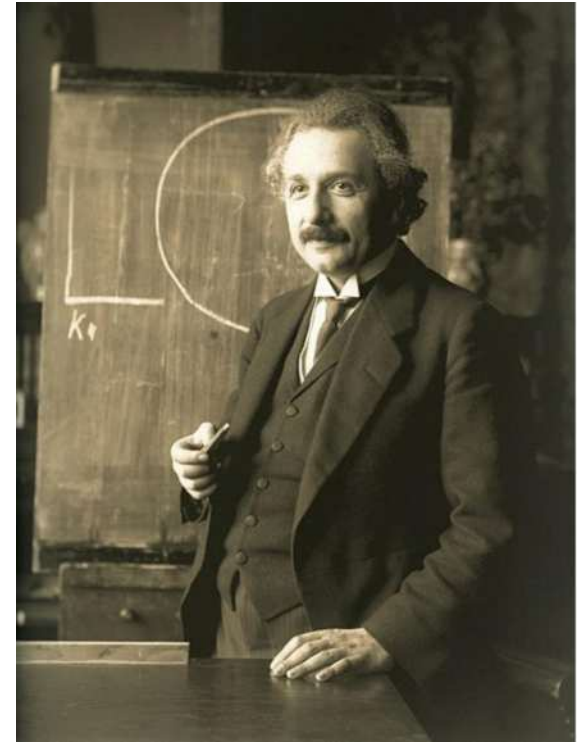


Photo: F. Schmutzer



@zlatko_minev



zlatko-minev.com